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Aerodynamic Optimization using CFD for a Hybrid eVTOL in Cruise

Tesi di Laurea Magistrale in
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Author: **Leonardo Ungureanu**

Student ID: 10730172
Advisor: Prof. Giuseppe Quaranta
Co-advisor: Prof. Wang Liang (Tsinghua University)
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Abstract

The development of electric Vertical Take-Off and Landing (eVTOL) aircraft promises to transform urban air mobility, yet their cruise efficiency remains a critical challenge. In hybrid quad-rotor configurations, such as the W90, front rotors operating in edgewise flow during cruise induce a local downwash effect over most of the wing span, degrading aerodynamic efficiency and increasing power consumption. This study investigates whether wing geometry modifications – specifically twist distribution and longitudinal positioning – can mitigate this interference and reduce cruise power demand.

A CFD-RANS framework is implemented in *STAR-CCM+* using the Virtual Disk Model (VDM) as the primary modeling tool. Wing twist distribution is optimized via a surrogate-based framework (Kriging surrogate, Genetic Algorithm) implemented in *Dakota*, and multiple design strategies are evaluated to assess their impact on vehicle power consumption.

The baseline analysis reveals that the front rotors induce a local downwash effect, reducing the wing and the total vehicle lift-to-drag ratio respectively by 16.9% and 10.6%, compared to the rotor-off condition. Twist distribution optimization alone yields only marginal improvements (+2.4% in wing L/D) when wing lift is fixed, with negligible changes in power consumption. Forward wing displacement amplifies front rotor interference and is detrimental. In contrast, the lift-based design strategy, combined with a four-parameter trim optimization (vehicle pitch angle, rotor shaft angle, front and rear rotor rotational speeds), achieves a 2.8% reduction in total power consumption according to VDM estimates. This demonstrates that, even though rotor-wing interference cannot be eliminated, redistributing the vertical force burden toward the wing proves effective in improving cruise efficiency. The alternative dual-system configuration shows potential, but the zero-pitch baseline case reveals inconsistencies and remains inconclusive.

A model uncertainty assessment reveals that VDM and SMM predictions are consistent in evaluating aerodynamic loads (deviations below $5-10\%$), though they differ substantially in rotor torque and power estimates ($15-25\%$). Since neither method has been validated against experimental data, results carry inherent uncertainty, and wind tunnel testing remains an essential next step.

Keywords: eVTOL, hybrid quad-rotor, rotor-wing interaction, aerodynamic optimization, surrogate-based optimization, genetic algorithm, Virtual Disk Model, Sliding Mesh Method, CFD, STAR-CCM+.

Abstract in lingua italiana

Lo sviluppo di velivoli eVTOL (electric Vertical Take-Off and Landing) promette di trasformare la mobilità aerea urbana, ma l'efficienza in crociera resta una sfida. Nelle configurazioni ibride quadrirotore, come il W90, i rotori anteriori che operano con flusso obliquo nel piano del rotore durante la crociera generano un *downwash* locale su gran parte dell'apertura alare, degradando l'efficienza aerodinamica e aumentando la potenza richiesta. Questo studio valuta se modifiche geometriche all'ala – in particolare la distribuzione di torsione alare e la posizione longitudinale – possano mitigare tale interferenza e ridurre la potenza necessaria in crociera.

È implementata una struttura CFD-RANS in *STAR-CCM+* con il *Virtual Disk Model* (VDM) come strumento di modellazione principale. La distribuzione di torsione alare è ottimizzata mediante un approccio basato sui modelli surrogati (surrogato Kriging, algoritmo genetico) implementato in *Dakota*. Parallelamente, diverse strategie progettuali sono valutate in termini di impatto sulla potenza richiesta dal velivolo.
L'analisi della configurazione di base mostra che i rotori anteriori inducono un *downwash* locale che riduce l'efficienza aerodinamica dell'ala e del veicolo rispettivamente del 16.9% e del 10.6% rispetto alla condizione senza rotori. L'ottimizzazione della sola distribuzione di torsione alare produce miglioramenti limitati (+2.4% nell'efficienza alare) con variazioni trascurabili della potenza richiesta. L'avanzamento dell'ala amplifica l'interferenza dei rotori anteriori ed è deleterio. Al contrario, una strategia progettuale basata sull'incremento della portanza alare, combinata con un'ottimizzazione dell'assetto a quattro parametri (incidenza del velivolo, angolo di calettamento del rotore, regimi di rotazione anteriori e posteriori), consente una riduzione del 2.8% della potenza necessaria, dimostrando che la ridistribuzione del carico verticale verso l'ala è efficace nel migliorare le prestazioni in crociera. La configurazione alternativa *dual-system* mostra potenzialità, ma la configurazione di base ad assetto nullo evidenzia incongruenze e risulta inconclusiva.
Una valutazione dell'incertezza della modellazione rivela che VDM e SMM (*Sliding Mesh Method*) forniscono previsioni coerenti riguardo ai carichi aerodinamici (deviazioni inferiori al $5-10\%$), mentre divergono ampiamente nelle stime di coppia e potenza ($15-25\%$). Poiché nessuno dei due metodi è stato validato sperimentalmente, i risultati sono intrinsecamente incerti e prove in galleria del vento rimangono un passo essenziale.

Parole chiave: eVTOL, quadrirotore ibrido, interferenza ala-rotore, ottimizzazione aerodinamica, ottimizzazione basata sul surrogato, algoritmo genetico, modello di disco virtuale, metodo sliding mesh, CFD, STAR-CCM+.

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1 | Introduction

1.1. Urban Air Mobility (UAM) and eVTOL aircraft

Nowadays, urban transportation is increasingly strained by traffic congestion, as cargo, personal, and taxi transport operate on the ground, hindering urban mobility. Future developments require the creation of a modern, integrated three-dimensional transportation system [23], driving the development of the low-altitude economy. This economic model comprises vehicles flying at altitudes between zero and one thousand meters, from manned to unmanned systems [23]. In an urban context, both airplanes and helicopters are ill-suited for transportation, as the former require dedicated take-off and landing infrastructures, while the latter are extremely noisy and raise safety concerns at very low altitudes. In this framework, electric Vertical Take-Off and Landing (eVTOL) vehicles have been developed to enable vertical take-off and landing and are fully electric or hybrid-electric. These vehicles are designed to enable safe and efficient air operations in metropolitan areas for both passenger-carrying and uncrewed aircraft systems [38], and they can be used for many other applications such as cargo delivery and urban management [23].

The origins of VTOL aircrafts lie in the first half of the twentieth century, when multiple non-conventional helicopter concepts were developed to overcome the inherent limitations of conventional helicopters [7], including limited cruise speed due to compressibility effects and retreating blade stall. This field evolved through compound and tilt-rotor configurations, designed to combine hover and vertical take-off and landing capabilities with the high-speed cruise performance of fixed-wing aircraft [7]. For a comprehensive review of the most relevant events in the twentieth-century history of unconventional helicopters, the reader is referred to [7]. Building on the knowledge gained during the previous century and on advances in electric or hybrid propulsion, the modern eVTOL concept emerged in 2009, when the U.S. startup Joby Aviation was founded [23]. In 2011, Italian helicopter manufacturer AgustaWestland formally introduced the eVTOL concept, while in 2016 Uber launched its “Uber Elevate” urban air mobility initiative, which proposed the use of eVTOL aircraft for urban aerial taxi services [23]. In recent years, airworthiness standards have been revised by the U.S. FAA and EASA to support eVTOL certification, paving

the way for a future in which eVTOLs will be integrated into ordinary transportation. However, regulations still represent the main obstacle to the full-scale implementation of eVTOLs. According to [40], hundreds of eVTOL concepts have been unveiled, but only a few have achieved first flight, reflecting the relative immaturity of this industry. It is notable that 70% of the concepts under development are conceived in China, the USA, and Europe [38].

The value of eVTOLs stems from their versatility, which makes them suitable for a wide range of applications such as air taxi, policing, firefighting, air ambulance missions, surveillance, and express logistics. Moreover, they could help relieve ground traffic congestion and improve the efficiency of goods distribution [23, 38]. The low-altitude economy is expected to grow rapidly, and the global market size for UAM (Urban Air Mobility) is estimated to exceed 80 billion dollars by 2035 [38].

In this context, energy efficiency, operational range, and acoustic pollution are of primary importance for the integration of eVTOL vehicles into everyday life. These aspects strongly depend on the rotor-airframe interactions, which degrade performance and increase noise. In this study, the W90 concept is investigated in terms of power consumption. Since it is a hybrid configuration, featuring eight rotors arranged on four axes and a fixed wing, rotor-wing interference is the main source of performance loss. Throughout the present work, this phenomenon is analyzed, and modifications to wing positioning and twist distribution are employed to improve the vehicle efficiency in cruise. Further details are provided in the following section.

1.2. W90 hybrid quad-rotor configuration and research objectives

A convertiplane is an aircraft that combines hovering and vertical take-off and landing capabilities, typical of helicopters, with high-speed cruise flight, typical of airplanes. In the literature, these hybrid aircraft are typically classified into four subtypes: tilt-rotor, tilt-wing, rotor-wing, and dual-systems [32]. Although this classification is proposed by [32] for miniature UAVs (Unmanned Aerial Vehicles), it also applies for full-scale VTOLs. Tilt-rotors are designed to provide lift in hover and thrust in cruise. In the latter condition, the aircraft operates as a conventional aircraft, with the fixed wing contributing to vertical equilibrium. Depending on the distribution of the propulsive system, they can be further subdivided into: bi-rotor, tri-rotor, and quad-rotor. The latter can be further divided into '× and '+' layouts depending on the arrangement of the rotors [32]. In tilt-wing

convertiplanes, the wing is partially or entirely deflected together with the rotors during the transition phase. Rotor-wing (or stop-rotor) configurations employ a rotary wing that spins to provide lift during vertical flight and stops during cruise, acting as a fixed wing. The last type is the dual-system convertiplane, which features two separate sets of propulsion systems: rotors to provide vertical thrust only, and a forward pusher for cruise.

The configuration differences among these convertiplanes are the result of decades of concepts developed to overcome the key helicopter limitations: bounded cruise speed due to compressibility effects on the advancing blade and large stall regions on the retreating blade. Moreover, UAM raised further concerns regarding efficiency, noise, and safety, rendering twentieth-century VTOL developments attractive even for lower-speed flight. The first conceived concept is the conventional tilt-rotor, which features two rotors mounted on wing tips that tilt to provide vertical and forward thrust. The main drawback of this configuration lies in hover performance. When the aircraft is hovering, the wing acts as a bluff body in the rotor-induced wake, generating negative lift with a magnitude of up to 10% of the generated thrust [22], or even higher [8]. The tilt-wing concept minimizes this phenomenon, known as download, even when only the outer part of the wing is deflected, drastically reducing the wing surface on which the rotor wake impinges. However, in addition to increased mechanical complexity, the tilted wing in hover configuration introduces a non-negligible longitudinal force, thus complicating stability in both hover and transition phases [8]. The quad-rotor (or quad-plane) design reduces rotor-wing interference in hover by introducing geometric separation between the wing and the rotors. However, cruise performance is degraded by the destructive interaction between the wing and the front rotors. These induce a downwash effect that alters the wing lift and drag distributions [9, 20].

The configuration considered in this study, the W90 (Figure 3.1), can be classified as a hybrid quad-rotor (or quad-plane) eVTOL, featuring eight rotors mounted on four axes arranged in '×' layout and a fixed wing. However, it is not a proper convertiplane, as it is not designed for high cruise speeds. Nevertheless, in this condition, rotors are highly pitched, operating in edgewise flow, providing both lift to support the wing and forward thrust. As a rare configuration [32], it has limited support in the literature for investigating the characteristics of this aircraft. A similar concept is analyzed in terms of aerodynamic interference between the rotors and the airframe [9], while an experimental investigation of a miniature quad-plane is provided by [20]. Finally, the impact of the relative positioning between the wing and the rotors is numerically assessed by [2].

The lack of investigations on this kind of aircraft is mainly due to the infancy of the

eVTOL industry and to cruise performance limitations caused by rotor-wing interference. Similar concepts address this issue through geometric vertical separation between the wing and the rotors [10, 27]. The scope of this study is to assess whether cruise performance can be improved by modifying the wing longitudinal position and twist distribution, without significantly altering the vehicle design. While twist distribution optimization in this specific configuration has not been investigated in the literature, the impact of the relative distance between the wing and the rotors has been studied by [2]. However, the W90 has a unique design and operates at a different angle of attack from that considered in [2], making it reasonable to revisit the investigation.

1.3. Methodology overview and thesis outline

The aim of the present study is to enhance the cruise performance of the W90. Due to the complexity of rotor-airframe interaction, low-fidelity and mid-fidelity methods are ill-suited for this analysis. Although a mid-fidelity approach can, in most cases, predict qualitative aerodynamic behavior, quantitative results show that the uncertainty associated with lift and drag estimations is generally high, reaching 30% deviations in some configurations [25]. Since this study requires optimization, the solver must be able to capture even minimal geometry variations while still providing reliable results. Therefore, CFD (Computational Fluid Dynamics) RANS (Reynolds-Averaged Navier-Stokes) modeling is selected, as it generally allows satisfactory load estimation.

The analysis is conducted using the CFD solver *STAR-CCM+* by *Siemens*, in which rotor modeling is performed through the Virtual Disk Model (VDM), which combines the actuator disk model with the blade element method. This model, whose reliability has already been widely demonstrated in the literature [6, 18], is validated in Section 3.3.1 against high-fidelity axial-flow results computed with the Sliding Mesh Method (SMM) by [41]. However, due to the lack of experimental data, no further validation of the VDM is conducted under operative conditions, i.e., in forward flight. Therefore, an uncertainty assessment study is carried out in Section 4.8 by comparing results obtained with VDM and SMM for three different configurations. Again, since no experimental data are available, it cannot be stated which method provides the most accurate estimate, and the aforementioned analysis is not to be interpreted as a validation of the VDM against the SMM, but rather as an uncertainty assessment.

In Section 3.4, the complete setup is described, and the flow-field characterization confirms that the primary cause of performance loss is the front rotor wakes interfering with the wing [9, 20], which locally modifies the effective angle of attack and reduces it over most

of the wingspan. Therefore, as anticipated in the previous section, twist distribution and wing longitudinal position are chosen as design variables, since twist is expected to be the most effective parameter for mitigating local downwash, while wing positioning varies the wing-rotor interference effects. These parameters are employed in Chapter 4 to diminish the overall vehicle power consumption through multiple strategies. The first approach consists of modifying the twist distribution to counteract the rotor-induced local downwash, thus restoring the wing lift distribution without rotors (Section 4.3). In the following section, Section 4.4, numerical optimization is used to minimize wing drag while maintaining the wing lift at the baseline value. The same optimization setup is used in Section 4.5 to design multiple wings with different lift requirements and to investigate whether a lift increase or decrease, which changes the thrust demand, can ultimately lead to a reduction in power consumption. Note that all the configurations tested require a cruise equilibrium search by adjusting the global vehicle pitch angle or the rotor shaft angle, and the front and rear rotor rotational speeds. Then, in Section 4.6, the impact of the wing's longitudinal position is assessed. Finally, the last design strategy consists of a low-fidelity investigation, where the W90 is modified into a dual-system by installing an additional forward pusher. In conclusion, three characteristic configurations encountered during the various design approaches are simulated with the SMM to assess the uncertainty associated with the VDM model, Section 4.8.

The aforementioned optimization setup, implemented within the *Dakota* framework, is discussed in Section 4.2. An offline surrogate-based method is selected to minimize computational costs while maintaining adequate accuracy. As a search method, the Genetic Algorithm (GA) is selected due to its flexibility and validated use in the literature [3, 4, 15]. Further details and motivations behind these choices are reported in the dedicated section.

2 | Theoretical Background

2.1. Rotor Modeling

Rotor modeling is particularly important because of the prohibitive computational cost required to resolve the flow field around the actual rotating blade. Besides the need for a time-accurate simulation, a dense body-fitted mesh is required near the blades, substantially increasing the mesh size. The actuator disk model represents the earliest approach, providing closed-form expressions for thrust and power under idealized assumptions; however, these assumptions considerably reduce its range of applicability and reliability. The blade element approach considers the rotor's influence in terms of the momentum it imparts to the surrounding fluid. In particular, the rotor is modeled as a distribution of momentum sources whose intensities and variations are determined from the rotor geometry and the flow properties [29].

In this section, the actuator disk theory is presented, as it establishes the physical foundation for the momentum source approach. Subsequently, the blade element method is described with particular attention to its implementation in *STAR-CCM+*, following the formulation of Rajagopalan [29]. Finally, the time-accurate sliding mesh method is discussed.

2.1.1. Actuator disk model

The actuator disk model represents the first attempt to describe the interaction between a rotor and the surrounding flow without resolving the detailed blade geometry. Introduced by Rankine in the nineteenth century and subsequently developed by Froude and Betz, this model replaces the rotor with a permeable disk of infinitesimal thickness, across which a pressure discontinuity is assumed while the normal velocity remains continuous [37].

Consider a steady, incompressible, inviscid flow with freestream velocity V_∞ normal to a disk of area A. Let $\Delta p = p^+ - p^-$ denote the pressure jump across the disk, where p^+ and p^- are, respectively, the upstream and downstream pressures relative to the disk, so

that the thrust generated by the rotor is given by:

$$T = \Delta p \cdot A. \tag{2.1}$$

Consider now a control volume bounded by the surface at infinity enclosing the entire flow field and by the disk A, and apply integral momentum conservation to it, yielding:

$$T = \dot{m}[(V_\infty + w_j) - V_\infty] = \dot{m} w_j, \tag{2.2}$$

where w and w_j are, respectively, the induced velocity at the disk and at infinity downstream, while $\dot{m} = \rho A(V_\infty + w)$ is the mass flow rate through the disk.
Applying Bernoulli's theorem separately upstream and downstream of the disk and subtracting the two expressions yields the pressure jump:

$$\Delta p = \frac{1}{2}\rho[(V_\infty + w_j)^2 - V_\infty^2] = \rho w_j \left(V_\infty + \frac{w_j}{2}\right). \tag{2.3}$$

Substituting this expression into (2.1) and equating it to (2.2), the fundamental relation of simple momentum theory is obtained:

$$w = \frac{w_j}{2}, \tag{2.4}$$

the induced velocity at the disk is therefore half of that at infinity downstream. Substituting (2.4) into (2.2), the generated thrust reads:

$$T = 2\rho A(V_\infty + w)w, \tag{2.5}$$

while the power absorbed by the rotor equals the rate of change in kinetic energy of the fluid:

$$P = \dot{m} \cdot \frac{1}{2}[(V_\infty + w_j)^2 - V_\infty^2] = T(V_\infty + w), \tag{2.6}$$

and the ideal rotor efficiency is therefore:

$$\eta = \frac{TV_\infty}{P} = \frac{V_\infty}{V_\infty + w} = \frac{1}{1 + a}, \tag{2.7}$$

where $a = w/V_\infty$ is the axial interference factor. Simple momentum theory provides a fundamental preliminary design criterion: for a given thrust, efficiency increases with rotor diameter, since $w = T/(2\rho A V_\infty)$ decreases as A increases.
The assumptions of actuator disk theory greatly limit its applicability to complex real-

world configurations. The induced velocity w is considered constant over the entire disk, which is not valid in applications such as a rotor in forward flight, due to the load asymmetry. Rotational induction is neglected by simple momentum theory, whereas general momentum theory extends the model to account for the tangential velocity component induced by blade rotation, as described in detail in [37]. Viscosity, along with stall and separation phenomena, is neglected, and blade geometry is not accounted for.
These limitations motivated the development of the Blade Element Method (BEM), which retains the actuator disk concept as the region where source terms are applied but computes local forces based on blade geometry and aerodynamic characteristics.

2.1.2. Blade Element Method (BEM)

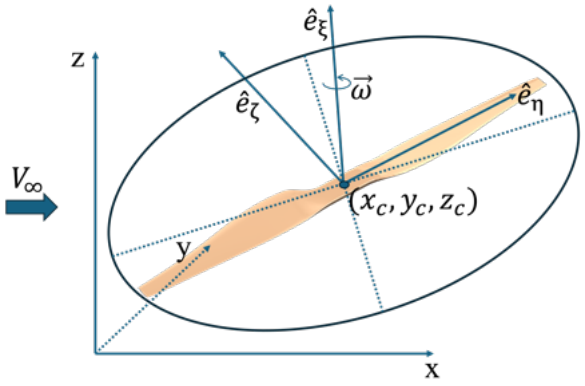
The blade element theory follows the same idea as the Prandtl lifting-line theory, modeling the rotor as a disk over which an infinite number of radial vortices of intensity $\gamma(r)$ are distributed. While the complete theoretical derivation is available in [37], a more practical discussion is presented here. In particular, the BEM implementation in *STAR-CCM+*, associated with the virtual disk model, follows the formulation of Rajagopalan [29], which is reviewed in this section.

The rotor is modeled as a distribution of momentum sources, whose strength is determined by the local flow field properties (computed by CFD), the rotor geometry, and the aerodynamic characteristics of the blade cross-section. An iterative process is employed to resolve the coupling between the disk and the surrounding flow field.
Considering steady, incompressible, laminar flow, the Navier-Stokes equations in the Cartesian coordinates of the computational domain read:

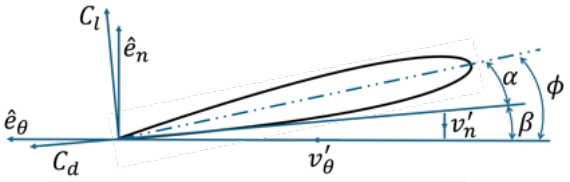
$$\begin{aligned}
&\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}+\frac{\partial w}{\partial z}=0,\\
&\rho\left(u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}+w\frac{\partial u}{\partial z}\right)=\mu\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2}\right)-\frac{\partial p}{\partial x}+S_x,\\
&\rho\left(u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}+w\frac{\partial v}{\partial z}\right)=\mu\left(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}+\frac{\partial^2 v}{\partial z^2}\right)-\frac{\partial p}{\partial y}+S_y,\\
&\rho\left(u\frac{\partial w}{\partial x}+v\frac{\partial w}{\partial y}+w\frac{\partial w}{\partial z}\right)=\mu\left(\frac{\partial^2 w}{\partial x^2}+\frac{\partial^2 w}{\partial y^2}+\frac{\partial^2 w}{\partial z^2}\right)-\frac{\partial p}{\partial z}+S_z.
\end{aligned} \tag{2.8}$$

The source terms $S_x,\ S_y,\ S_z$ in the momentum equations introduce the rotor's influence into the flow field. Their definition requires four coordinate systems:

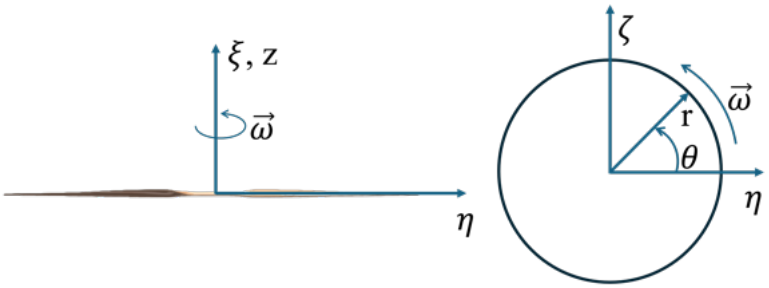
- Computational domain frame (x, y, z): the center of the disk is located at (x_c, y_c, z_c), and the disk rotates with angular velocity $\vec{\omega}$, Figure 2.1a;
- Rotor-based Cartesian frame (ξ, η, ζ): its origin coincides with the disk center, the ξ axis is aligned with $\vec{\omega}$, and the η and ζ axes lie in the rotor plane, Figure 2.1a;
- Rotor-based cylindrical frame (r, θ, z), defined as in Figure 2.1c;
- Blade deflection frame (n, θ, s), defined to account for blade flapping (Figure 2.1d).



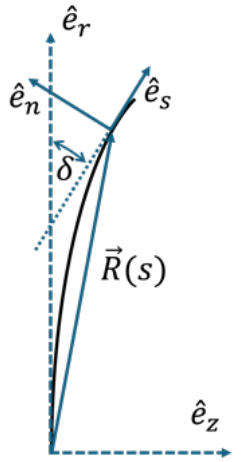
(a) Rotor-based cartesian (ξ, η, ζ) and computational domain (x, y, z) frames.



(b) Aerodynamic forces at blade section.



(c) Rotor-based cylindrical polar frame (r, θ, z).



(d) Blade curvature frame (n, θ, s).

Figure 2.1: Reference frames for the blade element method [29].

The transformations between the coordinate systems are:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} + \mathbf{M}_1^T \begin{bmatrix} \xi \\ \eta \\ \zeta \end{bmatrix}, \qquad \begin{bmatrix} \hat{e}_\xi \\ \hat{e}_\eta \\ \hat{e}_\zeta \end{bmatrix} = \mathbf{M}_2^T \begin{bmatrix} \hat{e}_r \\ \hat{e}_\theta \\ \hat{e}_z \end{bmatrix}, \qquad \begin{bmatrix} \hat{e}_r \\ \hat{e}_\theta \\ \hat{e}_z \end{bmatrix} = \mathbf{M}_3^T \begin{bmatrix} \hat{e}_n \\ \hat{e}_\theta \\ \hat{e}_s \end{bmatrix}, \tag{2.9}$$

where the complete expressions of matrices $\mathbf{M}_1, \mathbf{M}_2, \mathbf{M}_3$ are reported in [29].
Given the blade velocity due to rotation $\vec{V}_{bl} = \mathbf{M}_3(\mathbf{M}_2\mathbf{M}_1\vec{\omega} \times \vec{R}(s))$, where $\vec{R}(s)$ is the position vector, and the fluid velocity at any point s on a blade at angular position θ, $\vec{V} = (u, v, w)$, the relative blade velocity reads:

$$\vec{V}_{\text{rel}} = v'_s\hat{e}_s + v'_\theta\hat{e}_\theta + v'_n\hat{e}_n = \mathbf{M}_3\mathbf{M}_2\mathbf{M}_1\vec{V} - \mathbf{M}_3(\mathbf{M}_2\mathbf{M}_1\vec{\omega} \times \vec{R}(s)). \tag{2.10}$$

The aerodynamic forces on the airfoil section at s are determined by the component of $\vec{V}_{\text{rel}}$ lying in the plane normal to $\hat{e}_s$. The angle between this component and $\hat{e}_\theta$ is (see Figure 2.1b):

$$\beta = \arctan(-v'_n/v'_\theta). \tag{2.11}$$

The angle of attack experienced by the airfoil, given a twist angle ϕ of the section relative to the plane of rotation, is:

$$\alpha = \phi - \beta. \tag{2.12}$$

Given the angle of attack α and the magnitude of the relative velocity, the aerodynamic force coefficients C_l and C_d are determined from the airfoil polar, and lift and drag forces acting on the blade element read:

$$L = \frac{1}{2}\rho({v'_n}^2 + {v'_\theta}^2)C_l c ds, \tag{2.13}$$

$$D = \frac{1}{2}\rho({v'_n}^2 + {v'_\theta}^2)C_d c ds. \tag{2.14}$$

The aerodynamic loads can then be expressed in the (n, θ, s) frame and rotated through $\mathbf{M}_1, \mathbf{M}_2, \mathbf{M}_3$ to obtain the force vector $\vec{F}$ in the (x, y, z) frame, so that $-\vec{F}$ represents the instantaneous force acting on the fluid element at the (s, θ) location.
Finally, the time-averaged source terms (S_x, S_y, S_z) are obtained by multiplying this force by the fraction of time the blade spends in the $\Delta\theta$ sector during its complete revolution:

$$\vec{S} = \frac{B\Delta\theta}{2\pi}(-\vec{F}), \tag{2.15}$$

where B is the number of blades.

Although this derivation is presented for laminar, incompressible and steady flow, it can be generalized to be included in RANS solvers, whether steady or unsteady, compressible or incompressible. Referring to Eq. (2.15), the time-accurate source terms do not require any scaling. The source term from a given blade element is distributed over the corresponding cells in the finite volume mesh, proportionally to their cell volume [35].

2.1.3. Sliding Mesh Method (SMM)

Higher-fidelity rotor modeling is generally obtained with the Multiple Reference Frame (MRF) method or with dynamic mesh methods. The first approach consists of a steady-state calculation in which the user specifies the rotational speed in one or more mesh subdomains. Within these regions, the flow is solved using the moving reference frame equations [39], which account for Coriolis and centripetal accelerations due to rotation. The Sliding Mesh Method (SMM), instead, is a particular case of general dynamic mesh motion. It is a time-accurate computation in which one or more user-defined subdomains rigidly rotate relative to an enclosing stationary domain.
The generic transport equation for a scalar variable ϕ over a fixed control volume V reads:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V \rho\phi \,\mathrm{d}V + \int_{\partial V} \rho\phi\,\vec{u}\cdot\mathrm{d}\vec{A} = \int_{\partial V} \Gamma_\phi \nabla\phi\cdot\mathrm{d}\vec{A} + \int_V S_\phi \,\mathrm{d}V, \tag{2.16}$$

where $\vec{u}$ is the flow velocity, Γ_ϕ is the diffusion coefficient, and S_ϕ is the source term associated with ϕ. When a dynamic mesh method is employed, the transport equation must account for cell motion, yielding:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V \rho\phi \,\mathrm{d}V + \int_{\partial V} \rho\phi\,(\vec{u}-\vec{u}_g)\cdot\mathrm{d}\vec{A} = \int_{\partial V} \Gamma_\phi \nabla\phi\cdot\mathrm{d}\vec{A} + \int_V S_\phi \,\mathrm{d}V, \tag{2.17}$$

where $\vec{u}_g$ denotes the mesh velocity. For a sliding mesh, the motion is rigid inside each rotating cell zone, so that the shape and volume of individual cells remain constant in time, i.e., $V^{n+1} = V^n$. Under this assumption, and using the first-order backward difference formula, the temporal term can be discretized as:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V \rho\phi \,\mathrm{d}V = \frac{(\rho\phi)^{n+1} - (\rho\phi)^n}{\Delta t}\, V, \tag{2.18}$$

where n and $n+1$ indicate the quantities at the current and next time steps, respectively. Eq. (2.18), together with these simplifications, allows the solution in the sliding–mesh zones to be advanced in time once the rigid mesh motion has been prescribed for each zone. Since the grid itself is moving, the resulting flow field is inherently unsteady, as in any dynamic-mesh formulation [39].

2.2. Numerical Optimization: Surrogate Modeling and Genetic Algorithms

Engineering design problems often involve the minimization or maximization of one or more objective functions subject to constraints, where the objective and constraint functions are evaluated through computationally expensive simulations such as Computational Fluid Dynamics (CFD) or Finite Element Analysis (FEA). In these contexts, the choice of an appropriate optimization strategy is critical to balancing accuracy, computational cost, and the ability to explore complex design spaces.

This chapter presents the numerical optimization methods employed in the present work, covering three complementary aspects: sampling strategies, surrogate modeling, and genetic algorithms.

2.2.1. Latin Hypercube Sampling (LHS)

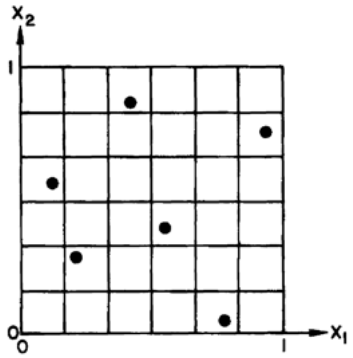


Figure 2.2: Latin Hypercube Sampling design for six points ($N_s = 6$) and two input variables (x_1, x_2) [28].

Latin Hypercube Sampling (LHS) is a stratified sampling technique introduced by McKay *et al.* [21], which ensures that all portions of each design variable x_k probability distribution are represented by input values [28]. To construct a sample of size N_s, the range of each variable is divided into N_s strata of equal marginal probability – i.e., the probability of each variable considered individually – of $1/N_s$, and a sample $x_k^{(j)}$ is retained from each stratum. For a design-variable space of dimension K, the multi-dimensional sample is assembled by randomly coupling one sample j from each variable x_k. Figure 2.2 shows an LHS design for six samples ($N_s = 6$) and two variables (x_1, x_2), illustrating how LHS ensures well-stratified coverage of the two-dimensional design space.

2.2.2. Surrogate modeling

Surrogate models are mathematical expressions constructed to replace the response of a computationally expensive function. They can be divided into parametric and non-parametric models: the former assume a fixed global functional form (e.g., a polynomial) with a limited number of parameters calibrated from the data, while the latter rely on local information and adapt their complexity to the data, yielding higher flexibility [42]. Among the most common parametric surrogate models are polynomial regression, Multivariate Adaptive Regression Splines (MARS), and Kriging (GP, Gaussian Process), while Radial Basis Functions (RBF) and Neural Networks (NN) are examples of non-parametric surrogate models. Within this section, only an overview of Kriging, RBF, and NN models is discussed; the reader is referred to [28, 42] for a detailed analysis of these and many other models.

Gaussian process regressions – Kriging

Kriging is a weighted linear combination of the training samples of the modeled function, in which the weights are determined by the spatial correlation between data points. The closer the samples are, the higher the correlation and the larger the weights [42]. Among the large number of Kriging interpolators used in the aircraft aerodynamics literature (see [42]), the following discussion concerns the Universal Kriging (UK) method, as implemented in *Dakota* [1].
Given the number of design variables n_{dv} defining a design space $\mathscr{X} \subset \mathbb{R}^{n_{dv}}$, assume the expensive modeled function $y(\mathbf{x})$, $\mathbf{x} \in \mathscr{X}$, to be a realization from a stochastic process:

$$y_{\mathrm{UK}}(\mathbf{x}) = \mathbf{f}^T(\mathbf{x})\boldsymbol{\beta} + Z(\mathbf{x}) = \mu(\mathbf{x}) + Z(\mathbf{x}), \tag{2.19}$$

where $\mu(\mathbf{x}) = \sum_{j=1}^{k} \beta_j f_j(\mathbf{x})$ is the global underlying model, representing the trend, with f_j and β_j being the basis functions and the corresponding weights, respectively. $Z(\mathbf{x})$ is the local deviation from the mean, modeling the correlation between nearby samples. In particular, $Z(\mathbf{x})$ is a zero-mean, stationary Gaussian stochastic process with covariance:

$$\mathrm{cov}\left(Z(\mathbf{x}^{(i)}), Z(\mathbf{x}^{(j)})\right) = \sigma^2 \mathbf{R}\left(\mathbf{x}^{(i)}, \mathbf{x}^{(j)}; \boldsymbol{\theta}\right), \tag{2.20}$$

where $\mathbf{R}$ is the Spatial Correlation Function (SCF), σ^2 is the variance of the process, and $\boldsymbol{\theta}$ is the set of Kriging hyperparameters.
A Universal Kriging predictor $\hat{y}_{\mathrm{UK}}$ of the objective function y, based on generalized least

squares, can be expressed as:

$$\hat{y}_{\mathrm{UK}}(\mathbf{x}) = \mathbf{f}^T(\mathbf{x})\hat{\boldsymbol{\beta}} + \mathbf{r}^T(\mathbf{x})\mathbf{R}^{-1}(\mathbf{Y} - \mathbf{F}\hat{\boldsymbol{\beta}}), \tag{2.21}$$

where $\mathbf{f}$ is the vector of basis functions, $\mathbf{R}$ is the spatial correlation matrix between all the training points, and $\mathbf{r}$ is the correlation vector between the evaluation point $\mathbf{x}$ and the training points. The remaining components are $\mathbf{Y}$, the vector of response values, $\hat{\boldsymbol{\beta}}$, the generalized least-squares estimates of the trend basis-function coefficients, and $\mathbf{F}$, the matrix of basis functions, containing the trend basis functions evaluated at all data points. Among all possible SCFs (see [42]), the most commonly preferred one – and the one implemented in *Dakota* GP – is the exponential form, which, for two training points $\mathbf{x}$, $\tilde{\mathbf{x}}$, reads:

$$R(\mathbf{x}, \tilde{\mathbf{x}}) = \prod_{j=1}^{n_{dv}} e^{\left(-|\mathbf{x}_j - \tilde{\mathbf{x}}_j|^{\rho_j}\theta_j\right)}, \tag{2.22}$$

where the components θ_j of $\boldsymbol{\theta}$ and $1 \leq \rho_j \leq 2$ are estimated using a Maximum Likelihood Estimation (MLE) procedure [1].

In summary, GP models are composed of a contribution designated to capture large-scale variations – the parametric trend function – and a realization of a stochastic process that locally stretches the surrogate toward the training data by quantifying the correlation between nearby points [42]. Besides fitting the data and yielding a highly reliable surrogate, the Kriging model incorporates variance estimation, a characteristic that allows for evaluating the uncertainty associated with each prediction and can be exploited to identify the poorly explored regions of the design space. Finally, it must be mentioned that when the number of variables or samples is excessively large, the correlation matrix $\mathbf{R}$ may become ill-conditioned, yielding numerical instability. However, specific strategies can be employed to overcome this issue; further details on the *Dakota* implementation are available in [1].

Radial Basis Functions (RBF)

The RBF model consists of approximating the target function as a linear combination of N_{RBF} radially symmetric basis functions, yielding:

$$\hat{y}_{\mathrm{RBF}}(\mathbf{x}) = \mathbf{w}^T\boldsymbol{\phi} = \sum_{i=1}^{N_{\mathrm{RBF}}} w_i \phi\left(\|\mathbf{x} - \mathbf{c}_i\|\right), \tag{2.23}$$

where w_i are the weights and $\phi\left(\|\mathbf{x} - \mathbf{c}_i\|\right)$ are the radial basis functions, whose values

depend on the distance of the evaluation point $\mathbf{x}$ from the center points, called centroids $\mathbf{c}_i$ [1]. In general, the response of radial basis functions decreases (or increases) monotonically with distance from the centroid [28]. Among all possible shapes, *Dakota* employs Gaussian radial basis functions, whose expression is:

$$\phi_i(\mathbf{x}) = \exp\left(-\frac{(\mathbf{x}-\mathbf{c}_i)^2}{\delta^2}\right) \tag{2.24}$$

The parameters of the function are the center $\mathbf{c}_i$ and the radius δ, which in *Dakota* are determined via linear least-squares solution approach [1].

Overall, an RBF interpolates the training data, with the training points often used as centroids. From Eq. (2.24), the radial basis functions yield a significant response only in the neighborhood of their centers, i.e., near the training data. Notably, RBF is particularly well-suited when the number of variables or data points is large, as, unlike GP, it does not involve matrix inversion, resulting in superior numerical stability. However, RBF does not provide intrinsic uncertainty quantification, limiting its use in adaptive sampling.

Neural Networks (NN)

Artificial Neural Networks (NNs) are non-parametric, non-linear regression models inspired by biological neural networks. They consist of a multilayered architecture of interconnected processing units (neurons), where each connection has an associated weight that is iteratively adjusted by a learning rule [42]. Among the various NN architectures, *Dakota* implements a stochastic layered perceptron (SLP) model [1], based on the direct training approach of Zimmerman [44], which allows for reduced training costs.
The form of the SLP artificial neural network can be written as:

$$\hat{y}(\mathbf{x}) \approx \tanh(\tanh(\mathbf{x}\mathbf{A}_0 + \theta_0)\mathbf{A}_1 + \theta_1), \tag{2.25}$$

where $\mathbf{A}_0$, θ_0, $\mathbf{A}_1$, and θ_1 are the matrices and vectors corresponding to the neuron weights and offset values in the NN model. These are computed during the training process via a singular value decomposition method [1].

The SLP artificial neural network is a non-parametric surface-fitting method that has been shown to be efficient in recovering steady and unsteady flow features, even with gappy data. It can also model data trends characterized by slope discontinuities, as well as multiple maxima and minima [1, 42]. However, unlike GP and RBF, the NN surface is not an interpolator, and its response at the training points may differ from the training values [1]. Moreover, neural networks are prone to overfitting, and their black-box nature

prevents the quantification of prediction uncertainty and the assessment of correlations between input variables and responses [42].

2.2.3. Genetic Algorithm (GA)

Genetic Algorithms (GAs) are derivative-free search methods employed in numerical optimization that mimic the process of biological natural selection. Originally developed by John Holland [14], their features and implementation are comprehensively discussed by Goldberg [11]. The robustness of GAs, coupled with their ability to tackle problems without restrictive assumptions about the search space and their simple implementation, renders them suitable for a wide range of applications in many different fields. Analogously to many other works in the literature [3, 4, 15], the present study employs a genetic algorithm for wing-geometry optimization aimed at maximizing the lift-to-drag ratio, subject to linear and nonlinear constraints. Connecting back to the previous theoretical background, further examples of applications include retrieving the hyperparameters $\boldsymbol{\theta}$ when constructing a Kriging surrogate or searching for candidates in infill optimization strategies [13].
This section is meant to provide a brief overview of genetic algorithms, focusing on their most important features. For a complete discussion of GA characteristics and implementation, the reader is referred to Goldberg [11].

Prior to entering the optimization process, the classical procedure in genetic algorithms requires encoding the design variables x_i, typically by transforming the values into binary strings. Subsequently, the binary strings are concatenated to form chromosomes. For example:

Binary encoding example		
Variable	**Value**	**Binary string (4-bit)**
x_1	10	1010
x_2	6	0110
Chromosome	10100110	

Table 2.1: Example of binary encoding for two design variables.

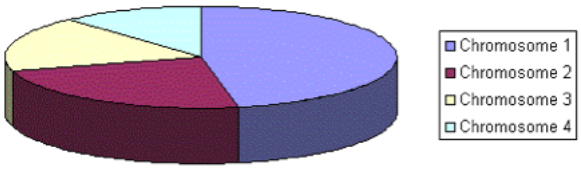
In *Dakota*, binary encoding is employed for specific genetic algorithm configurations only. In general, the software supports real-valued encoding (floating-point representation) as the standard approach for continuous optimization problems [33], in which design variables

are treated directly as real numbers rather than being converted to binary strings [26]. However, for the sake of simplicity, the following discussion is based on classical theory with binary encoding.

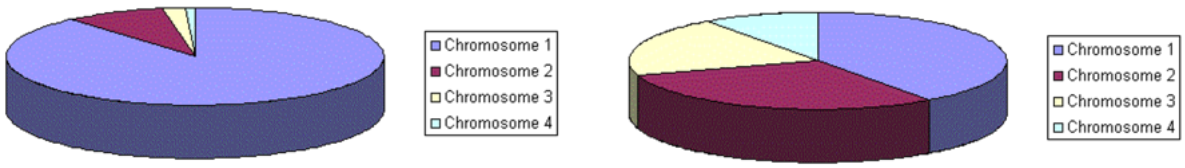
Mimicking the natural selection process, starting from a random set of chromosomes (the population), a simple genetic algorithm is composed of three parts: reproduction, crossover, and mutation [11].

Reproduction

In reproduction, exact copies of individuals (chromosomes) from the initial population are made according to their fitness function f. The function f is a measure of merit that quantifies the quality of each chromosome based on the objective value, typically adjusted to account for constraint enforcement. Copying chromosomes according to f means that strings with a higher fitness value have a higher probability of contributing one or more offspring in the next generation [11].



(a) Roulette wheel selection method.



(b) Rank-based selection method. Wheel before (left) and after (right) ranking.

Figure 2.3: Examples of selection methods in genetic algorithms: (a) roulette wheel selection, (b) rank-based selection [24].

There are several ways to select individuals for reproduction. The simplest is to create a biased roulette wheel where each chromosome in the population is assigned a slot whose size is proportional to its fitness, Figure 2.3a. By spinning the wheel multiple times, individuals occupying larger portions of the wheel will be selected more often, generating more copies to enter the crossover and mutation phases. When the fitness values of the chromosomes differ substantially (Figure 2.3b, left), individuals other than the best have only a marginal chance of being selected. To overcome this limitation, the population can

be ranked by assigning rank 1 to the worst chromosome, rank 2 for the second worst, and so on up to N (the population size) for the best one. This procedure leads to the situation depicted in Figure 2.3b (right), so that the selection probability is redistributed. Note that this redistribution causes a reduction in the convergence speed of the algorithm. A further noteworthy approach is *elitism*, which consists of copying the best chromosomes to the next generation without modification. This avoids the risk that high-fitness individuals are lost during crossover and mutation, which are random processes.

Crossover

After the reproduction phase, the selected individuals enter a mating pool to produce the next generation through crossover and mutation. Crossover consists of recombining the chromosomes of two or more randomly selected parents to generate new offspring.

Operator	Parent A	Parent B	Offspring
Single-point	**11011** | 111	11001 | **011**	11011 011
Two-point	**11** | 011 | **011**	00 | **101** | 110	11 101 011
Uniform	**1 1** 0 0 1 0 **1 1**	1 1 **0 1 1 1** 1 0	1 1 0 1 1 1 1 1
Arithmetic (*and* operator)	11001011	11011111	11001001

Table 2.2: Binary crossover operators. The vertical bar (|) indicates the crossover point(s). Retained bits are in bold.

For binary encoding, crossover can be performed in multiple ways, such as single-point, two-point, uniform, and arithmetic crossover (Table 2.2). After randomly selecting parents A and B, single-point crossover divides the chromosomes at one random position (indicated as | in Table 2.2), and generates an offspring by joining parts of the genes of the two parents. Analogously, two-point crossover randomly divides the chromosomes into three parts, while uniform crossover randomly recombines the genes of the parents. Finally, arithmetic crossover is obtained by performing an arithmetic operation between the chromosomes (the *and* operator is used in Table 2.2).
Crossover techniques depend on the encoding, and for the real-encoded variables case, the reader is referred to [26].

Mutation

Mutation plays a secondary role in genetic algorithms and in biological processes. An effective mutation rate should affect about one in a thousand bits [11], and it is even rarer in nature. However, its implementation is compulsory, as it allows for recovering potential losses in genetic material. For example, in the limit case, it may happen that, over generations, a bit in a specific position becomes zero or one for all the chromosomes. Consequently, neither crossover nor reproduction would be able to reintroduce that genetic feature into future populations. Mutation consists of a random change in the chromosome in a random position, which avoids these scenarios.

Operator	Original offspring	Mutated offspring
Bit inversion	11**0**01001	11**1**01001

Table 2.3: Bit inversion mutation operator. The mutated bit is highlighted in bold.

Table 2.3 reports an example of bit-inversion mutation, where the value of one bit is inverted. As with crossover, the mutation operators depend on the encoding. For real-encoded variables, it can be implemented as a small variation in the value of one of the variables composing the chromosome.

In summary, the genetic algorithm is an iterative method schematized as follows [24]:

1. **Start** Generate a random population of N individuals;
2. **Fitness** Evaluate the fitness function f for all chromosomes;
3. **New population** Create a new population:
    i. **Reproduction** Select the parents to generate the next generation;
    ii. **Crossover** Choose a crossover probability (e.g., 80%) and perform crossover between the selected individuals;
    iii. **Mutation** Perform mutations on a number of offspring according to a chosen mutation probability (e.g., 1%);
    iv. **Accepting** Place the new offspring in the new population;
4. **Replacement** Replace the old population with the new one;
5. **Test** Check the convergence condition (e.g., percentage change in the objective) and repeat 2 until the condition is satisfied.

The implementation of a simple GA is straightforward and extremely powerful. It does not work directly with the variables themselves, but manipulates them to assemble the chromosomes. These are further grouped into a population, which evolves through generations by means of a reproduction-crossover-mutation cycle guided by probabilistic rules. Thanks to the fact that genetic algorithms evaluate an objective function without auxiliary information (e.g., derivatives), they are highly flexible and applicable to most of the problems, regardless of the specific features of the solution space. Unlike other methods, the final output of the GA is not a single optimal solution but a population, rendering it possible to explore other suboptimal variable combinations. Finally, it should be noted that genetic algorithms are computationally expensive, although they allow extensive parallelization, and that the tuning procedure is largely case-sensitive and complicated, since it requires a trial-and-error approach.

3 | Setup and Case Analysis

In this chapter, the numerical framework is presented and applied to characterize the baseline configuration. First, the target vehicle and its rotor geometry are introduced. The computational setup is then described for both the Virtual Disk Model and the Sliding Mesh Method, covering mesh topology, domain definition, and solver configuration. The VDM is validated against reference SMM results, and mesh and time-step independence studies are conducted to ensure solution reliability. Finally, the validated framework is employed to analyze the baseline flow field, quantifying the aerodynamic impact of rotor-wing interaction on both local and integrated loads. The main findings motivate the design strategies investigated in the following chapter.

3.1. Target Vehicle

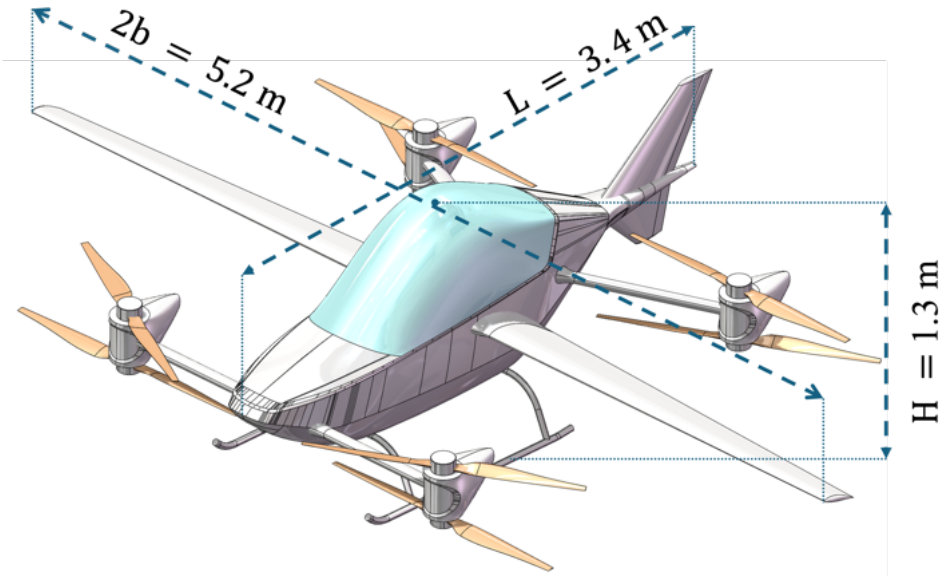


Figure 3.1: W90 target vehicle geometry.

Figure 3.1 shows the target vehicle, the W90. This ultralight eVTOL features a nominal take-off mass of 210 kg. The propulsion system consists of eight two-bladed propellers (Figure 3.2) arranged on four axes. The aircraft comprises the fuselage, vertical stabilizer, multirotor system, and a high-lift fixed wing.
During cruise, the wing provides approximately 50% of the total lift, substantially reducing the required propulsive power. Although control is mainly provided by the propulsive system, the vertical tail provides directional stability, thereby reducing rotor loading and assisting with yaw equilibrium. The blade has a radius of 0.7112 m; its twist and normalized chord distributions are reported in Figure 3.3.



Figure 3.2: D5620 rotor blade geometry.

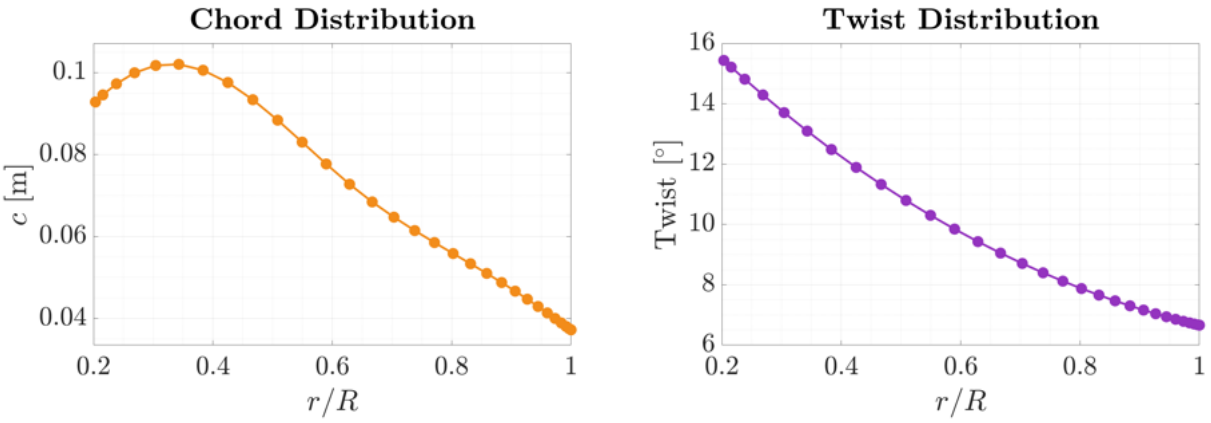


Figure 3.3: D5620 chord and twist distributions against normalized radial position.

3.2. Numerical Setup

This section describes the numerical setup. It covers the configuration of the Virtual Disk Model (VDM) and the Sliding Mesh Method (SMM), mesh topology, and solver settings. These aspects are implemented within the CFD (*Computational Fluid Dynamics*) software *STAR-CCM+* by *Siemens*.

3.2.1. Virtual disk model

The Virtual Disk Model (VDM) is a *STAR-CCM+* tool that represents propellers, turbines, rotors, and similar devices as actuator disks [35]. It allows for investigating the influence that the rotors have on the flow at a modest computational cost. Whenever the specific interaction between the fluid and the blades is not of primary interest, the virtual disk model provides reliable results without requiring high-fidelity methods such as the sliding mesh method.
As discussed in Section 2.1.2, the VDM relies on the Blade Element Method (BEM). The rotor is modeled as a distribution of momentum sources. Their magnitudes and variations are computed from the local flow field and the rotor geometry, although the blade shape is not explicitly resolved. Instead, the setup requires chord and twist distributions along the span, while the airfoil characteristics are represented solely by two-dimensional lift and drag polar data.
The VDM configuration requires the following parameters:

- **Disk Location and Orientation** The disk center and its normal vector must be specified;
- **Disk Radius and Thickness** The disk thickness need not match the physical rotor thickness; mesh requirements are the primary concern. The solver distributes the momentum source terms across cells in the direction normal to the disk via a Gaussian shape function [35]. The accuracy of the Gaussian shape function depends on the number of cell layers along the disk thickness. As their number increases, numerical noise is reduced. Regarding accuracy, only a minor impact on results is observed beyond two cell layers;
- **Chord and Twist Distribution** The chord and twist distributions along the blade span must be provided;
- **Lift and Drag Coefficients** The lift and drag coefficients of the two-dimensional airfoil are required to define the BEM. These data must be computed over a wide range of angles of attack and Reynolds numbers. Moreover, within the operating range, their accuracy has a significant impact on model fidelity. Therefore, they should be computed with high-fidelity solvers (RANS). Furthermore, the software allows these tables to be specified for multiple sections along the disk radius;
- **Tip Loss Correction** *STAR-CCM+* provides two tip-loss correction models: the *constant* model applies a uniform correction factor beyond a user-defined radial location, while the *cosine* model distributes the attenuation according to a half-

cosine function, yielding smoother and more physically realistic tip loading.

The baseline configuration features eight rotors with a radius of 0.7112 m. During cruise, the rotors lean toward the advancing direction with a $-5.6°$ shaft angle. The disk thickness is set to 0.02 m. This value approximates the physical rotor thickness while accommodating two cell layers in the disk normal direction. Thus, the stability and accuracy of the solution are guaranteed without a significant increase in the overall cell count .
Given the extensive calculations required to generate lift and drag polar data, an existing database is utilized. These values cover an angle of attack range of $[-22.5°,\ +22.5°]$ and a Reynolds number range of $[2.8 \cdot 10^5, 7 \cdot 10^5]$ for the NACA64907 airfoil. Section 3.3.1 demonstrates that the aerodynamic performance of the NACA64907 and that of the actual blade airfoil are comparable. Since the rotor features a uniform airfoil profile along its entire span, the virtual disk model is built by providing only the NACA64907 characterization.
Finally, tip-loss correction employs the *cosine* model, with a starting normalized radial position of $r/R = 0.95$.

3.2.2. Mesh topology – VDM

STAR-CCM+ offers tetrahedral, polyhedral, and trimmed-cell volume meshers. Fixed the number of cells, the tetrahedral mesher is the fastest volume mesh generator and the most efficient in terms of memory usage. However, it exhibits lower accuracy compared to the others [35]. The polyhedral and trimmed-cell mesh tools yield comparable accuracy. Between these two, the choice of mesher is driven by convergence behavior and mesh quality. The trimmed cell mesh is predominantly Cartesian, built of regular elements (predominantly hexahedral cells with minimal skewness), yielding optimal solution stability.
Moreover, the *STAR-CCM+* documentation [35] indicates that it provides good mesh quality across a wide range of cases. This reliability and flexibility make the trimmed cell mesher the preferred choice in an optimization environment, where the aircraft geometry varies substantially. Some configurations may be highly unconventional, and the robustness of the overall setup is therefore of primary importance.

Figure 3.4 shows the surface mesh generated through the automated mesh tool in *STAR-CCM+*. Certain features of the fuselage present unnecessary refinement. However, although local cell sizing is available within the software, it must be arranged manually. To use the setup for multiple configurations, full automation of the meshing process is essential. The optimal compromise lies in prioritizing workflow simplicity at the cost of a

moderate increase in mesh size.

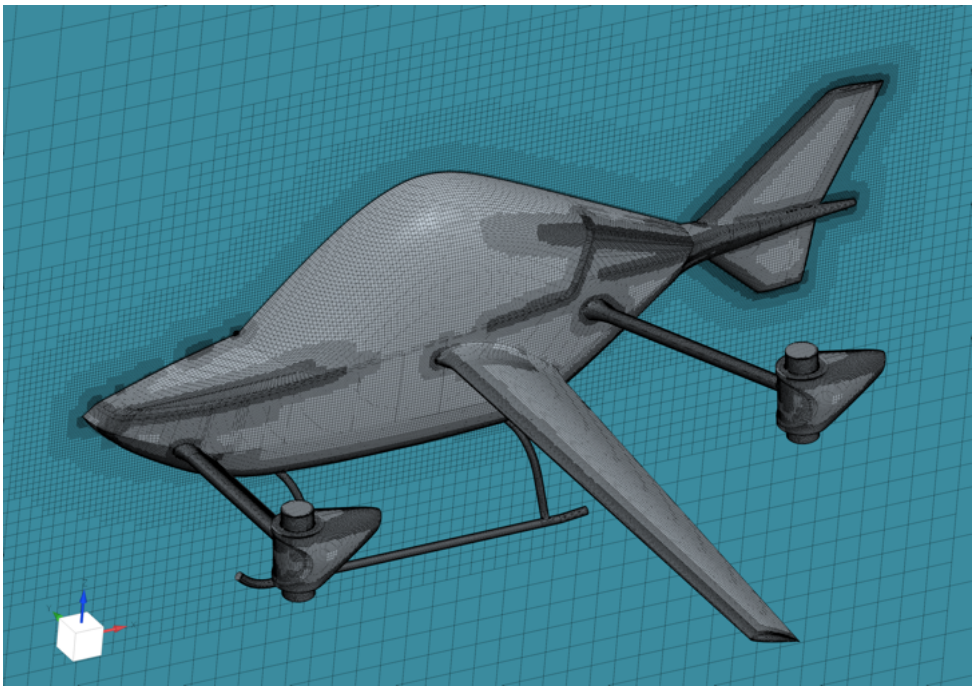
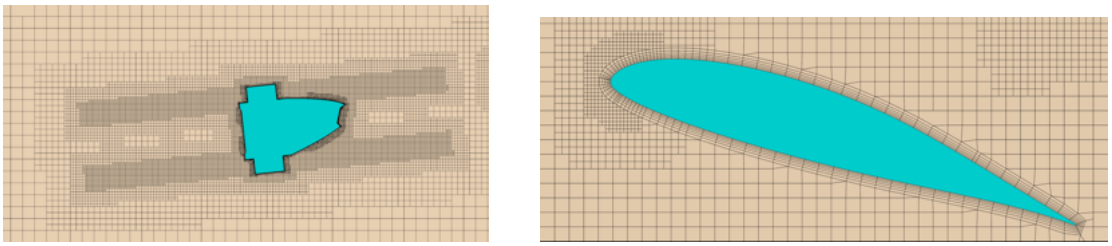


Figure 3.4: Surface mesh of the complete vehicle.



(a) Rotors refinement. (b) Boundary layer focus.

Figure 3.5: Mesh refinements and boundary layer.

The boundary layer region is discretized with orthogonal prismatic cell layers. The first-layer height is selected to satisfy y^+ requirements, while the total thickness is designed to capture the bulk of the near-wall velocity variation. The growth rate is kept below 1.2 to accurately resolve the logarithmic layer.
The interface between the prismatic cell layers and the hexahedral background mesh is handled through a trimmed transition layer. These polyhedral cells are generated by pruning hexahedral cells at one or more faces or edges (Figure 3.5b).
To provide an accurate description of the near-rotor regions, local refinement zones are defined. Figure 3.5a shows that, in the vicinity of the virtual disks, the mesh size is

constrained to maintain two cell layers within the disk thickness (see Section 3.2.1). Additionally, the cell size is limited in the rotor wake regions to ensure adequate resolution of key flow features.
Finally, the computational domain is box-shaped. Its dimensions, reported in Figure 3.6, are chosen to minimize the influence of the far-field on the solution.

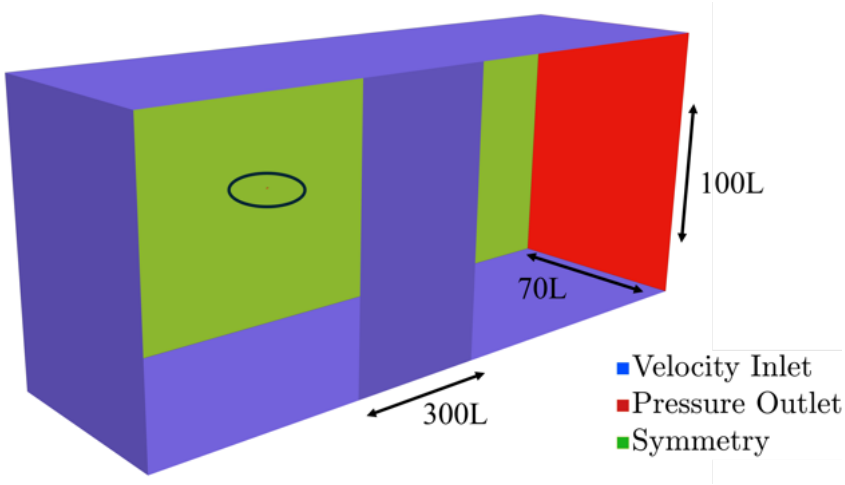


Figure 3.6: Far-field boundary conditions and computational domain dimensions, expressed in vehicle lengths. The vehicle location within the domain is highlighted.

3.2.3. Mesh topology – SMM

The SMM consists of simulating the actual propeller geometry in rotation. A dedicated subdomain is defined for each rotating component, meshed independently and moving relative to the enclosing stationary domain.

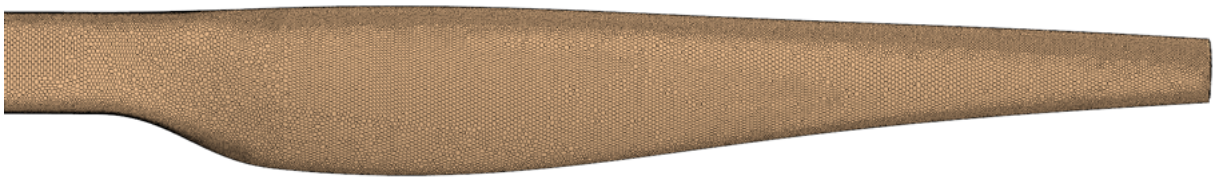


Figure 3.7: Blade surface mesh.

Away from the rotors, the mesh topology is kept as in Section 3.2.2, preserving the same y^+ requirements over the vehicle and in the refinement regions. Near the rotors, rotating subdomains are defined and meshed as follows. Within the rotating subdomains, polyhedral cells are employed in preference to trimmed cells, owing to their superior geometric flexibility and favorable convergence properties. The trimmed-cell mesher produces

pruned, low-quality cells both near the blade surface and at the rotating-stationary interface, precisely where the flow field is most complex, thereby degrading the accuracy of information transfer across it. Figure 3.7 shows the rotor surface mesh; cell density is increased near the blade leading edge to resolve the higher local surface curvature.
The volume mesh structure is illustrated in Figure 3.8. The first-layer height is set to satisfy $y^+ \lesssim 1$ over the rotor tips, where dynamic pressure and local Reynolds number reach their peak values. Beyond the boundary layer region, the cell size grows progressively toward the interface with the stationary domain. The subdomain extent is minimized while still fully enclosing the blade and providing a sufficient buffer for smooth polyhedral cell-size growth toward the interface. The cell size is enforced to assume the same value on the two sides of the interface, in the rotating and stationary domains.

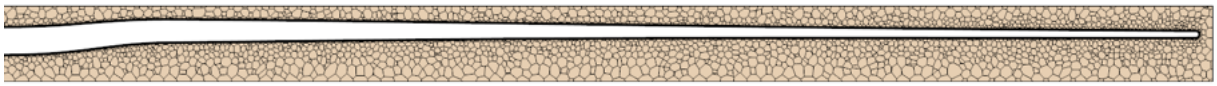


Figure 3.8: Blade volume mesh.

3.2.4. Solver configuration – VDM

The solver configuration is guided by accuracy, stability, and computational-cost considerations. In RANS (*Reynolds-Averaged Navier-Stokes*) modeling, the $k-\omega$ SST model is widely used. It combines an excellent description of the boundary layer and the outer flow with good separation-prediction capabilities. However, it has strict mesh requirements to achieve $y^+ \sim 1$. This affects the first-cell height, increasing the mesh size and complicating the mesh topology. Thus, solution stability becomes more critical. The $k-\varepsilon$ model employs wall functions to describe the viscous sublayer and buffer layer. Only the logarithmic region is resolved. The y^+ value must be within the range $[30, 300]$, allowing for a smaller mesh size and simpler topology with superior numerical stability.
As previously stated, stability and computational cost are of primary importance in the present work. Extensive calculations are required, involving multiple geometries. The $k-\varepsilon$ model yields the best compromise. This study does not involve boundary-layer or separation analysis, but it focuses on global flow phenomena. Thus, this method fulfills the accuracy requirements while maintaining a modest cell count and excellent stability.

Beyond the choice of the turbulence model, the solver is configured as follows:

- **Incompressible flow** The freestream velocity is fixed at $V_\infty = 30$ m/s, which does not justify considering compressibility effects. Although the rotor blade tips reach a Mach number Ma $\gtrsim 0.3$, the related effects are accounted for within the virtual

disk model when computing the momentum sources. Thus, the actual flow field has no regions where compressibility effects are expected to be relevant;

- **Segregated flow** The segregated flow algorithm solves the conservation equations of mass and momentum sequentially, whereas the coupled algorithm solves them simultaneously. The segregated algorithm is preferred for its lower computational cost, despite the coupled solver's superior robustness;
- **Steady-state RANS** Unsteady RANS simulations are prohibitively computationally expensive. The flow field is expected to be highly unsteady due to the mutual interaction between the front rotors, the rear rotors, and the vehicle body. However, the computational cost would severely limit the number of feasible simulations. Steady results still enable the description of the main flow features and a meaningful comparison between multiple configurations;
- **Boundary Conditions** Velocity inlet ($V_\infty = 30$ m/s), pressure outlet, and symmetry boundary conditions are displayed in Figure 3.6;
- **Symmetry** This study targets cruise conditions exclusively. The symmetry of the flight condition and the aircraft is exploited to halve the computational domain and significantly reduce computational costs;
- **Wall Treatment** *STAR-CCM+* implements a *two-layer all-y^+ wall treatment*. It uses blended wall functions to accurately resolve both high-y^+ and low-y^+ regions, modeling the viscous sublayer and buffer layer.

As a direct consequence of the symmetry assumption, unless otherwise stated, all aerodynamic and performance quantities reported hereafter pertain to the half-vehicle configuration. Full-vehicle values are obtained by applying a factor of two to the reported results.

3.2.5. Solver configuration – SMM

The solver setup for the SMM requires substantial modifications compared to the configuration described in the previous section.
The rotors operate in forward flight, leading to extensive separation over the retreating blade. As discussed in Section 3.2.4, compared with $k-\varepsilon$, the $k-\omega$ SST model is better suited for stall characterization. Hence, although it is not possible to determine conclusively the best method without validation against experimental data, the latter turbulence model is selected.
Following Section 3.2.4, the solver is configured as follows:

- **Compressible flow*** Although the freestream velocity, $V_\infty = 30$ m/s, is within the incompressible regime, the rotor blade tips reach a Mach number Ma $\gtrsim 0.3$. In this region, compressibility effects cannot be neglected. Thus, the solver is configured to allow for density variation in the flow field, under the ideal-gas assumption;
- **Coupled Flow*** The coupled algorithm solves the momentum and continuity equations simultaneously, providing more robust pressure-density coupling than the segregated approach. Hence, it is preferable in compressible flow simulations and is therefore selected;
- **Unsteady RANS*** The sliding mesh method is time-accurate; a time step corresponding to $\sim 1°$ of rotor rotation is selected to balance temporal resolution and computational cost (see Section 3.3.3). Eight inner iterations per time step are employed, ensuring convergence within each time interval;
- **Boundary Conditions** Velocity inlet ($V_\infty = 30$ m/s), pressure outlet, and symmetry boundary conditions are shown in Figure 3.6;
- **Symmetry** This study targets cruise conditions exclusively. The symmetry of the flight condition and the aircraft is exploited to halve the computational domain and significantly reduce the computational cost;
- **Wall Treatment** *STAR-CCM+* implements a *two-layer all-y^+ wall treatment*. It uses blended wall functions to accurately resolve both high-y^+ and low-y^+ regions, capturing or modeling the viscous sublayer and buffer layer.

The SMM further requires the definition of the rotating domains, the axes of rotation, and the rotation rates. Moreover, additional boundary conditions are required: one interface per rotor is defined between the rotating and stationary domains.

3.3. Validation

This section presents the validation of the VDM against reference SMM results reported in [41]. The mesh size is then selected based on a mesh-independence study. Finally, a time-step independence analysis is performed for the SMM, which shares part of the mesh infrastructure with the VDM setup.

*Modified with respect to the VDM solver configuration, Section 3.2.4.

3.3.1. Virtual disk model validation

Section 3.2.1 describes the virtual disk model configuration. This configuration is now employed to validate the VDM against SMM results from [41]. It should be noted that the SMM setup employed for the fidelity assessment in Section 4.8 differs from the reference and has not been independently validated.

As previously stated, the VDM definition requires lift and drag coefficient data. These are calculated for the NACA64907 airfoil rather than for the actual blade airfoil (the two airfoils are shown in Figure 3.10). Due to the high computational cost required to generate complete airfoil polars, an existing database is employed. Given a Reynolds number of $\mathrm{Re} = 4.75 \cdot 10^5$, the performance of NACA64907 is compared with that of the actual airfoil (Figure 3.9). Despite significant camber differences between the two blade sections (Figure 3.10), the linear region and stall angle are compatible. Thus, the database is deemed satisfactory for VDM implementation, and its suitability is further confirmed by the validation presented hereafter.

All data in Figure 3.9 are computed using the $k-\omega$ SST model to ensure the best estimates of load coefficients and stall characteristics.

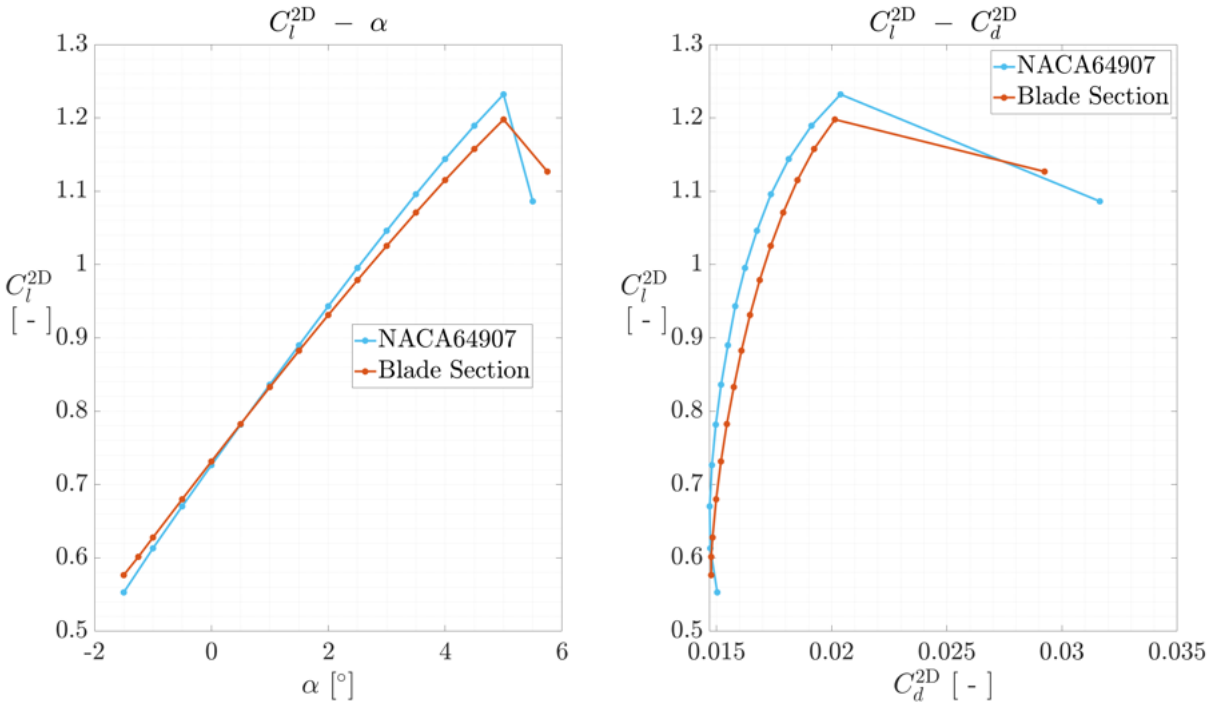


Figure 3.9: VDM airfoil polar curves. Performance of the NACA64907 and the actual blade airfoil are compared. $\mathrm{Re} = 4.75 \cdot 10^5$.

Prior to the validation of the VDM model against the SMM, the impact of the number of

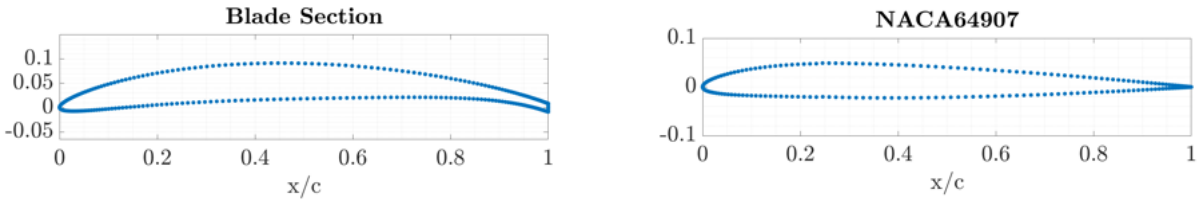


Figure 3.10: Blade section and NACA64907 airfoils.

cell layers along the disk thickness is investigated. Figure 3.11 shows the VDM thrust and torque convergence history under axial-flow conditions. Specifically, the mesh is refined to yield two or four cells across the disk thickness. The two-cell-layer simulation exhibits numerical noise due to poor approximation of the Gaussian shape function (see Section 3.2.1) in the direction normal to the disk. Although the four-cell-layer result converges more smoothly, the discrepancy in thrust and torque between the two cases do not exceed $\sim 0.5\%$. In summary, increasing the refinement along the disk thickness suppresses numerical noise while yielding negligible improvement in accuracy at a significantly higher cell count. Therefore, the use of only two cells across the disk normal direction is justified.

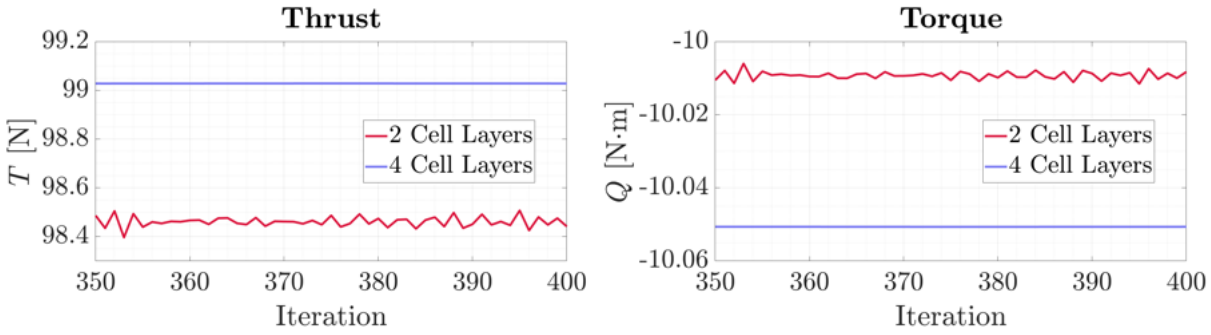


Figure 3.11: VDM thrust and torque convergence histories for two and four cell layers across the disk thickness, under axial flow conditions.

The available results [41] are computed in ANSYS Fluent using the Sliding Mesh Method (SMM). The SMM entails a time-accurate simulation, in which the domain enclosing the propeller rotates at each time step with respect to the surrounding stationary far-field. This method has been shown to exhibit excellent agreement with experimental data [36]. Torque and thrust data are provided for the single-rotor configuration [41], with the freestream velocity aligned with the rotor axis. The SMM has been employed to evaluate multiple rotor rotational speeds and freestream velocities, after being validated against experimental data. The time-resolved simulations employed the $k-\omega$ SST turbulence model. Therefore, VDM results must include calculations with both the $k-\varepsilon$, chosen for the present work, and the $k-\omega$ SST turbulence models. Thus, any deviation from the

SSM data can be unambiguously attributed either to the VDM implementation or to the turbulence model selection.

Figure 3.12 displays the comparison between the SMM and VDM results. The freestream velocity is 10 m/s, close to the expected velocity component normal to the rotor plane during cruise. First, it is notable that no significant differences are observed when comparing VDM results obtained with the two turbulence models. Thus, the virtual disk model is only slightly affected by the choice of turbulence model. Second, both thrust and torque curves show that the error increases with rotational speed. While thrust values are generally overestimated, torque values are underestimated. Finally, the focus should be placed on the expected operating rotational speed of $\sim$ 1300 rpm. Here, the VDM results are in excellent agreement with SMM results.
It should be noted that the validation condition is far from the operational configuration. As previously stated, validated SMM results are available only for axial flow and the single-rotor configuration. Instead, the target vehicle employs coaxial rotors in forward flight. Nevertheless, the agreement observed under these simplified conditions does not guarantee the same accuracy in the actual operational configuration. Yet, the overall consistency of the results justifies proceeding with the VDM, using the aforementioned configuration, as the modeling approach.

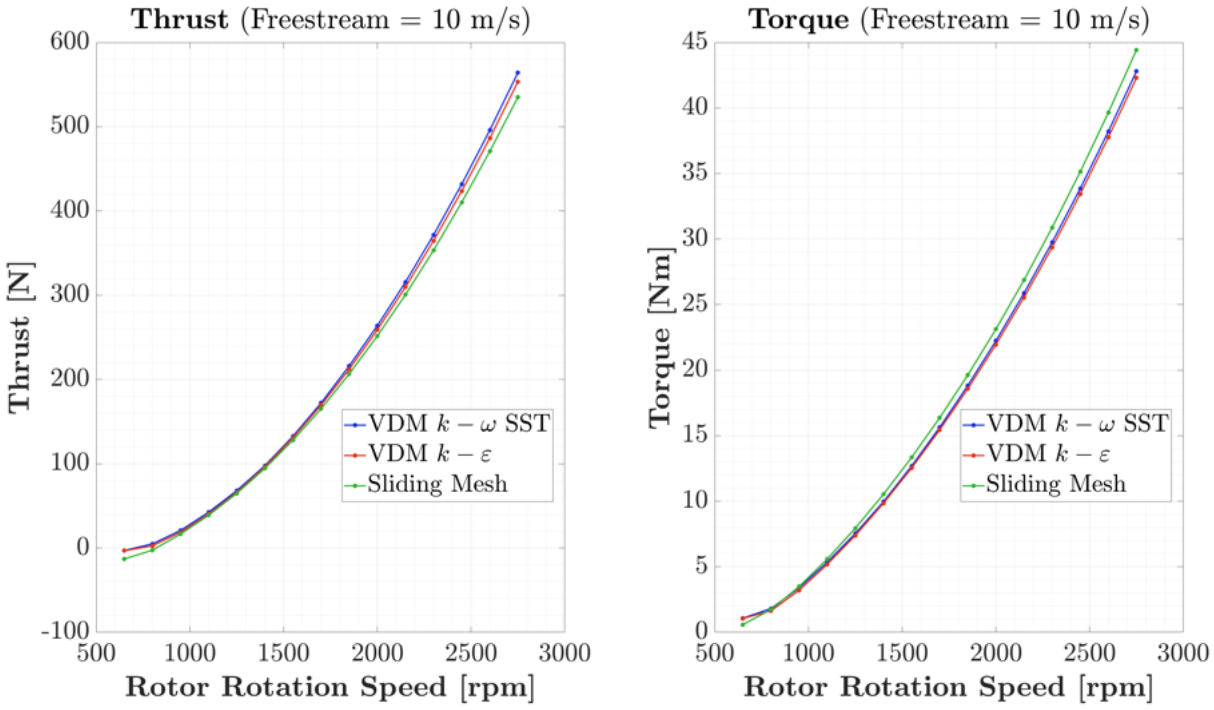


Figure 3.12: VDM Validation. Comparison between SMM and VDM results. VDM data are computed both with the $k-\varepsilon$ and $k-\omega$ SST turbulence models.

3.3.2. Mesh independence – VDM

The mesh topology is discussed in Section 3.2.2. *STAR-CCM+* enables all mesh parameters to be defined as fractions of a base size. Thus, decreasing the latter refines the mesh uniformly without altering its structure. By sequentially dividing the base size by a factor of 1.5, four different meshes are obtained. Their sizes are approximately [4, 9, 13, 24] million cells.

Figure 3.13 shows the lift and drag deviations of each mesh relative to the finest one. Given the high computational demand of this study, the 9-million-cell mesh offers the optimal compromise. It exhibits less than 5% deviation in both drag and lift values from the 24-million-cell mesh, while keeping an acceptable cell count.
Figure 3.14 displays the y^+ contours. The y^+ range is wide due to the complex flow field. However, the y^+ values lie between 30 and 100 nearly everywhere, ensuring a valid application of the wall-function approach.

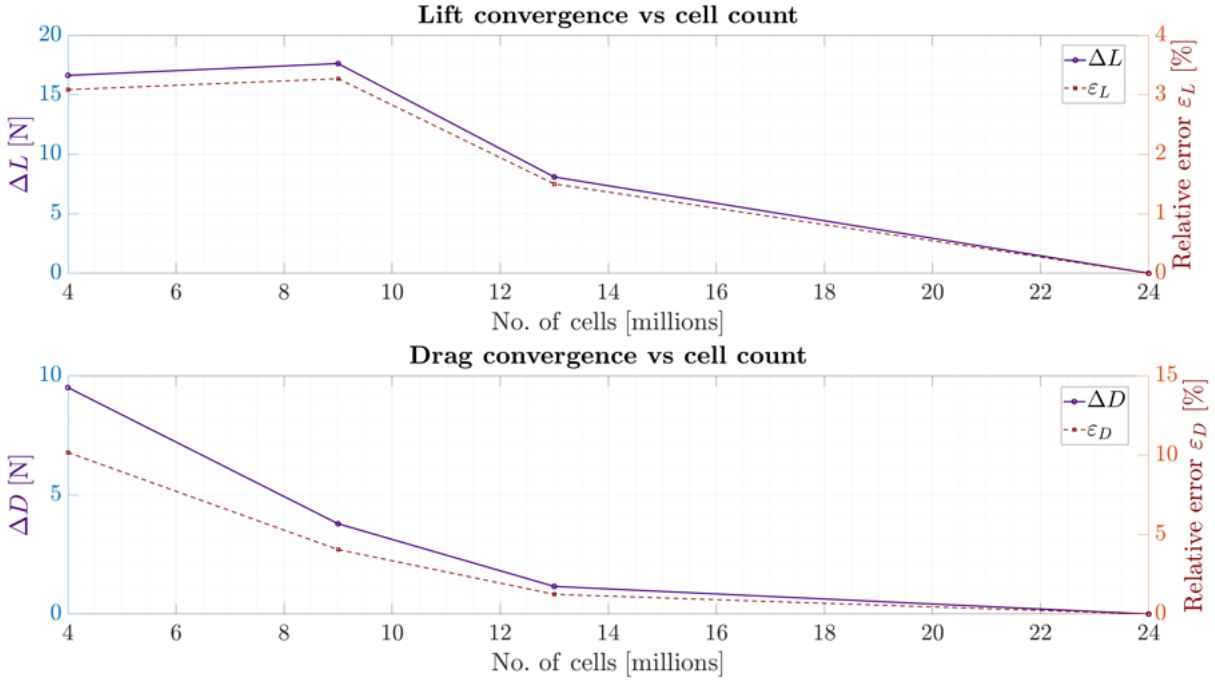


Figure 3.13: Mesh independence results. Reference values from finest mesh (24 million cells).

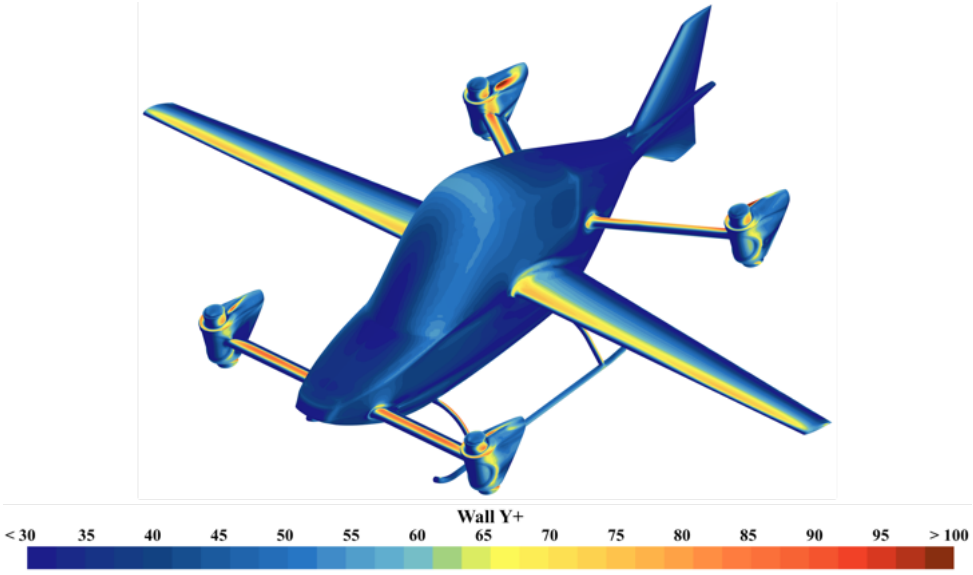


Figure 3.14: y^+ contours of the 9-million-cell mesh.

3.3.3. Timestep independence – SMM

Analogously to Section 3.3.2, mesh independence is studied by varying the mesh size near the rotors and evaluating the differences in thrust, torque, and thrust tilt angle for the four rotors across the different meshes. The base size is selected to keep the maximum error below $\sim 2\%$ in order to ensure an adequate flow description at a moderate computational cost. The detailed results are not reported here.

In addition to the mesh-independence study, a time-resolved simulation requires a time-step independence study. Figure 3.15 shows thrust, torque, and thrust tilt angle values for multiple time steps Δt. Starting from $\Delta t = 2.5 \cdot 10^{-4}$ s, the time step is halved twice, and the impact on the aforementioned values is analyzed. It is notable that $\Delta t = [2.5,\ 1.25,\ 0.625] \cdot 10^{-4}$ s correspond approximately to blade rotations of $[2,\ 1,\ 0.5]^\circ$ per Δt.

The results show that time resolution strongly affects the computed values. Due to the highly complex unsteady interactions between the rotors, the time step must ensure no more than one degree of blade rotation per time step. Figure 3.15 reveals that $\Delta t = 2.5 \cdot 10^{-4}$ s yields more than five percent error in the thrust magnitude for all rotors and up to 1° deviation of the thrust tilt angle $\Delta\delta_{Ti}$ relative to the finest time discretization tested. The torque shows only minor sensitivity to the time step.

Considering that the computational cost doubles when the time step is halved, $\Delta t = 1.25 \cdot 10^{-4}$ s yields the best compromise, as it ensures less than 5% error in thrust magnitude compared with $\Delta t = 0.625 \cdot 10^{-4}$ s, along with less than 0.5° discrepancy in thrust tilt

angle.

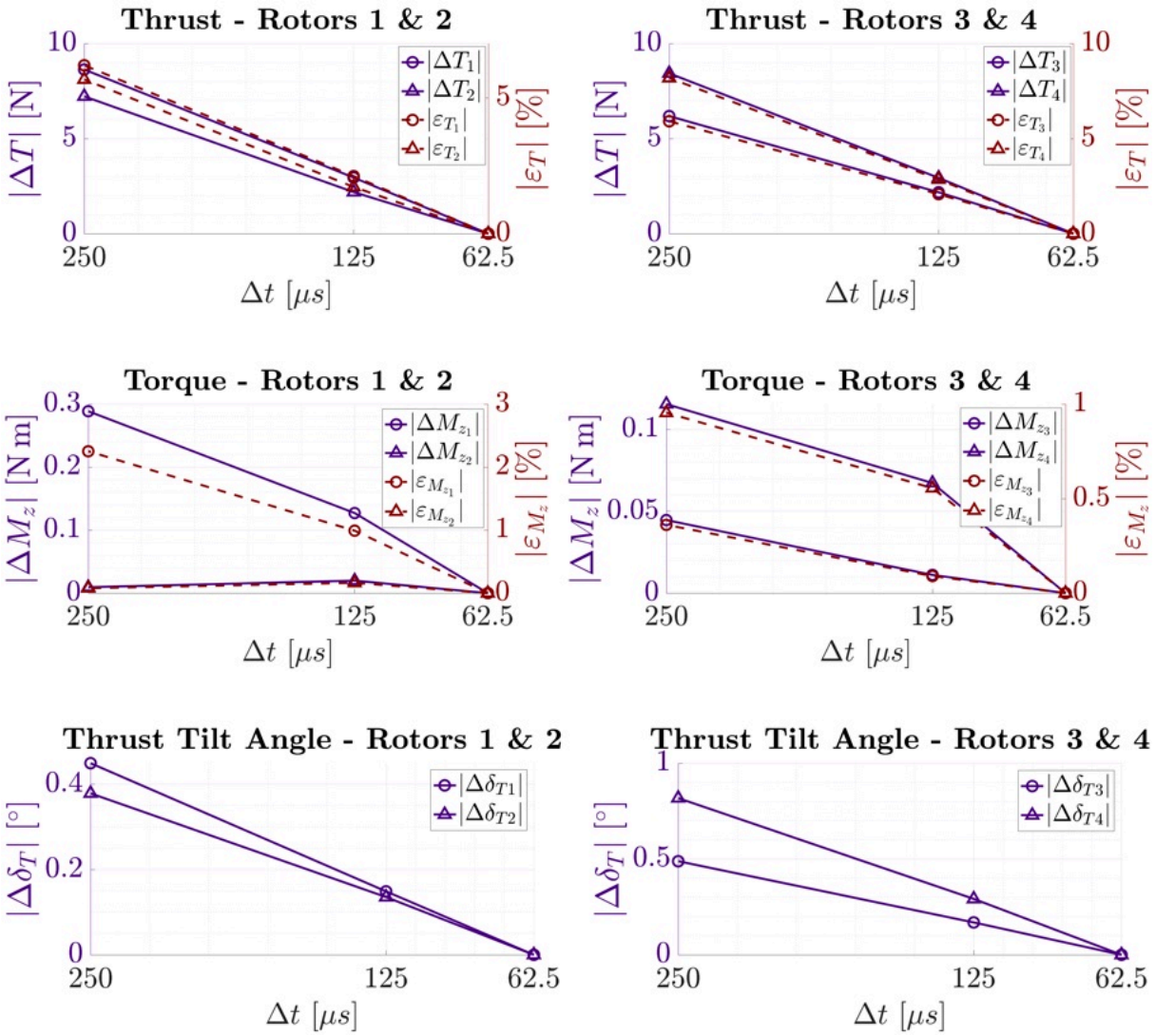


Figure 3.15: SMM timestep independence. Thrust magnitude, torque, and thrust tilt angle variations, relative to the finest time discretization, computed with different time step values Δt. The results are averaged over one period.

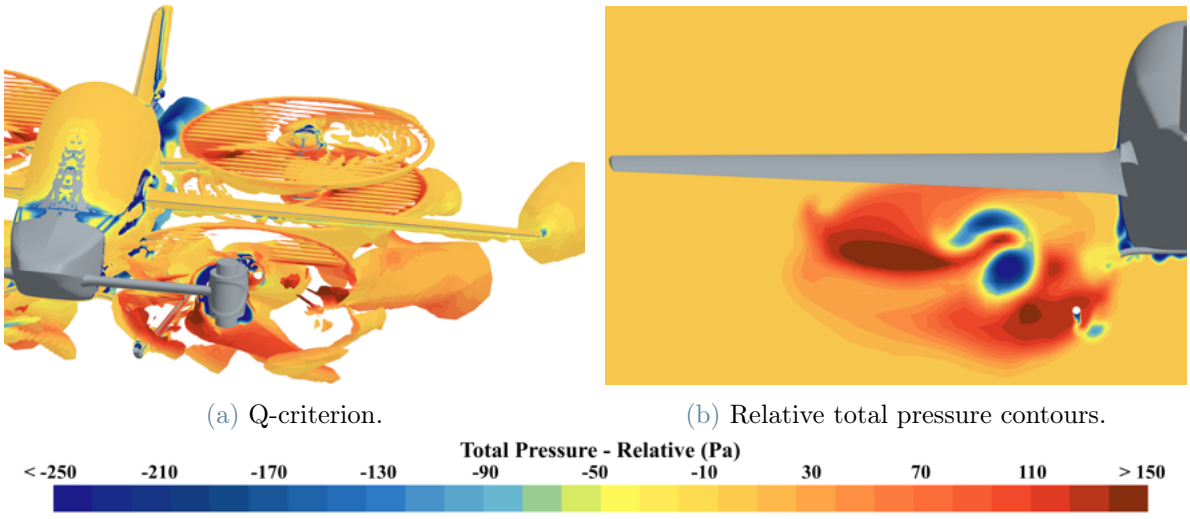
3.4. Case Analysis

The analysis presented in this section is conducted using the VDM framework described in Section 3.2.4, focusing on vehicle performance under cruise conditions. The freestream velocity is fixed at $V_\infty = 30$ m/s, and trim is achieved by adjusting the rotation rates of the front (Ω_{front}) and rear (Ω_{back}) rotors and the vehicle angle of attack α. Table 3.1 lists the values required to satisfy vertical, horizontal, and pitching equilibrium.

Trimming parameters	
$\boldsymbol{\Omega}_{\text{front}}$	1282.9 rpm
$\boldsymbol{\Omega}_{\text{back}}$	1320.5 rpm
α	-10.59°

Table 3.1: Trimming parameters for the baseline configuration.

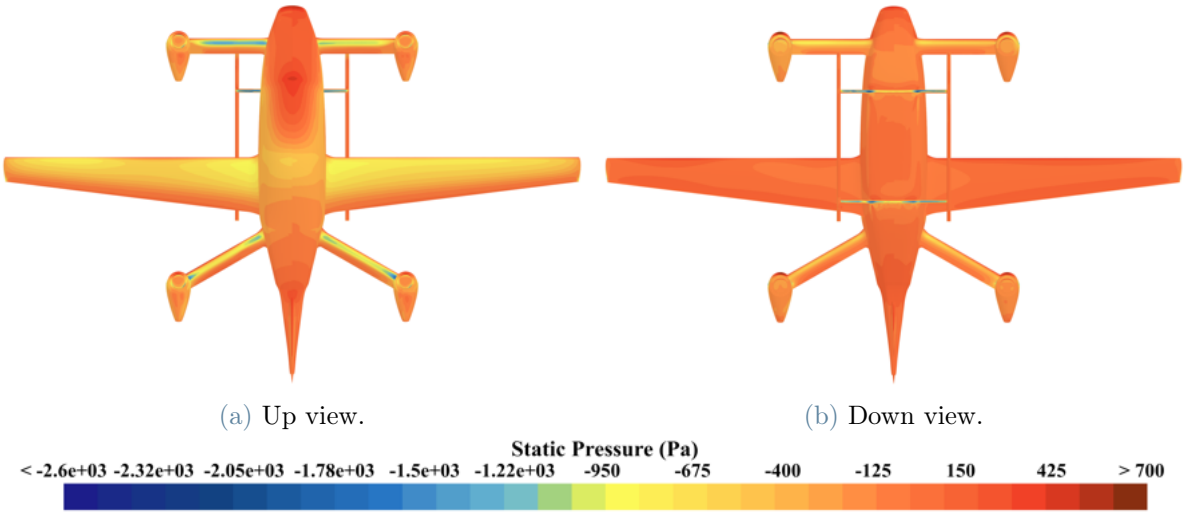
The main phenomenon governing the flow field lies in the interference between the rotors and the vehicle airframe. A reasonable first step is to provide a qualitative overview of the flow. The Q-criterion is a well-suited tool for this purpose, as its iso-surfaces enable the clear identification and visualization of coherent vortices. Figure 3.16a shows a Q-criterion iso-surface colored by relative total pressure (the reference pressure is the freestream total pressure). The fluid domain is clipped in the longitudinal direction for clarity. Notably, there is no direct interaction between the front-rotor slipstreams and the vehicle airframe, except for the landing gear. The most intense vortex structure detaches from the front rotor mounting and comprises two main vortices. The first is oriented along the freestream direction. The second is deflected downward by the rotor wake and is consequently ingested and accelerated by the lower rotor.



(a) Q-criterion. (b) Relative total pressure contours.

Figure 3.16: The Q-criterion iso-surface is colored by relative total pressure contours (3.16a); relative total pressure contours are displayed on the vertical plane (body frame) located one mean chord upstream of the wing leading edge (3.16b). The pressure reference is the freestream total pressure. Moving-rotors configuration.

Although the rotor vortex cores do not impinge on the wing, they still convect a region of higher total pressure toward its leading edge. Figure 3.16b shows total pressure contours on a vertical plane (body frame) located one mean chord upstream of the wing leading edge. Here, both of the aforementioned vortices and their locations are clearly visible. The higher total-pressure region is confined to the inboard half of the wingspan. Thus, the main aerodynamic effects on performance are expected to occur in this region.



(a) Up view. (b) Down view.

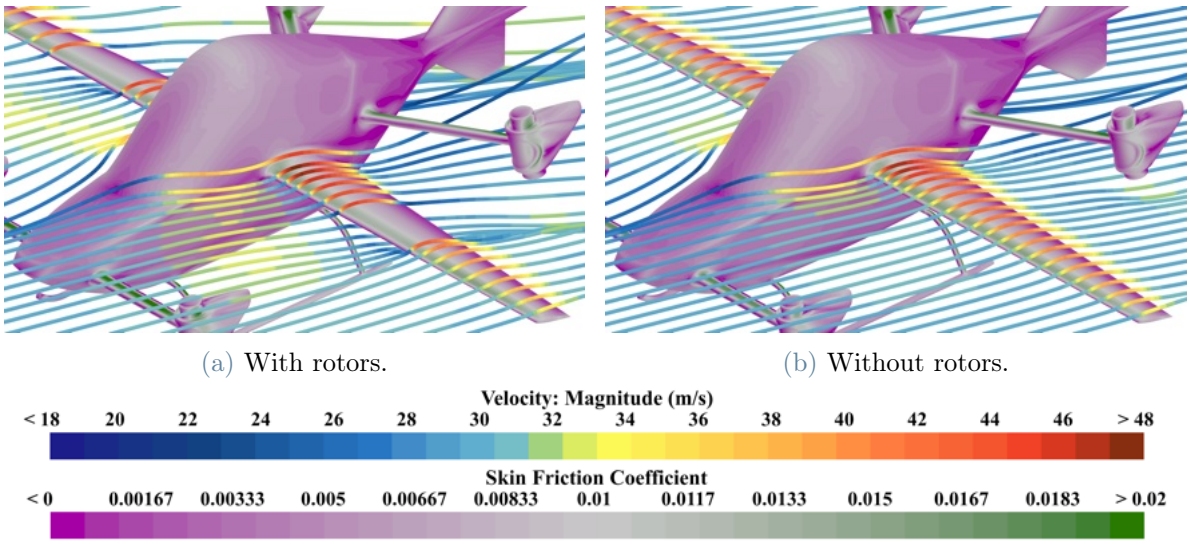
Figure 3.17: Static pressure contours in the up and down views are displayed. For each sub-figure, the same flight condition is analyzed with (left-hand side) and without (right-hand side) moving rotors.

The following approach consists of analyzing the same flight condition with and without implementing the virtual disk model. This enables the isolation of the effects due to rotor-vehicle interaction. The present work aims to modify the wing geometry to improve its performance. Nevertheless, the ensuing investigation focuses on the interaction between the wing and the rotors.

Figure 3.17 displays static pressure contours. For each sub-figure, the left and right halves show results obtained with and without rotating rotors. The fuselage static pressure is not strongly affected by the rotor implementation. The front rotor arms, however, experience a considerably lower pressure value, as they are immersed in the rotor wakes. Regarding the wing, the main focus is on the mid-span. As discussed in the previous paragraph, this is the region where total pressure is increased. The contours show that, on the upper side of the wing (Figure 3.17a), static pressure follows the same trend as total pressure, while a marginal decrease is observed on the lower surface. Conversely, toward the tip, static pressure is slightly reduced on the upper side of the wing and increased on

the lower side. Were this phenomenon solely due to rotor-wake interference, a broadly symmetric response across the intrados and extrados would be expected. Instead, these effects are attributed to a local change in effective angle of attack induced by the rotor wakes, affecting both the local velocity and pressure fields.

A further qualitative description is needed to understand the change in local angle of attack. Figure 3.18 again compares the same flight condition with and without rotating rotors. The streamlines are colored by velocity magnitude, while skin-friction-coefficient contours are shown on the vehicle airframe. Figure 3.18a clearly shows that, at mid-span, streamlines are directed downward, while those toward the tip are directed upward. Accordingly, it can be stated that the rotor wakes induce both downwash and upwash effects. This is consistent with the static-pressure distribution modification observed in the previous paragraph and with the skin-friction-coefficient contours. Skin friction is decremented at the mid-span and slightly incremented toward the tip, which is compatible with an increase and a decrease in local angle of attack, respectively. Finally, examining the streamline velocity contours near the wing mid-span, it can be noticed that velocity magnitude is increased in the case with rotating rotors, consistent with the total-pressure increment observed in Figure 3.16b.



(a) With rotors.

(b) Without rotors.

Figure 3.18: Streamlines colored by velocity magnitude and skin-friction-coefficient contours over the vehicle's surface. The same flight condition is analyzed with (3.18a) and without (3.18b) rotating rotors.

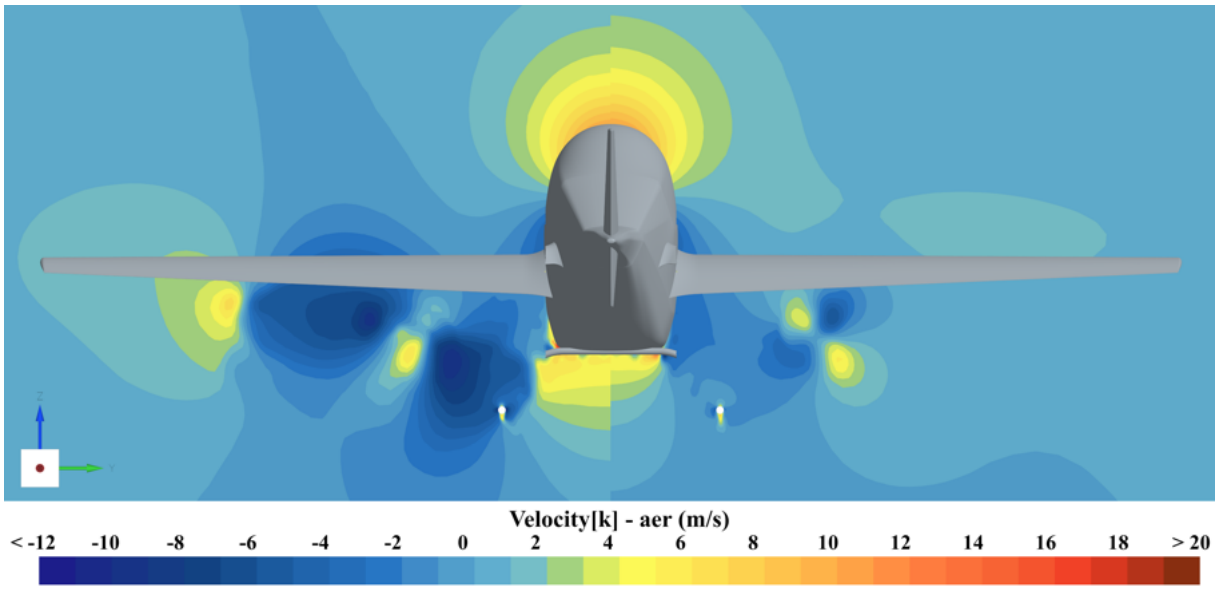
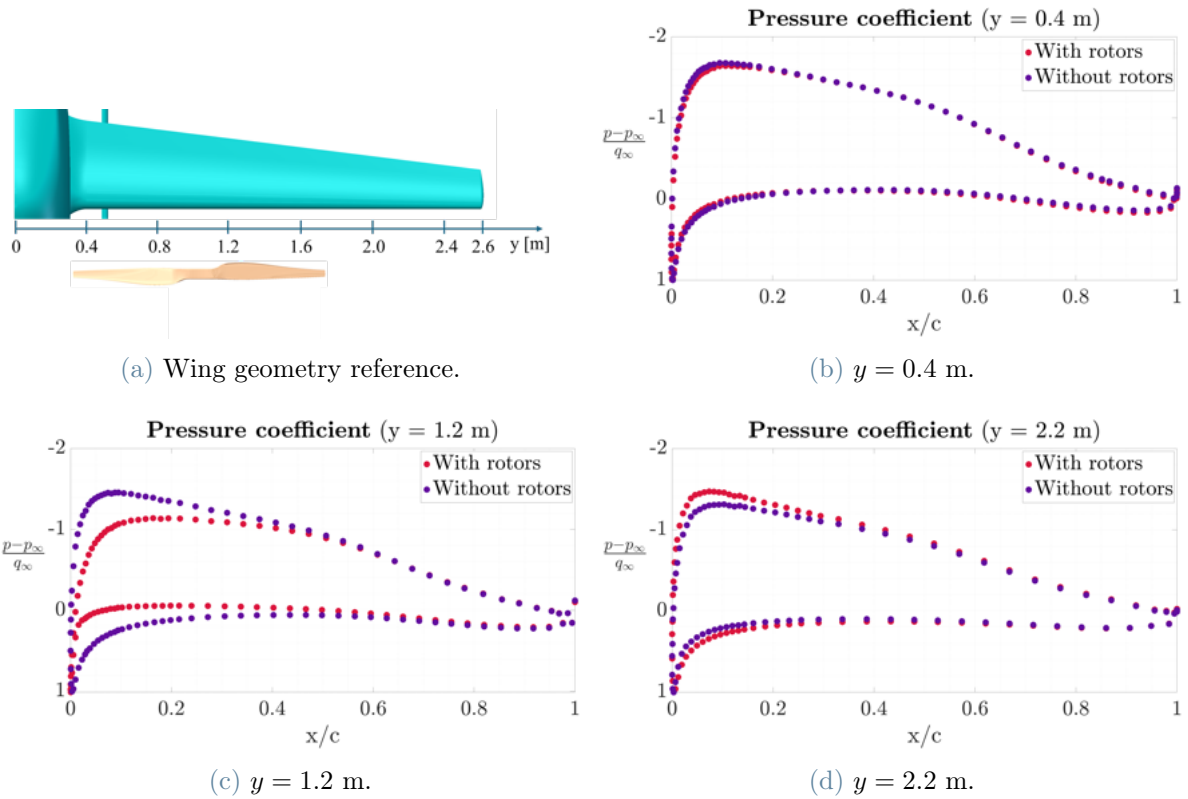


Figure 3.19: Velocity contours. Vertical velocity component (aerodynamic frame) is displayed over the vertical plane (body frame) located one mean chord upstream with respect to the wing leading edge. The same flight condition is analyzed with (left hand side) and without (right hand side) moving rotors.

Downwash and upwash effects can be further analyzed by inspecting the vertical velocity component. Figure 3.19 displays contours of the vertical velocity component in the aerodynamic frame, plotted on the vertical plane in the body frame located one mean chord upstream of the wing leading edge. The left and right halves of the image show results obtained with and without rotating rotors. Here, the downwash and upwash regions are clearly distinguishable. The rotor wakes induce a reduction in vertical velocity over approximately the inboard half of the wing and an increase toward the tip. According to these contours, the downwash effect is expected to peak slightly inboard of the wing mid-span.

The previous paragraphs focused on the main phenomena characterizing the flow field. The following discussion investigates the impact of the rotors on the loads and overall vehicle performance.
Figure 3.20a provides a geometric reference for the wing. The rotor blade is displayed to clarify its spanwise region of influence. Figure 3.20 again compares the results obtained with and without rotating rotors. The pressure coefficient, defined using the freestream static and dynamic pressures, is plotted against the normalized chord at multiple spanwise locations. At $y = 0.4$ m (Figure 3.20b), near the fuselage, there is virtually no difference between the two distributions. Seemingly, at this location, rotor effects are of minor

significance compared with the fuselage interference. At $y = 1.2$ m, the rotor-wing interaction is expected to be at its maximum. Due to the increase in total pressure observed in Figure 3.16b, the pressure-coefficient peak would be expected to exceed 1 when the rotors are rotating. Since this effect is barely noticeable in the plot, it is evident that the increase in total pressure does not play a major role in aerodynamic loads. Comparing the pressure difference over the chord, it is significantly reduced when the rotors are moving. This further confirms the reduction in local angle of attack, which translates into a lift loss. Conversely, the pressure-coefficient distribution at $y = 2.2$ m (Figure 3.20d) exhibits an increase in the pressure difference, which is consistent with an increment of the local angle of attack and results in a lift gain.



(a) Wing geometry reference.

(b) $y = 0.4$ m.

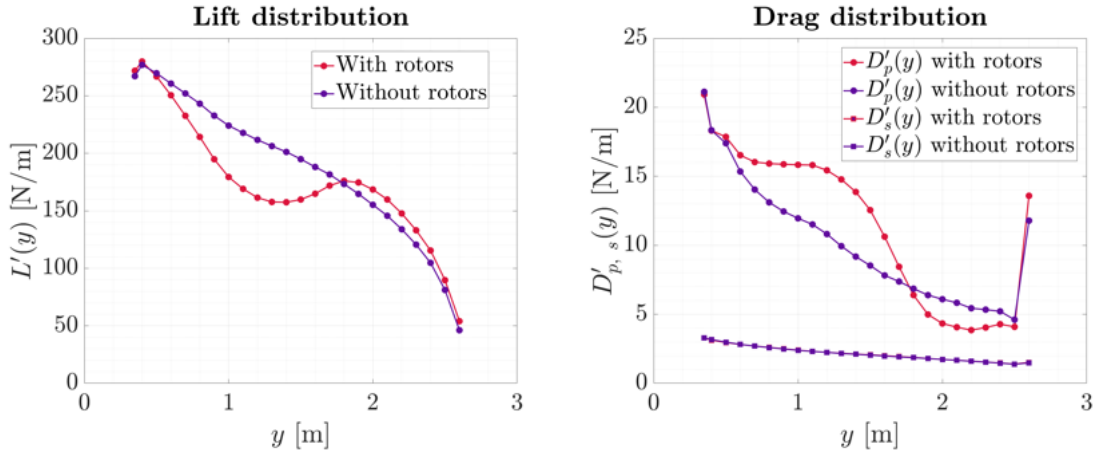
(c) $y = 1.2$ m.

(d) $y = 2.2$ m.

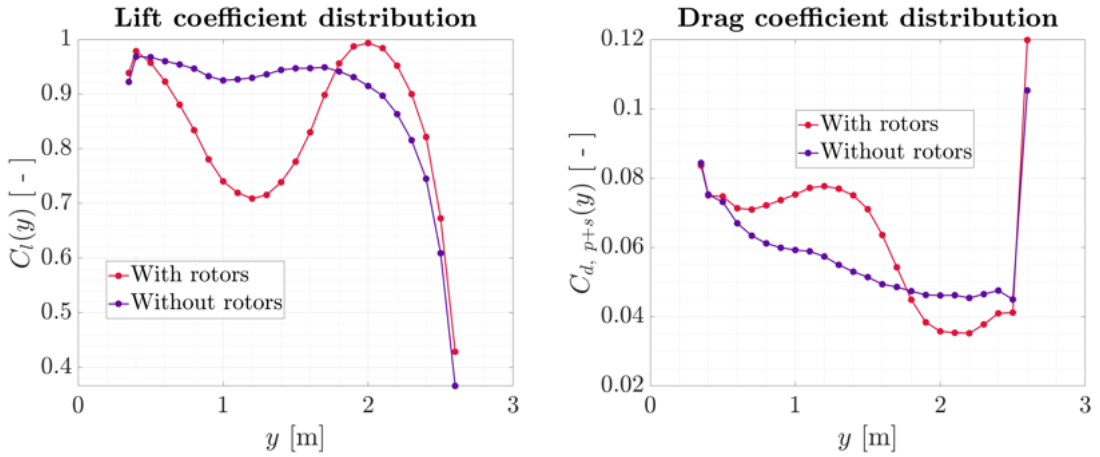
Figure 3.20: Pressure-coefficient distributions versus normalized chord at different wingspan locations, panel 3.20a provides a geometric reference of the wing. The same flight condition is analyzed with and without moving rotors.

Finally, the load distributions are displayed in Figure 3.21. The lift distribution resembles the behavior inferred from the observations in the previous paragraphs. The two main phenomena are the increase in velocity magnitude and the change in the local angle of attack. As observed in Figure 3.20b, near the fuselage, rotor effects are secondary com-

pared with the wing-fuselage interference. The induced downwash affects more than half of the wing, up to a spanwise location of $\sim y = 1.8$ m. Here, the reduction in angle of attack outweighs the velocity increase, resulting in a net reduction in lift. Toward the tip, instead, the rotors induce an upwash effect, which increases lift. At this spanwise location, there is only a minor velocity difference between the two configurations (see Figure 3.18). Figure 3.21b shows the loads normalized by the freestream dynamic pressure and the local chord. Although the lift-coefficient magnitude is altered by the local velocity variation, its distribution reveals the location of maximum interference. At $\sim y = 1.2$ m, the distribution reaches a local minimum. Thus, the downwash effect is at its maximum in this region.



(a) Load distributions.



(b) Load-coefficient distributions. Non-dimensionalization with respect to freestream conditions.

Figure 3.21: Spanwise lift and drag distributions. Subscripts $_p$ and $_s$ refer to pressure and shear contributions, respectively. The same flight condition is analyzed with and without moving rotors.

The drag distributions (Figure 3.21a) are decomposed into shear and pressure contributions. The shear-drag distributions of the two configurations are nearly identical. This is mainly attributable to the choice of turbulence model, as $k-\varepsilon$ model is not expected to capture small shear-drag variations accurately. However, as observed in Figure 3.16b, the high-velocity rotor wake does not impinge directly on the wing. Thus, it is reasonable that the shear drag is not significantly affected, rendering the associated error negligible. The pressure drag, instead, exhibits significant sensitivity to rotor implementation. It is considerably increased in the downwash region and slightly reduced toward the tip. The drag variation is consistent with the change in local angle of attack induced by the rotor wakes. A variation in the angle of attack rotates the local lift vector, altering its streamwise component. In the downwash region, this rotation increases the streamwise projection of the lift vector, resulting in higher pressure drag. Toward the tip, the upwash effect acts in the opposite direction, yielding a slight reduction in pressure drag.

In conclusion, Table 3.2 reports the integrated values of lift, drag, and lift-to-drag ratio for both the wing alone and the complete vehicle. The configurations with and without rotating rotors are compared. As expected, aerodynamic performance degrades when rotors are implemented. The wing lift decreases by ~ 30 N and the wing drag increases by ~ 3 N. However, total vehicle lift decreases by only ~ 12 N and the drag increases by ~ 8 N. Thus, while the wing contribution to lift decreases, that of the remaining vehicle components increases, likely due to rotor wake impingement on the mounting supports. Overall, the most important result is that rotor implementation significantly reduces the lift-to-drag ratio of both the wing and the complete vehicle by 16.9% and 10.6%, respectively. This demonstrates that rotor-airframe aerodynamic interference represents a non-negligible contribution to the overall vehicle drag budget and that its effect on lift extends beyond the wing to the remaining vehicle components.

Configuration	L^w [N]	D_p^w [N]	D_s^w [N]	L^w/D_{p+s}^w [-]	L^{Tot} [N]	D_{p+s}^{Tot} [N]	$L^{\mathrm{Tot}}/D_{p+s}^{\mathrm{Tot}}$ [-]
Without rotors	439.4	21.4	5.18	16.53	536.9	87.3	6.15
With rotors	408.7	24.6	5.17	13.74	524.9	95.5	5.50
$\boldsymbol{\Delta}_{\%}$	-7.0%	$+14.8\%$	0%	-16.9%	-2.2%	$+9.4\%$	-10.6%

Table 3.2: The same flight condition is analyzed with and without moving rotors. Superscripts ^{w} and ^{Tot} refer to wing-only and complete vehicle contributions, respectively. Subscripts $_{p}$ and $_{s}$ refer to pressure and shear contributions, respectively. The last row reports percentage change relative to the without-rotors configuration†.

Based on these observations, the simplest strategy consists of restoring the baseline lift distribution obtained without rotating rotors (see Figure 3.21a), which leads to a more elliptic spanwise distribution. This is achieved by adjusting the twist distribution to compensate for the rotor-induced downwash.
The second approach focuses on minimizing wing drag while maintaining the same lift as the baseline. An optimizer is employed to determine the best twist distribution, subject to geometric, stall-margin, and tip-loading constraints.
Finally, the effect of the wing streamwise position relative to the rotors is investigated. This parameter is coupled with its incidence to explore a broader range of configurations. Further strategies are introduced progressively as the analysis progresses, based on the results of the aforementioned approaches.

[†]All the results reported here and in the following tables are for the half vehicle, as symmetry is exploited.

4 | Design Strategies and Performance Evaluation

This chapter constitutes the core of the present study. First, the strategies to improve vehicle performance are outlined. Second, the optimization framework and tools are presented, and their adoption is justified. These are then applied to the design strategies presented in the first section, and further studies stemming from the results are progressively introduced; the potential of an alternative design is briefly explored. Finally, an uncertainty assessment study is conducted by comparing VDM and SMM results.

4.1. Performance Improvement Strategies Overview

Section 3.4 provided a qualitative and quantitative description of the flow field and its modification due to rotor interference. This chapter explores different strategies to improve vehicle performance. The main phenomenon governing the flow field is the change in the local angle of attack. Thus, the design parameters are the wing twist distribution and its streamwise position relative to the rotors. The wing planform and airfoil shape are kept fixed.

Based on these observations, the simplest strategy consists of restoring the baseline lift distribution obtained without rotating rotors (see Figure 3.21a), which leads to a more elliptic spanwise distribution. This is achieved by adjusting the twist distribution to compensate for the rotor-induced downwash.

The second approach focuses on minimizing wing drag while maintaining the same lift as the baseline. An optimizer is employed to determine the best twist distribution, subject to geometric, stall-margin, and tip-loading constraints.

Finally, the effect of the wing streamwise position relative to the rotors is investigated. This parameter is coupled with its incidence to explore a broader range of configurations. Further strategies are introduced progressively as the analysis progresses, based on the results of the aforementioned approaches.

4.2. Optimization environment

The target vehicle, described in Section 3.1, combines the thrust of four pairs of coaxial rotors with its aerodynamic lift. As this work focuses only on the cruise condition, the lifting burden is distributed among the wing ($\sim 40\%$), the body ($\sim 10\%$), and the rotors ($\sim 50\%$). Equilibrium in the advancing direction is achieved by forward thrust balancing the drag of the body and wing.

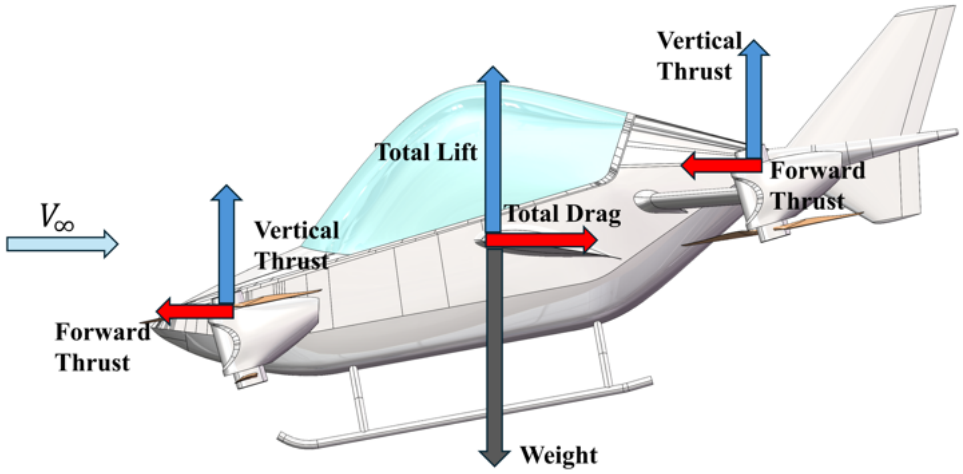


Figure 4.1: Visualization of the forces acting in cruise condition. Force magnitudes and application points are shown for illustrative purposes only and do not represent actual values.

The ultimate goal is to reduce the power required by the rotors in trimmed flight. Considering modifications to the wing alone, this objective can be pursued through two main strategies:

1. **Lift-to-drag ratio increase** The wing geometry is modified to achieve the same lift as the baseline configuration, but with reduced drag. Consequently, less forward thrust is required, and a new trim condition is obtained. This results in a higher nose-up pitch angle, increasing wing lift and diverting more thrust to the vertical equilibrium (Figure 4.1 may help visualize this mechanism). Finally, the required thrust, and consequently the rotational speed of the rotors, decreases, reducing the required power;

2. **Lift increase** The rationale behind this vehicle concept is that the wing provides lift more efficiently than the rotors. Thus, redistributing the vertical force requirement by increasing wing lift may yield a reduction in the rotational speed of the rotors and, consequently, in the required power. However, unlike strategy 1, this involves

a circular trimming effect: higher lift leads to higher drag, demanding a more nose-down trimmed configuration to satisfy the increased forward-thrust requirement. Hence, pitching reduces wing lift and augments the load on the rotors, partially offsetting the initial benefit. Consequently, the net effect on rotor power remains highly unpredictable, and this strategy must be applied with caution.

Both strategies require optimization. Although the goal is power minimization, the power requirement can be evaluated only when the vehicle is trimmed. Introducing cruise-equilibrium search into the optimization process would demand prohibitive computational resources. Thus, the study is driven by the lift-to-drag ratio. Once an optimal configuration satisfying the drag and constraint objectives is found, it is trimmed, and the resulting impact on power consumption is analyzed.
As previously stated, the optimal wing design is investigated by means of the twist distribution, as the dominant phenomenon is the rotor-induced change in the local angle of attack. The choice of the design variables and their number is discussed in Section 4.2.1. Accordingly, the optimization problem is expressed as follows:

$$\begin{gathered} \max L^w/D^w \\ \text{w.r.t.} \\ L^w \geq L^w_{\text{ref}}, \quad C_l(y) \leq \bar{C}^{\text{2D}}_{l,\ \max}, \quad C_l(y) \leq 1.15\ C_l(y=0.4), \\ \mathbf{b}_1 \leq A\mathbf{x} \leq \mathbf{b}_2. \end{gathered} \tag{4.1}$$

The minimum-drag configuration is obtained in compliance with nonlinear (imposed lift, stall-margin, and tip-loading constraints) and linear (geometric relations ensuring a feasible twist distribution) constraints.

4.2.1. Design variables

The number of variables used to control the twist distribution results from a compromise between the computational cost and the design-space flexibility. Bons *et al.* and Mader & Martins conducted wing-optimization studies that included twist-distribution management [5, 19]. In their works, eight and nine twist variables were used, respectively. Although these choices were mainly guided by their overall problem setups, they enabled the desired lift distribution to be achieved. Moreover, Bons *et al.* performed a sensitivity study by varying the number and spacing of the twist variables along the span, consistently obtaining the same optimal result regardless of the chosen parametrization [5].
In the following sections, the optimization setup is presented. It employs a surrogate

model to limit computational cost. An increased number of variables not only degrades the surrogate-model accuracy, but it also requires a larger dataset for training. Thus, although the aforementioned works suggest the use of eight variables or more, the most prudent procedure consists of starting from a reduced design space. In practice, five variables are employed to define the twist distribution. This number provides sufficient flexibility to assess the impact of twist on wing performance. Should the results prove promising, additional computational resources can be allocated to refine the design process.

4.2.2. Geometry modeling

The geometry of each vehicle configuration is generated through a two-step process. First, the wing shape is generated and modified within the OpenVSP framework. Then, it is exported and combined with the remaining vehicle components in SolidWorks.
OpenVSP, an open-source parametric geometry modeling tool, enables efficient wing parameterization. Given the spanwise airfoil distributions, wing planform parameters, and incidence angle, the resulting geometry is shown in Figure 4.2. The parameterized shape can be easily manipulated via the built-in Python API, making it suitable for managing the large number of configurations required by the optimization process.
Remarkably, the reconstructed wing geometry, when integrated into the complete vehicle model, faithfully reproduces the actual geometry in terms of integral aerodynamic values and load distributions, with a discrepancy below 0.5%.

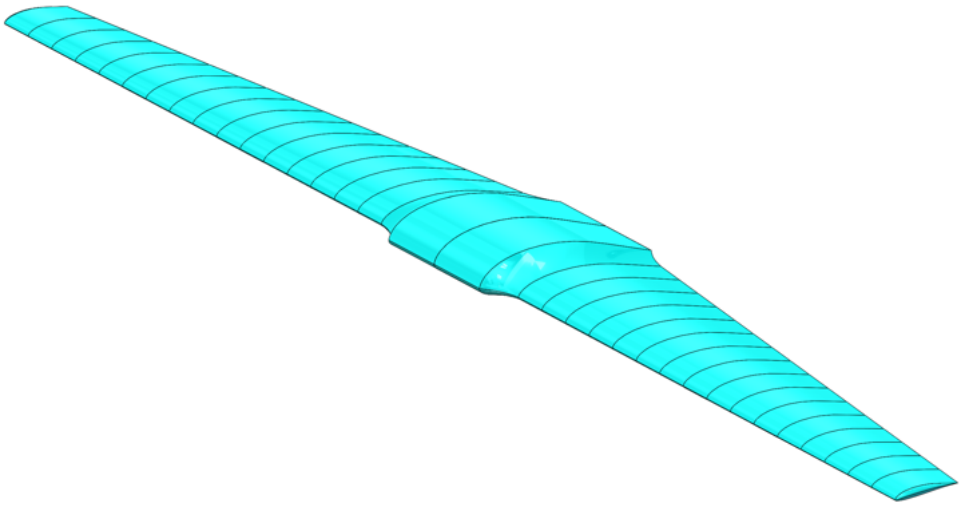


Figure 4.2: Parametric wing representation. The geometry is modeled within OpenVSP.

4.2.3. Optimization algorithm

The low dimensionality of the optimization problem and the high cost of objective-function evaluations naturally drive the choice toward surrogate-based methods [42]. Two main approaches are considered:

1. **Offline surrogate-based optimization** Often referred to as *one-shot solution* [28], the simplest strategy consists of building a surrogate model, validating its accuracy, and using it in place of the real high-fidelity model (CFD simulation);

2. **Bayesian Optimization** The other procedure involves training an initial surrogate model on a limited dataset, which is employed to provide efficient guidance in the choice of further training data. Among the most widely used Bayesian methods, the EGO (*Efficient Global Optimization*) algorithm stands out [16]. It consists of updating the surrogate, built using the Kriging method, at each iteration. The updated interpolated model is used to identify the subsequent function evaluation, balancing the performance improvement promised by the surrogate (exploitation) and the need for exploration in regions of high model uncertainty [16].

In the present study, the offline approach is employed. Although Bayesian optimization may yield better results, it requires multiple additional expensive function evaluations upon any minor change in the problem formulation. Conversely, although the offline strategy entails a higher initial cost, it allows multiple constraint variations to be investigated without the need for further CFD computations. A good practice is to employ the latter method to fabricate the ideal setup and obtain an estimate of the optimum. Should the preliminary results be promising, Bayesian optimization can be applied as a refinement step.

Based on the offline surrogate-based approach, an optimization algorithm is required to query the interpolant. Since surrogate function evaluation is inexpensive, computational cost plays only a secondary role in algorithm selection.
Undoubtedly, the method must enable a global search for the optimum. Although finding the global minimum can never be guaranteed, global algorithms ensure wide exploration of the design space, reducing the probability of converging to suboptimal solutions.
Global algorithms can be distinguished on the basis of the search method [1]. Gradient-based optimizers achieve the best convergence rates when the problem is smooth, unimodal, and well behaved. However, they are intrinsically local and must be coupled with multi-start strategies to ensure adequate design-space coverage. Moreover, they are highly sensitive to the accuracy of the derivatives, which is particularly critical in the context of a surrogate-based approach. Derivative-free methods, instead, are more robust and bet-

ter suited for global optimization. Their drawback lies in significantly slower convergence rates, making them more computationally demanding than gradient-based methods [1]. Given their robustness and global nature, the present work relies on a derivative-free method. As previously stated, the increased computational cost is rendered negligible by the surrogate-based approach. Furthermore, their inherent parallelizability enables concurrent function evaluations, significantly accelerating the search.

Among the most common derivative-free global algorithms are the Genetic Algorithm (GA), DIRECT (DIviding RECTangles), and Particle Swarm Optimization (PSO). Pattern Search (PS), while intrinsically local, may be extended to a global search through multi-start strategies.
The selection of the optimization algorithm depends not only on its properties, but also on its optimization implementation within the chosen framework and on ease of use. In the present study, the *Dakota* optimization framework is employed. Within *Dakota*, PSO is not implemented and, among the remaining cited algorithms, GA is the only method with built-in support for linear constraints.
Beyond software compatibility, the choice of the Genetic Algorithm is further supported by its extensive use and validation in the aerodynamic shape optimization literature [3, 4, 15]. GA has therefore demonstrated robustness and reliability in this context. For these reasons, the Genetic Algorithm is selected as the method for the present work.

4.2.4. Linear constraints

Constraints in optimization problems are classified as either linear or nonlinear. Linear constraints consist of linear relationships among the design variables x_i. As introduced in Equation 4.1, they are expressed as

$$\mathbf{b}_1 \leq A\mathbf{x} \leq \mathbf{b}_2, \tag{4.2}$$

where $A \in \mathbb{R}^{m \times n}$ denotes the constraint coefficient matrix and $\mathbf{b}_1$, $\mathbf{b}_2 \in \mathbb{R}^m$ are the prescribed lower and upper bound vectors, respectively.
Given the five twist variables (see Section 4.2.1), the linear constraints can be employed to bound the magnitude of the twist-per-unit-span gradient along the wing span. This structural requirement must be accounted for to ensure design feasibility and is typically assessed using a structural model, thus yielding a nonlinear constraint. In the present work, such a model is unavailable. Therefore, the twist-per-unit-span gradient is constrained based on practical considerations. No established guidelines exist in the literature to determine reasonable values for $\mathbf{b}_1$ and $\mathbf{b}_2$. Nevertheless, in existing aircraft

configurations, this slope seldom exceeds $3°/m$ [31]. Hence, the latter bound is enforced in this study to ensure structural feasibility. Throughout the optimization process, the activity of this linear constraint is monitored to assess whether it limits the achievable performance improvement.

4.2.5. Nonlinear constraints

Nonlinear constraints involve nonlinear relationships between the objective and the design variables and cannot be directly enforced on the latter. Instead, they are commonly incorporated as penalties within the objective function.
As introduced in Equation 4.1, the present study dictates the following nonlinear constraints:

$$L^w \geq L^w_{\text{ref}}, \quad C_l(y) \leq \bar{C}^{\text{2D}}_{l,\,\max}, \quad C_l(y) \leq 1.15\, C_l(y = 0.4) \tag{4.3}$$

where L^w is the wing lift, which is enforced to be no less than the reference value (i.e., the baseline value), and $C_l(y)$ represents the sectional lift-coefficient distribution along the wingspan, extracted from the three-dimensional CFD simulation. The second constraint ensures an adequate stall margin at each spanwise station, while the third prevents excessive tip loading. Consistent with the optimization framework, both the lift and the sectional lift coefficients are evaluated via dedicated surrogates.
The maximum lift coefficient allowed at each spanwise station, $\bar{C}^{\text{2D}}_{l,\,\max}$, is determined by the *critical section theory* [17]. This approach provides a preliminary estimate of three-dimensional wing stall. Given its inherent limitations, it is suitable only for early design stages. The method requires evaluating the stall lift coefficient of the two-dimensional section at each wing spanwise station. $\bar{C}^{\text{2D}}_{l,\,\max}$ is then defined as the maximum sectional lift coefficient of the most critical section, reduced by a prescribed reasonable safety margin (typically 0.2 [17]). The limitations of this theory are indisputable, as three-dimensional effects are neglected. Hence, although this theory provides useful stall-management guidelines, it must be applied with caution.
Figure 4.3 displays lift and drag coefficient polars for the two-dimensional airfoils at five spanwise stations between the root and the tip. While the tip airfoil exhibits the highest $C^{\text{2D}}_{l,\,\max}$, the root airfoil yields the most critical value. Therefore, $\bar{C}^{\text{2D}}_{l,\,\max}$ is computed as $C^{\text{2D}}_{l,\,\max}$ @ $y = 0.4$ m reduced by 0.2. That is, $\bar{C}^{\text{2D}}_{l,\,\max} = 1.15$. This value represents the maximum allowable sectional lift coefficient for all spanwise stations during the optimization process, thereby influencing the spanwise lift distribution. Furthermore, considering Figure 3.21b, the maximum of $C_l(y)$ occurs where the local freestream velocity is augmented by rotor interference. Thus, since $C_l(y)$ is normalized by the freestream dynamic

pressure, its actual value is locally overestimated, thereby providing a conservative stall margin in the outboard region.

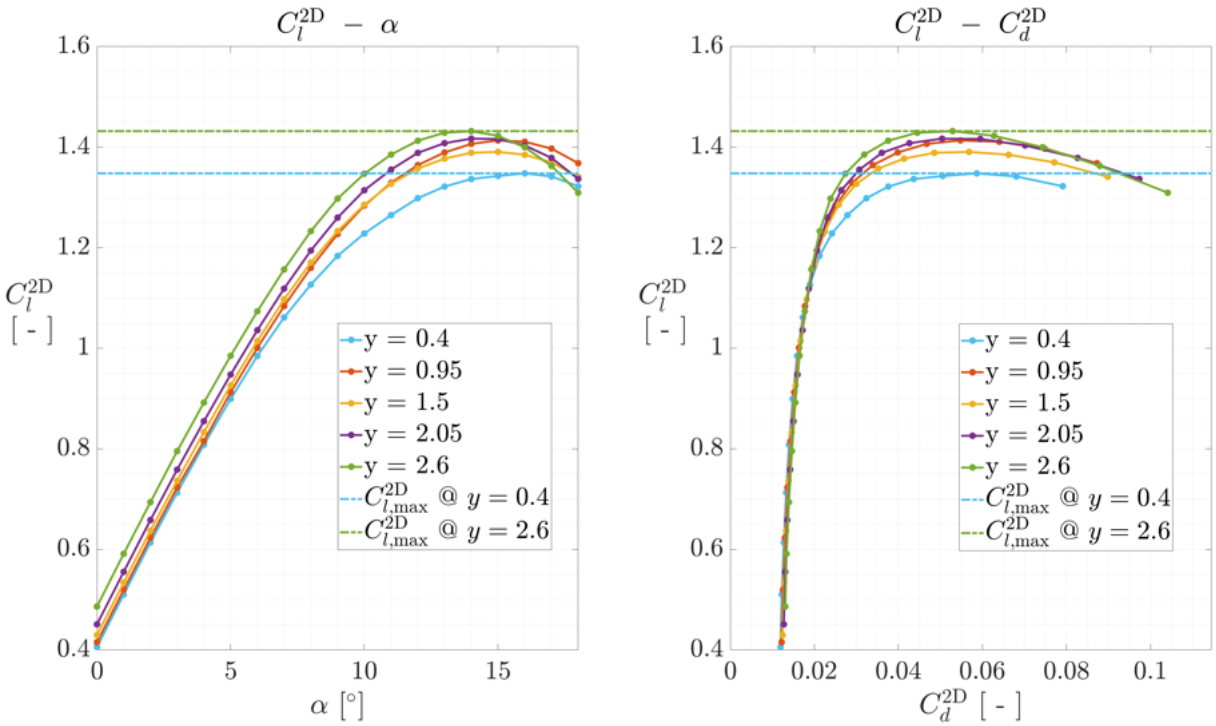


Figure 4.3: Lift and drag polar curves at multiple wing spanwise stations (see Figure 3.20a for reference). Two-dimensional airfoil results. The maximum lift coefficient for the root and tip airfoils is highlighted.

4.3. Downwash Compensation Strategy

From Figure 3.21b, it can be inferred that the rotors-wing interference culminates in the vicinity of $y = 1.2$ m, and Section 3.4 established that the dominant aerodynamic phenomenon is a local change in angle of attack. Hence, prior to initiating the optimization process, a preliminary analysis aimed at compensating the rotor-induced downwash is conducted to gain relevant insights. Restoring the local angle of attack through a geometric twist increment is expected to drive the spanwise lift distribution closer to an elliptical shape. While the local profile drag increases due to the higher effective angle of attack, the redistribution of circulation may reduce the induced-drag component, potentially yielding a net aerodynamic benefit.

Temporarily disregarding the structural considerations addressed in the linear-constraints section (Section 4.2.4), the spanwise twist distribution is adjusted to mitigate the local downwash effect. Figure 4.4 shows the baseline twist distribution compared with three

alternative configurations designed as follows. The twist is increased only within the region where the local angle of attack is reduced, i.e., 0.4 m $< y <$ 1.8 m. In particular, the twist value at $y \sim 1.2$ m (where the rotors-wing interference peaks) is imposed to be 2°, 3°, or 4°, while matching the baseline geometry at and beyond the boundaries of the described region. Finally, these values are connected by a cubic polynomial to ensure a smooth distribution. It is noted that the interpolation produces a limited overshoot at $y \sim 1.1$ m. Since this artifact is not expected to significantly alter the overall results of this simplified analysis, its correction was deemed unnecessary.

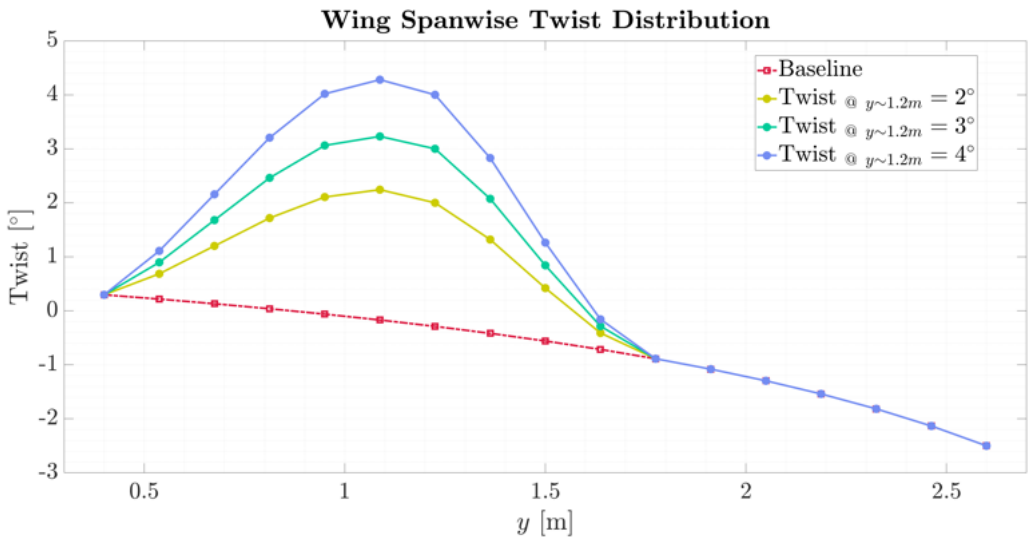


Figure 4.4: Spanwise twist distributions for the downwash compensation strategy, compared with the baseline geometry.

The modified twist distributions produce an increase in wing lift and drag. Thus, to compare the results, all configurations must be trimmed. Cruise equilibrium is achieved by calibrating the rotation rates of the front (Ω_{front}) and rear (Ω_{back}) rotors, and the vehicle angle of attack α. The trimming parameters are reported in Table 4.1, and the aerodynamic loads, pitching moment, and rotor power consumption are reported in Tables 4.2 and 4.3.

As previously discussed in the opening of Section 4.2, an increase in lift and the associated increase in drag result in a more nose-down equilibrium configuration. Consequently, the wing lift is decreased. Overall, the wing lift-to-drag ratio deteriorates as the geometry modification becomes more pronounced, while $L^{\text{Tot}}/D^{\text{Tot}}_{p+s}$ remains nearly unchanged. This behavior has two main reasons. First, wing lift and drag contribute by $\sim 75\%$ and $\sim 30\%$ to the total loads, respectively. Hence, when the wing lift-to-drag ratio is high, the benefit of increased wing lift outweighs the penalty of the associated drag increment,

even though the lift-to-drag ratio deteriorates. Second, the more pitched configuration causes the vehicle body to exhibit degraded aerodynamic performance. These two effects essentially compensate each other, resulting in an unchanged $L^{\text{Tot}}/D^{\text{Tot}}_{p+s}$.
Regarding rotor performance, since L^{Tot} shows only a minor increase (Table 4.2), the required vertical force is only marginally reduced, while the forward-force demand increases due to the higher drag. Overall, the thrust magnitude decreases with increasing twist (Table 4.3). Hence, as expected, the lift increment reduces the burden on the rotors in terms of the required force. However, trimming results in a more nose-down configuration, which reduces rotors efficiency, requiring higher rotational speeds (Table 4.1) even to produce a lower thrust output. In terms of the thrust-to-power ratio, Table 4.3 shows that T/P diminishes with increasing twist.

Configuration	α [°]	Ω_{front} [rpm]	Ω_{back} [rpm]
Base$^{\text{With rotors}}$	-10.59	1281.9	1320.5
$\text{T}^{\text{wist}}_{@\ y\sim 1.2m} = 2°$	-10.99	1290.3	1334.5
$\text{T}^{\text{wist}}_{@\ y\sim 1.2m} = 3°$	-11.20	1291.3	1341.9
$\text{T}^{\text{wist}}_{@\ y\sim 1.2m} = 4°$	-11.45	1296.8	1346.5

Table 4.1: Downwash compensation strategy, for the baseline configuration and for the modified twist distributions shown in Figure 4.4.

Configuration	L^w [N]	D^w_{p+s} [N]	$M^w_{y,\,CG}$ [Nm]	L^w/D^w_{p+s} [-]	L^{Tot} [N]	D^{Tot} [N]	$M^{\text{Tot}}_{y,\,CG}$ [Nm]	$L^{\text{Tot}}/D^{\text{Tot}}_{p+s}$ [-]
Base$^{\text{With rotors}}$	408.7	29.7	-31.8	13.74	524.9	95.5	-77.4	5.50
$\text{T}^{\text{wist}}_{@\ y\sim 1.2m} = 2°$	423.5	31.7	-31.5	13.35	535.8	97.9	-76.6	5.48
$\text{T}^{\text{wist}}_{@\ y\sim 1.2m} = 3°$	430.2	32.7	-31.5	13.15	538.4	99.0	-77.0	5.44
$\text{T}^{\text{wist}}_{@\ y\sim 1.2m} = 4°$	435.4	33.6	-31.5	12.94	542.7	99.7	-77.7	5.45

Table 4.2: Downwash compensation strategy: aerodynamic loads and pitching moment. Results are reported for the baseline configuration (with moving rotors) and for the modified twist distributions shown in Figure 4.4. All cases correspond to trimmed cruise conditions.

In summary, the greater the modification of the twist distribution, the higher the lift generated and the lower the required thrust. However, the degraded rotor efficiency in

the resulting trimmed configuration offsets any expected gain in terms of power, ultimately leading to increased required power (see Table 4.3).

Configuration	T [N]	$\Delta P_{\%}$	T/P [N/W]
Base$^{\text{With rotors}}$	508.9	/	0.089
$\text{T}^{\text{wist}}_{@\ y\sim 1.2m} = 2°$	504.8	+0.7%	0.087
$\text{T}^{\text{wist}}_{@\ y\sim 1.2m} = 3°$	501.2	+1.2%	0.086
$\text{T}^{\text{wist}}_{@\ y\sim 1.2m} = 4°$	497.8	+1.8%	0.085

Table 4.3: Downwash compensation strategy: rotor performance. Total thrust magnitude, thrust-to-power ratio, and percentage variation in required power relative to the baseline configuration for the modified twist distributions shown in Figure 4.4. All cases correspond to trimmed cruise conditions.

The load distributions are displayed in Figure 4.5. From the lift distributions, it is observed that the $\text{T}^{\text{wist}}_{@\ y\sim 1.2m} = 4°$ configuration nearly achieves the goal of this simplified strategy. The rotor-induced downwash is nearly fully compensated, while the upwash effect in the outboard region is marginally enhanced. Examining the drag distributions, a substantial local increase is observed in the rotor-wing interference region, consistent with the higher effective angle of attack. However, in the outboard region, a slight increase in lift and a reduction in drag are noted, compatible with a decrement in induced drag due to a more uniform lift distribution. Nevertheless, the inboard drag penalty largely exceeds the outboard benefit, resulting in the degraded performance reported in Table 4.2.
In conclusion, the downwash compensation strategy proves ineffective. Nevertheless, it provides useful insights:

- An increase in wing lift reduces the required thrust magnitude and may, in principle, improve the vehicle lift-to-drag ratio;
- The same increase in wing lift entails a drag increment, requiring a more nose-down trim condition and ultimately reducing rotor efficiency.

Given these two observations, it is reasonable to expect that an optimal compromise exists between lift augmentation and rotor performance degradation. This is addressed in Section 4.5.

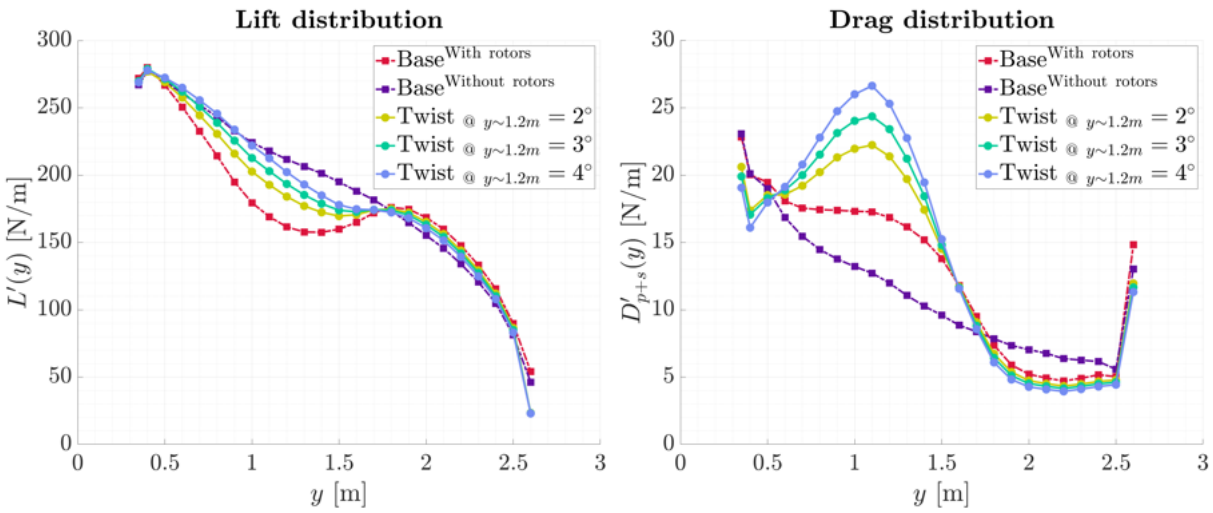


Figure 4.5: Downwash compensation strategy lift and drag distributions. Results comparison between multiple spanwise twist distributions (see Figure 4.4) and the baseline configuration with and without moving rotors is displayed. Except for $\text{Base}^{\text{Without rotors}}$, all results correspond to cruise-trimmed conditions.

Thus far, the trimmed configuration has been achieved by tuning the global pitch angle of the vehicle α and the rotational speeds of the rotors, Ω_{front} and Ω_{back}. Furthermore, the previous analysis demonstrated that an increase in lift reduces the thrust demand on the rotors, while an increment in $|\alpha|$ leads to a less efficient propulsion system. It is therefore of interest to assess which of the two effects prevails upon the introduction of a new trimming parameter, namely the rotor shaft angle γ_m. γ_m is defined as the angle between the body x-axis and the rotor plane. It quantifies the rotor pitch relative to the vehicle body frame. Accordingly, the absolute rotor pitch angle γ defines the rotor pitch relative to the freestream direction. These two angles are related through α as $\gamma = \gamma_m + \alpha$. In the baseline configuration, the rotors are mounted with a forward pitch of 5.6°, yielding $\gamma_m = -5.6°$.

Considering the most lift-enhancing configuration, $\text{T}^{\text{wist}}_{@\ y\sim1.2m} = 4°$, trimming can be achieved by adjusting either α, Ω_{front}, and Ω_{back}, or γ_m, Ω_{front}, and Ω_{back}. Finally, the two trimming strategies can be combined to identify the minimum-power configuration.

Table 4.4 compares the performance of the $\text{T}^{\text{wist}}_{@\ y\sim1.2m}$ configuration trimmed with fixed γ_m and modifying α, Ω_{front}, and Ω_{back}, and with fixed α while varying γ_m, Ω_{front}, and Ω_{back}. Predictably, the total lift is higher when α is kept constant, so is $L^{\text{Tot}}/D^{\text{Tot}}_{p+s}$. Furthermore, the thrust-to-power ratio is adversely affected by a higher $|\gamma|$. Notably, the new trimming parameter allows achieving approximately the same power consumption as the baseline. Thus, in this specific configuration, the (γ, Ω_{front}, Ω_{back}) parameter set yields

superior performance compared with (α, Ω_{front}, Ω_{back}), despite the significant decrease in T/P. Overall, compared with the baseline, this new trimming strategy closely balances the aerodynamic benefit from reduced thrust demand and the rotor performance penalty from increased γ.

Configuration	α [°]	γ [°]	L^{Tot} [N]	D^{Tot} [N]	$L^{\text{Tot}}/D^{\text{Tot}}_{p+s}$ [-]	T [N]	$\Delta P_{\%}$ [-]	T/P [N/W]
Base$^{\text{With rotors}}$	-10.59	-16.19	524.9	95.5	5.50	508.9	/	0.089
$\text{T}^{\text{wist}}_{@\ y\sim1.2m} = 4°$ (α, Ω_{front}, Ω_{back})	-11.45	-17.05	542.7	99.7	5.45	501.2	+1.8%	0.085
$\text{T}^{\text{wist}}_{@\ y\sim1.2m} = 4°$ (γ, Ω_{front}, Ω_{back})	-10.59	-19.58	600.3	103.4	5.81	442.5	0%	0.077

Table 4.4: Downwash compensation strategy: trimming strategies comparison. The baseline configuration is compared with the $\text{T}^{\text{wist}}_{@\ y\sim1.2m} = 4°$ configuration trimmed using two different parameter sets: (α, Ω_{front}, Ω_{back}) and (γ, Ω_{front}, Ω_{back}). Notation follows Table 3.2.

In conclusion, this simplified study demonstrates both the potential for performance improvement through lift augmentation and its limitations. The trimming procedure renders the outcome of a design modification difficult to predict, owing to the degradation in rotor thrust-to-power ratio and the α-induced load variation. Moreover, rotor-wing interaction is highly sensitive to the rotor pitch angle γ and the rotational speeds. Nevertheless, even a non-optimized design, the $\text{T}^{\text{wist}}_{@\ y\sim1.2m} = 4°$ configuration, demonstrated that an appropriate trimming strategy can reduce the required power. This is further addressed in Section 4.4, where multiple optimized designs and trimming strategies are examined to achieve a substantial reduction in required power.

4.4. Twist Distribution Optimization

This section investigates the feasibility of reducing wing drag while matching the baseline wing lift. As noted previously, wing lift and drag contribute by $\sim$ 75% and $\sim$ 30% to the complete vehicle aerodynamic loads, respectively. Hence, any wing drag reduction has a limited effect on overall vehicle performance, and only a substantial aerodynamic improvement would be considered meaningful.

Optimization Setup	
Objective	$\max L^w/D^w$
Variables	Twist at $y \sim$ [0.4, 1, 1.2, 1.8, 2.6] m
Constraints	Spanwise twist gradient ≤ 3 °/m
	$L^w \geq L_{\text{ref}} = 408$ N
	$C_l(y) \leq \bar{C}^{\text{2D}}_{l,\,\max} = 1.15$
	$C_l(y) \leq 1.15\ C_l(y = 0.4)$
Algorithm	Genetic Algorithm
Evaluation	Surrogate-based (Kriging)
Bounds	
Twist at y ∼	[**min**, **max**] [°]
0.4 m	[-2.5, 1.5]
1 m	[-2.0, 3.5]
1.2 m	[-2.5, 3.0]
1.8 m	[-3.5, 2.0]
2.6 m	[-4.0, 0.5]

Table 4.5: Optimization setup for wing twist distribution. A surrogate-based genetic algorithm maximizes L^w/D^w subject to aerodynamic and geometric constraints. Design variable bounds and algorithm settings are reported.

The optimization setup is summarized in Table 4.5. As described in Section 4.2, the objective function is the maximization of the wing lift-to-drag ratio, and the search process is subject to linear and nonlinear constraints. The geometric constraints impose a maximum spanwise twist gradient of 3°/m to ensure structural feasibility, while nonlinear constraints govern lift magnitude and distribution requirements. First, the wing lift of the optimal design is constrained to match the baseline value. Second, an adequate stall margin is enforced spanwise through *critical section theory*. Third, to prevent excessive tip loading, the sectional lift coefficient at each spanwise station is constrained to be at most 15% higher than the root-section value.

The search method employed is a Genetic Algorithm, with each function evaluation querying a Kriging surrogate model. While linear constraints and the minimum wing-lift constraint are handled by the *Dakota*'s built-in features, the remaining constraints are enforced as penalties through an external Python script. The objective function, wing lift, and sectional lift coefficients are all computed via dedicated surrogate models.

Table 4.5 summarizes the design variables and their bounds. As described in Section 4.2, five twist variables are employed in the first optimization stage. The bounds are inten-

tionally wide to ensure an adequately broad design space. The control sections are shown in Figure 4.6. Given the limited number of variables, non-uniform spacing is adopted, placing higher control-point density where the largest twist variation is expected. In addition to the root and tip control points, the remaining three are located near the maximum rotor-wing interference region ($y \sim 1.2$ m), in its vicinity ($y \sim 1$ m), and at the interface between the downwash and upwash regions ($y \sim 1.8$ m).
The twist distribution is reconstructed via piecewise interpolation using quadratic polynomials. Although this choice may degrade surrogate performance, it ensures a smooth spanwise twist distribution.

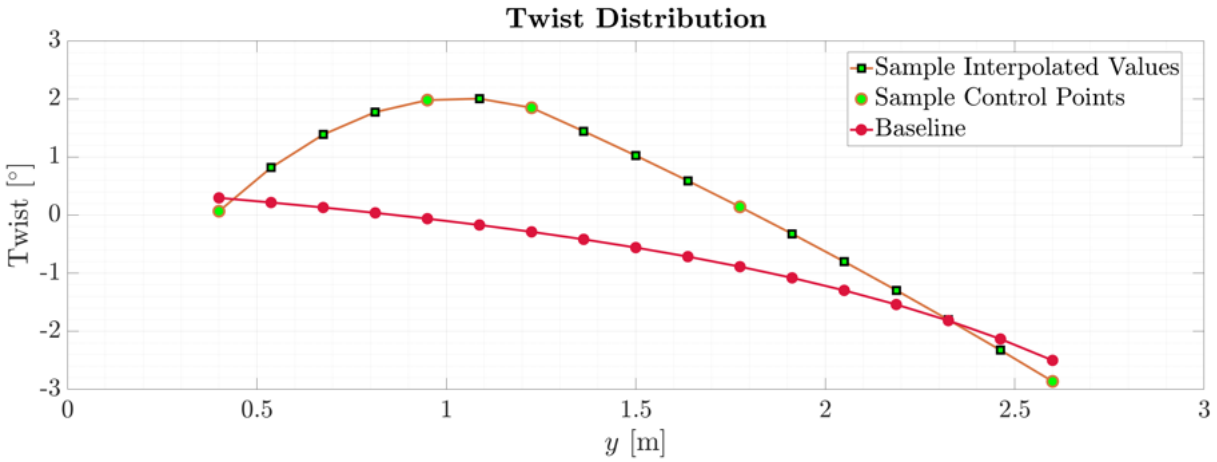


Figure 4.6: Baseline and representative sample twist distributions. Optimization control points and interpolated values are highlighted.

Prior to the optimization process, the surrogate model must be built and validated. The *Dakota* documentation recommends that the training set should comprise at least ten times the number of design variables in terms of true function evaluations.
The most common sampling strategies for training-set construction include the full-factorial, random, and Latin Hypercube Sampling (LHS) approaches. Full-factorial sampling consists of constructing a multidimensional grid over the design space. Although it ensures the most uniform design-space coverage and the best surrogate performance, it requires a large number of evaluations to achieve adequate resolution. It is therefore unsuitable for the present five-dimensional problem, where function evaluations are computationally expensive [28]. For a fixed sampling budget, random and LHS methods yield superior surrogate accuracy.
Random sampling draws points independently from a uniform distribution over the design space, resulting in regions of high sample density while leaving other regions poorly sampled. In contrast, LHS partitions each variable's range into N_s equal-probability intervals

and ensures that exactly one sample falls within each interval for every dimension (see Section 2.2.2). This stratification leads to more uniform coverage of the design space, thereby improving surrogate accuracy. For this reason, LHS is particularly well suited to expensive, high-dimensional problems where the number of true function evaluations must be limited.

Given the training set, the interpolation method must be selected. The available methods within the Dakota framework include Gaussian Process (GP, Kriging), Neural Networks (NN), and Radial Basis Functions (RBF). While the characteristics of these models are discussed in Section 2.2.2, their performance is assessed here. For validation purposes, the LHS dataset is partitioned. The surrogate model is trained on 80% of the data and tested on the remaining 20%. Surrogate predictions are compared with the high-fidelity evaluation to quantify the prediction error. The mean and maximum absolute errors are then computed, normalized by the objective range over the dataset, and reported in Table 4.6. Surrogate accuracy is assessed for wing lift, wing drag, and sectional lift coefficient at a representative spanwise station, $y = 1.8$ m, with sectional loads reported in the body reference frame).

Method	L^w		D^w_{p+s}		$F'_x(y)$ @ y=1.8		$F'_z(y)$ @ y=1.8	
	E	E_M	E	E_M	E	E_M	E	E_M
GP	0.23%	1.67%	0.31%	1.58%	0.17%	0.82%	0.17%	0.76%
NN	1.60%	10.02%	1.47%	6.18%	1.85%	8.72%	1.77%	8.16%
RBF	1.78%	12.47%	1.40%	9.26%	1.49%	10.99%	1.17%	6.73%

Table 4.6: Surrogate-model accuracy assessed via mean and maximum absolute percentage errors, $E = (\overline{|\varepsilon|})_{\%}$ and $E_M = (\max|\varepsilon|)_{\%}$, normalized by the objective range over the training dataset. Results are reported for the Gaussian Process (GP), Neural Network (NN), and Radial Basis Function (RBF) surrogate models for L^w, D^w_{p+s}, $F'_x(y)$ and $F'_z(y)$ at $y = 1.8$ m.

Notably, the Kriging model (GP) substantially outperforms NN and RBF and is therefore selected. Since the mean and maximum absolute errors never exceed 0.5% and 2%, respectively, the use of the surrogate model built via GP in place of the high-fidelity model is justified.
In summary, a Kriging model is built over an LHS-generated training set. Although Table 4.6 reports validation metrics for the 80% subset, the full dataset is used for the final construction.

Finally, the Genetic Algorithm is configured as follows:

- **Population size** The population size is set to 1000 individuals;
- **Generations** The number of generations is determined by a convergence criterion. Convergence is declared when the objective function does not improve by more than 0.5% over 100 consecutive generations;
- **Initialization type** A random initial population is adopted to ensure broad exploration of the design space;
- **Crossover type and rate** Crossover between randomly selected parents is adopted, with a crossover rate of 0.9;
- **Mutation type and rate** The selected mutation type is *replace uniform*, which selects a design variable from a randomly chosen individual and reassigns it to a value sampled within the variable bounds. The mutation rate is set to 1% of the population size;
- **Fitness type** Fitness is assessed via a merit function that balances objective improvement and constraint enforcement.

The literature offers limited universal guidelines for GA parameter selection, as optimal values are highly problem-dependent and parameters are typically determined through a systematic calibration procedure. Here, each parameter is varied independently while holding the others fixed, and the value yielding the best convergence behavior is retained. Population size and generation count are set conservatively high, as the surrogate-based evaluation renders the computational cost negligible. Beyond 500 individuals and 50 generations, only marginal improvements are observed. A crossover rate of 0.9, with two parents generating two offspring, is found to yield the best convergence. A low mutation rate is essential to preserve solution quality, as excessively high values degrade the algorithm into a random search.

Applying the setup described above, the optimization yields a preliminary optimal twist distribution. CFD validation of the optimal design reveals that the surrogate model slightly underestimates the lift and, consequently, the lift-to-drag ratio, each by approximately 1%. The surrogate is retrained on the augmented dataset, and the optimization is repeated. Consistent with the general SBAO (Surrogate-Based Analysis and Optimization) strategy presented by Queipo *et al.* [28], this iterative process is terminated after only two iterations, as no further improvement is observed.
The resulting twist distribution is shown in Figure 4.7, while aerodynamic loads and their distributions are reported in Table 4.7 and Figure 4.8, respectively. The optimized twist

distribution is unconventional, as it exhibits wash-in up to $y \sim 1.7$ m and washout toward the tip. In particular, it tends to reduce the effective angle of attack inboard while increasing it outboard, relative to the baseline. In terms of load distributions, this causes a reduction in lift and drag over the first half of the wing, up to $y \sim 1.2$ m, and an increase over the remaining wingspan. This is consistent with Section 4.3, which shows that an increase in twist over the rotor-wing interaction region leads to higher profile drag and a minor reduction in induced drag. However, although an optimal trade-off might be expected between the inboard drag increase and the outboard drag decrease, the optimization process tends to reduce aerodynamic loads inboard at the cost of increased tip loading. This suggests that a lift increase toward the tip incurs a lower drag penalty than in the inboard sections. The net effect amounts to a spanwise drag redistribution rather than an overall drag reduction.

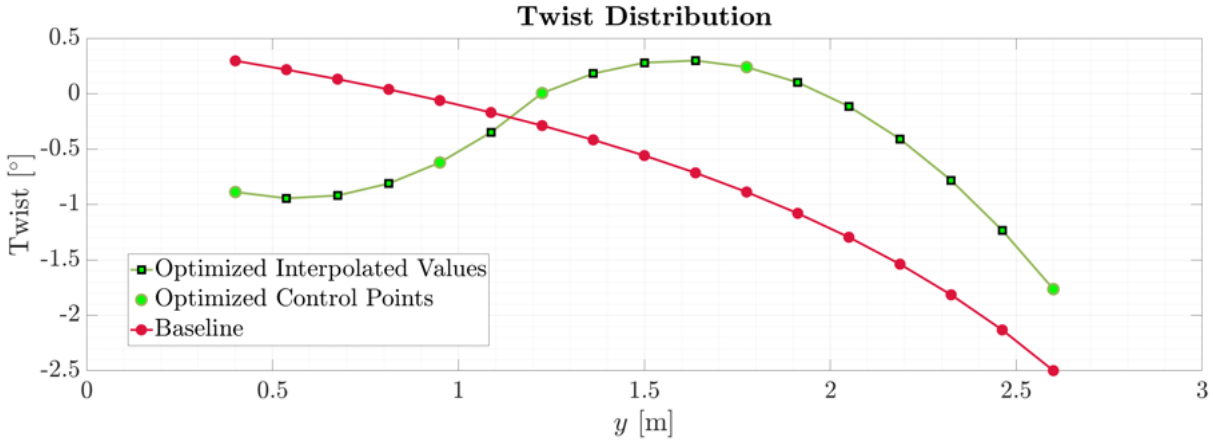


Figure 4.7: Baseline and optimized twist distributions. Optimization control points and piecewise quadratic interpolated values are highlighted.

The optimization results in a lift-to-drag ratio improvement of only 2.4% and a minimal wing pitching-moment variation, both translating into negligible changes in trim condition and power consumption. Among the active constraints, the tip-loading constraint proves to be the most restrictive. Its relaxation yields only a minor performance enhancement, after which the stall-margin constraint becomes binding. Nevertheless, the improvement is too marginal to justify constraint relaxation. Linear constraints are satisfied in nearly all evaluated designs.

These findings suggest that, within the constraints imposed, twist distribution alone offers limited room for improvement. This motivates the lift-based design strategy investigated in the following section, where the lift requirement itself is treated as a design parameter.

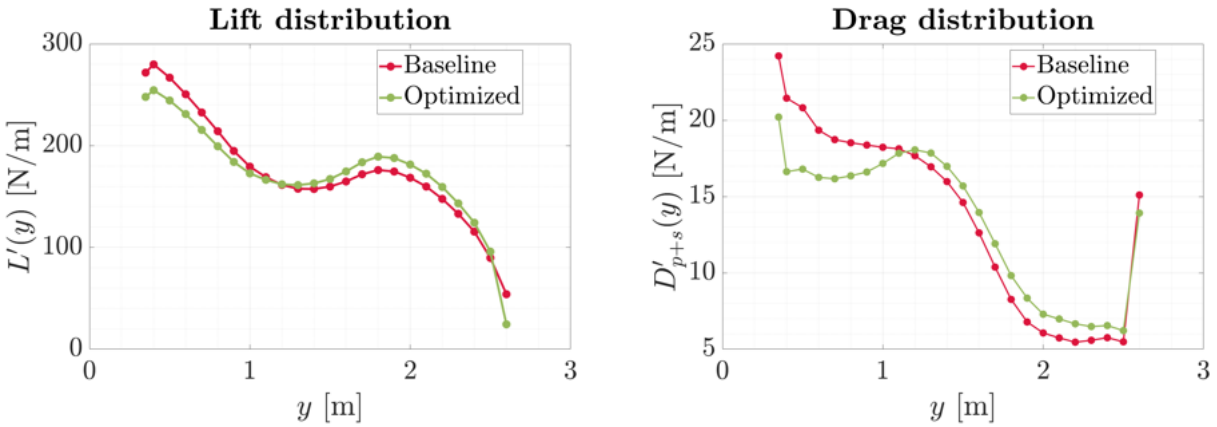


Figure 4.8: Spanwise lift and drag distributions of the baseline and twist-optimized configurations. Corresponding twist distributions are shown in Figure 4.7.

Configuration	L^w [N]	D^w_{p+s} [N]	$M^w_{y,\,CG}$ [Nm]	L^w/D^w_{p+s} [-]	$\max(C_l(y))$ [-]
Baseline	408.7	29.7	-31.8	13.74	0.99
Optimized	408.8	29.0	-30.9	14.07	1.07
$\boldsymbol{\Delta}_{\%}$	/	-2.4%		$+2.4\%$	

Table 4.7: Aerodynamic loads of the twist-optimized and baseline configurations. Corresponding twist distributions are shown in Figure 4.7.

In conclusion, the surrogate-based approach provides an accurate performance estimate. The solution can be effectively refined via an iterative process that retrains the surrogate model by augmenting the training set with high-fidelity evaluations of successive optimal designs. The optimization process demonstrates that only a minor wing drag reduction can be achieved by adjusting the twist distribution. Moreover, the optimal design does not tend toward a more uniform or elliptical lift distribution. This indicates that, in the presence of strong rotor-wing interference, spanwise lift non-uniformity is a less significant contributor to total drag than the local profile-drag increase induced by the modified angle of attack, as noted in Section 4.3.

Although surrogate-model performance has been proven satisfactory, further refinement may yield minor improvements. Moreover, the marginal sensitivity of wing aerodynamic performance to the twist distribution does not justify an increase in the number of design variables.

Overall, despite the modest aerodynamic improvements obtained, the surrogate-based op-

timization framework is considered validated. The same methodology is therefore adopted in the following section.

4.5. Lift-Based Wing Design Strategy

The previous sections demonstrate that, by adjusting the twist distribution only, no substantial increase in lift-to-drag ratio is achievable when lift is fixed (Section 4.4). However, Section 4.3 shows that a wing-lift increment may benefit the complete-vehicle performance, although the wing lift-to-drag ratio is degraded.
To this end, a systematic lift-based wing design strategy is investigated. Employing the validated setup introduced in Section 4.4, six new configurations are obtained via optimization. For each prescribed minimum lift requirement $L \geq L_{\text{ref}}$, the twist distribution is optimized for the maximum lift-to-drag ratio, subject to stall-margin, tip-loading, and geometric constraints. The resulting configurations are then compared under the two trim strategies introduced in Section 4.3: (α, Ω_{front}, Ω_{back}) and (γ, Ω_{front}, Ω_{back}). Since the ultimate goal of the present study is to reduce net power consumption, the optimal design is selected based on the minimum required power in cruise. Should both trim strategies prove promising for one or more configurations, they may be combined to identify the minimum power requirement across the four parameters (α, γ, Ω_{front}, Ω_{back}).

All configurations are obtained following the same optimization setup as in Table 4.5. The only modification concerns the value of L_{ref}. Specifically, six configurations are generated imposing

$$\mathbf{L}_{\text{ref}} = (\mathbf{1} \pm [5\%,\ 10\%,\ 15\%])L_{\text{base}}, \quad L_{\text{base}} = 408 \text{ N}. \tag{4.4}$$

These configurations are hereafter denoted as $[-15\%,\ -10\%,\ -5\%,\ +5\%,\ +10\%,\ +15\%]$, and the corresponding twist distributions are shown in Figure 4.9. It can be noted that, as the required L_{ref} decreases, the twist-distribution gradient in the inboard sections region is reduced. Furthermore, the configurations designed to generate the lowest wing lift, -15% and -10%, exhibit peak twist at the root, relaxing the lift requirement over the rotor-wing interference region. In particular, the -15% configuration features a nearly conventional wash-out twist distribution.

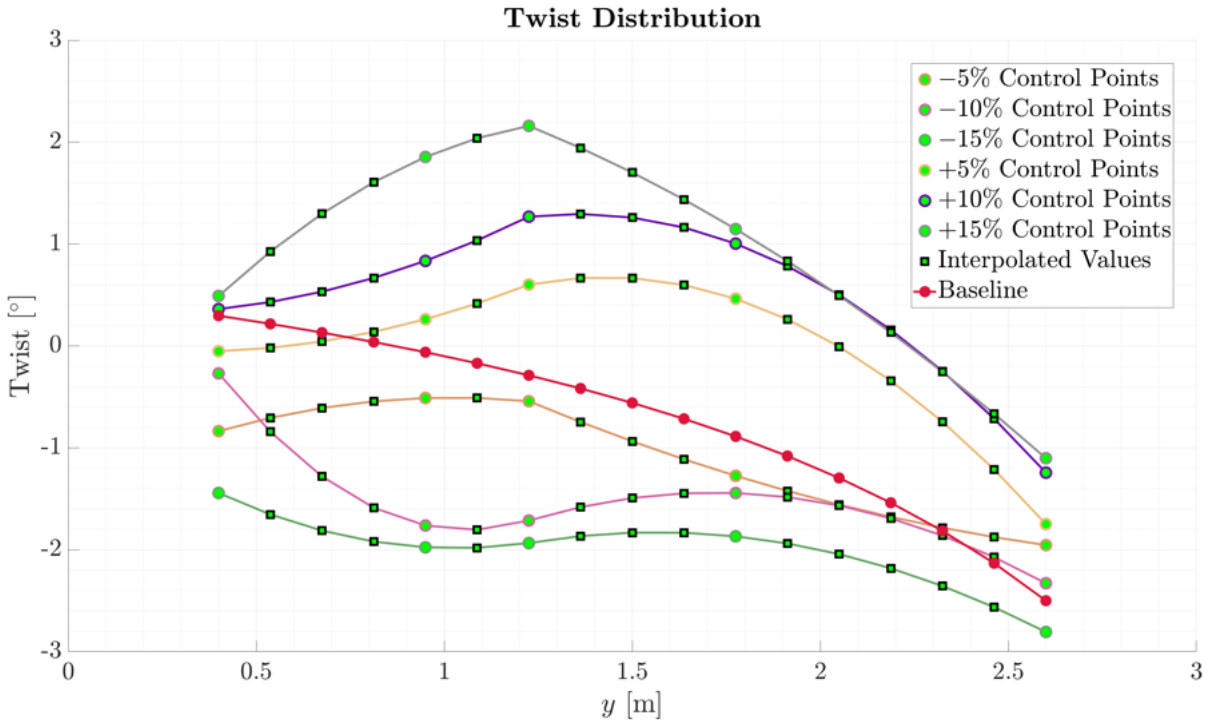


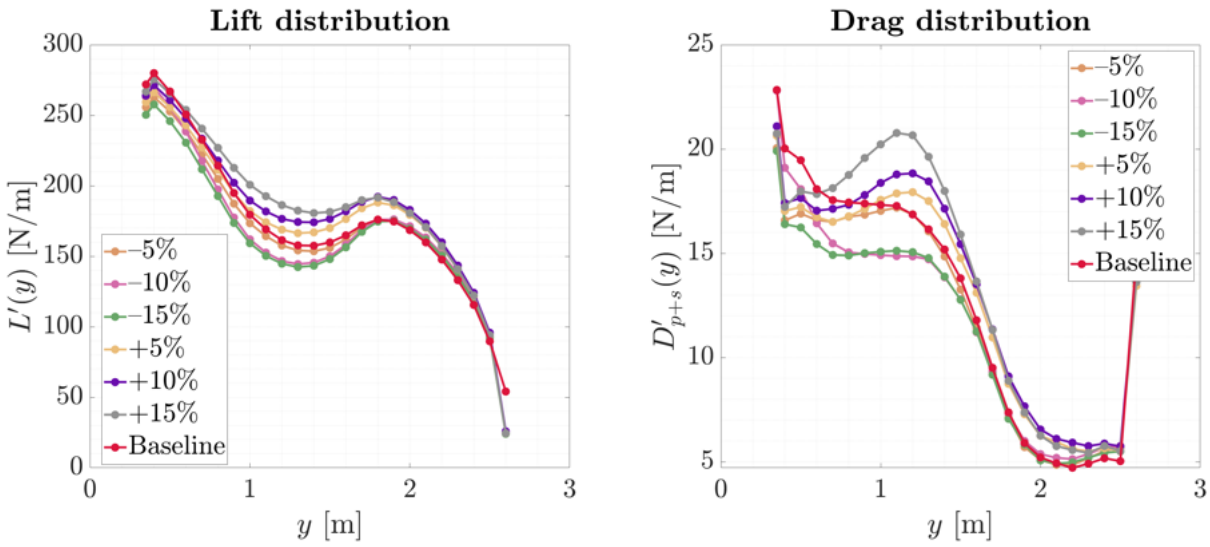
Figure 4.9: Optimized spanwise twist distributions for the six lift-constrained configurations [−15%, −10%, −5%, +5%, +10%, +15%] compared to the baseline.

4.5.1. Global *vs.* rotor pitch trimming

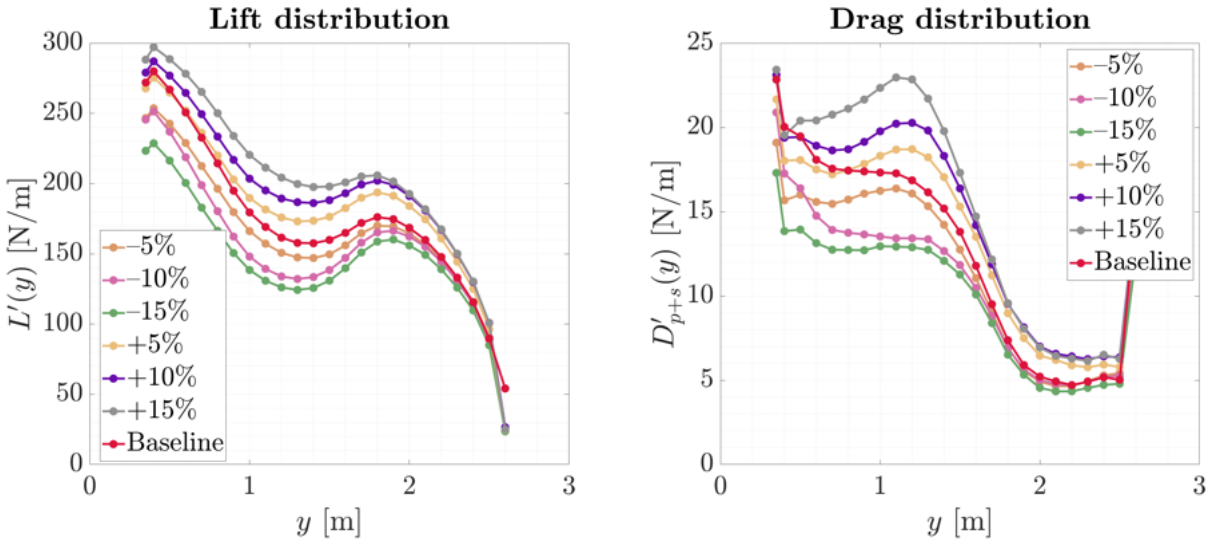
Figure 4.10 shows the load distributions for the six configurations compared with the baseline. Cruise equilibrium is achieved via either global pitch trimming (α, Ω_{front}, Ω_{back}), or rotor pitch trimming (γ, Ω_{front}, Ω_{back}), as shown in Figure 4.10a and Figure 4.10b, respectively.

As noted in Section 4.3, global pitch trimming drives the lift distributions toward the baseline, as α adjusts to satisfy the trim condition. Rotor pitch trimming, instead, largely preserves the optimized lift distributions, resulting in more distinct curves in Figure 4.10b, as α remains unchanged.

Consistent with Section 4.4, the optimization process tends to redistribute the load toward the tip. Considering Figure 4.10b, when the lift requirement is lower, the local lift deficit in the rotor-wing interference region deepens, yielding a smoother drag distribution across the affected spanwise section. This further demonstrates that, in wing lift-to-drag-ratio maximization, the optimum is not achieved by augmenting lift in the rotor-wing interference region, but by a twist distribution that redistributes drag toward a more uniform spanwise profile.



(a) Global pitch trimming (α, Ω_{front}, Ω_{back}).



(b) Rotor pitch trimming (γ, Ω_{front}, Ω_{back}).

Figure 4.10: Spanwise lift and drag distributions for the six lift-constrained configurations [−15%, −10%, −5%, +5%, +10%, +15%] and the baseline. Corresponding twist distributions are displayed in Figure 4.9.
All results are reported in trimmed cruise condition, achieved via either the (α, Ω_{front}, Ω_{back}), or the (γ, Ω_{front}, Ω_{back}) variable set.

Figure 4.11 and Table 4.8 report the results of this study. Prior to the analysis, it is noted that the stall-margin constraint is satisfied for all configurations, i.e., $\max(C_l(y)) \leq 1.15$ (Table 4.8).
Considering Figure 4.11, the wing-lift results show that, as expected, global pitch trimming attenuates the effect of the twist-distribution optimization, reducing the lift gap

between the configurations. This is consistent with wing-lift distributions being driven closer together by the α adjustment (Figure 4.10a). Considering the rotor-pitch curve, instead, it is inferred that γ modification has only a minor impact on wing lift, as its deviation from the optimized conditions is small. The wing and complete-vehicle lift-to-drag ratios exhibit opposite trends. As expected, an increased lift requirement leads to degraded L^w/D^w. However, recalling that wing lift and drag account for $\sim 75\%$ and $\sim 30\%$ of the total vehicle aerodynamic loads, respectively, the increase in wing lift outweighs the drag penalty from the complete-vehicle perspective.

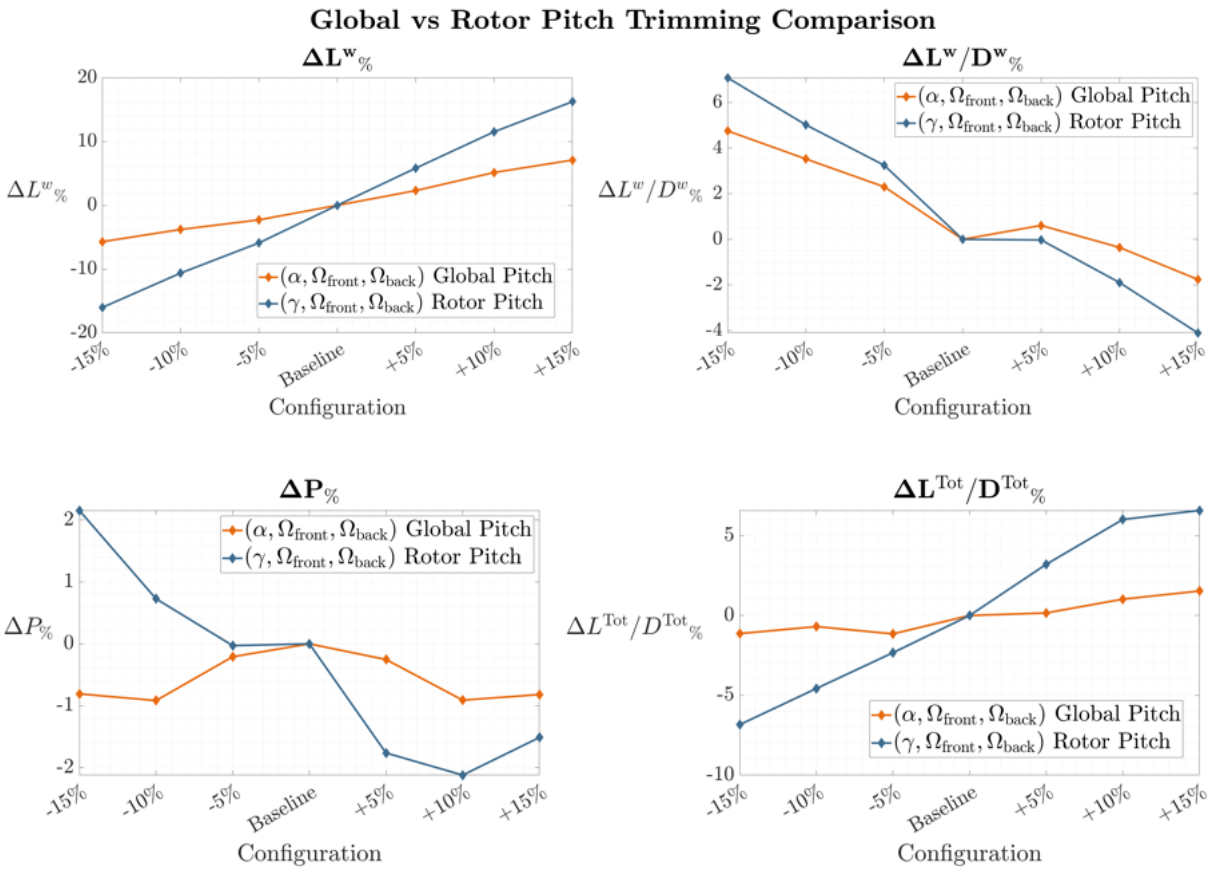


Figure 4.11: Wing lift, wing and complete-vehicle lift-to-drag ratios, and percentage variation in required power for the six optimized configurations [−15%, −10%, −5%, +5%, +10%, +15%] and the Baseline. Percentages are relative to the baseline. Corresponding twist distributions are displayed in Figure 4.9.
All results are reported in trimmed cruise condition, achieved via either the (α, Ω_{front}, Ω_{back}), or the (γ, Ω_{front}, Ω_{back}) variable set.

Remarkably, the $L^{\text{Tot}}/D^{\text{Tot}}$ rotor-pitch curve exhibits a quasi-linear trend, except for the +15% configuration, which yields a smaller lift-to-drag-ratio improvement than expected from the linear trend. This may be attributed to the significant twist modification altering the wing-fuselage interference and seemingly increasing fuselage drag.
Regarding the $L^{\text{Tot}}/D^{\text{Tot}}$ global-pitch curve, the α modification mitigates the effect of the twist-design changes on performance, resulting in only marginal variations in $L^{\text{Tot}}/D^{\text{Tot}}$.

Configuration		α [°]	γ [°]	$\max(C_l(y))$ [-]	$L^{\text{Tot}}/D^{\text{Tot}}$ [-]	T [N]	$\Delta P_{\%}$ [%]	T/P [N/W]
Global Pitch Trim	-15%	-9.70	-15.30	1.00	5.43	529.8	-0.81	0.093
	-10%	-10.00	-15.60	1.01	5.46	523.3	-0.9	0.092
	-5%	-10.29	-15.89	1.00	5.43	518.6	-0.2	0.090
	+5%	-10.91	-16.51	1.06	5.51	505.3	-0.3	0.088
	+10%	-11.16	-16.76	1.08	5.55	497.4	-0.9	0.087
	+15%	-11.42	-17.02	1.07	5.58	491.5	-0.8	0.086
Baseline		-10.59	-16.19	0.99	5.50	508.9	/	0.090
Rotor Pitch Trim	-15%	-10.59	-13.52	0.92	5.12	581.3	+2.2	0.099
	-10%	-10.59	-14.29	0.96	5.25	558.4	+0.7	0.096
	-5%	-10.59	-15.04	0.97	5.37	537.4	/	0.094
	+5%	-10.59	-17.11	1.08	5.67	487.4	-1.8	0.086
	+10%	-10.59	-18.37	1.13	5.83	461.3	-2.1	0.082
	+15%	-10.59	-19.41	1.14	5.86	443.3	-1.5	0.078

Table 4.8: Trim parameters, stall margin, and complete-vehicle and rotor performance for the six optimized configurations [−15%, −10%, −5%, +5%, +10%, +15%] compared with the baseline. Power variations are expressed as percentage deviations relative to the baseline. Corresponding twist distributions are shown in Figure 4.9.
All results correspond to trimmed cruise conditions, achieved either by the (α, Ω_{front}, Ω_{back}), or the (γ, Ω_{front}, Ω_{back}) variable set.

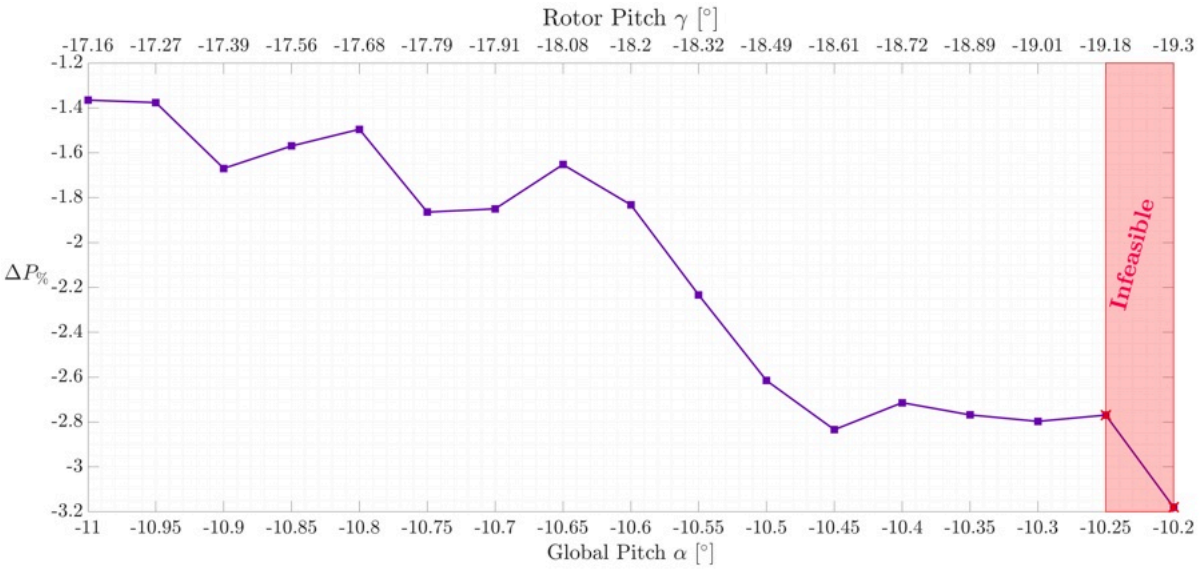
Finally, the power-consumption variation results are presented in Figure 4.11. For each configuration, the power requirement is governed by two competing effects. First, when wing lift is increased, the total thrust demand decreases (see Table 4.8), but the thrust in the forward direction must increment because of higher drag. Thus, the rotor pitch $|\gamma|$ must increase to divert more thrust in the advancing direction, thereby degrading the

rotor thrust-to-power ratio. Conversely, a lower wing lift reduces drag, while the total thrust demand increases. However, the required forward thrust decreases, and so does $|\gamma|$, resulting in improved rotor performance. The net balance between these two effects determines whether the power consumption increases or decreases relative to the baseline. Considering global pitch trimming, each configuration leads to a reduction in power consumption. This implies that, where thrust demand increases, the associated rotor performance gain dominates; where it decreases, the thrust reduction prevails over the efficiency penalty.
Regarding rotor pitch trimming, among the -15%, -10%, and -5% configurations, the increased thrust demand outweighs rotor-performance improvement, leading to increased or unchanged power consumption compared with the baseline. The remaining configurations, instead, exhibit a degraded T/P. However, the reduction in required thrust prevails, yielding a net decrease in the power requirement.

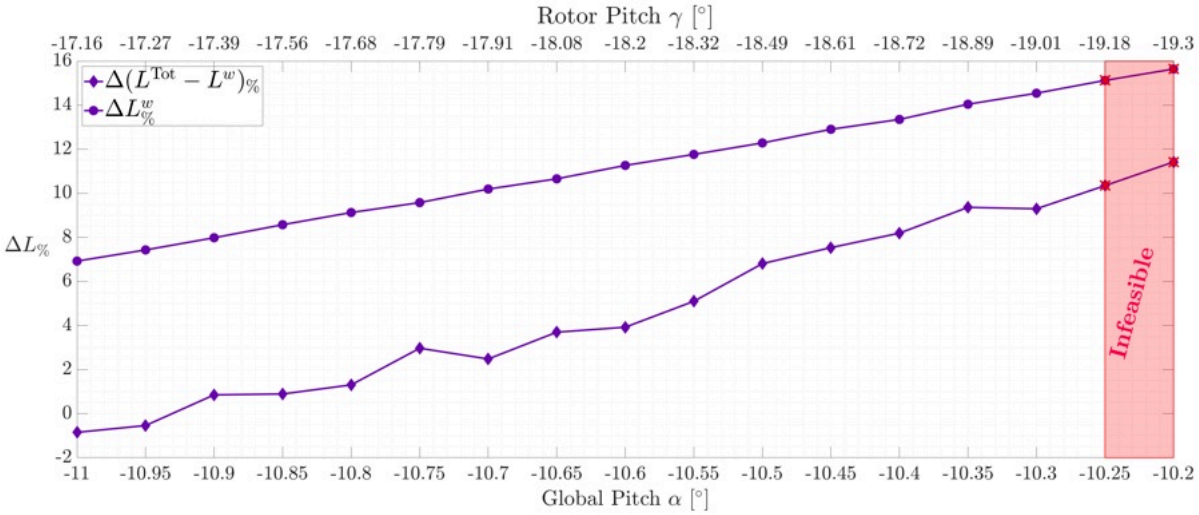
Overall, most of the configurations lead to a reduction in power consumption, albeit modest. The most promising design is $+10\%$, yielding $\Delta P_{\%} = -0.9\%$ and $\Delta P_{\%} = -2.1\%$ with global and rotor pitch trimming, respectively. Since both trim strategies produce a reduction in required power, combining them is expected to yield a further improvement. In the following section, the combination of the four parameters (α, γ, Ω_{front}, Ω_{back}) is investigated to identify the minimum power requirement for the $+10\%$ configuration.

4.5.2. Four-Parameter trim optimization

Since only three parameters are needed to retrieve a trim condition, the present approach relies on a sweep over α. For each imposed value of α, the remaining parameters γ, Ω_{front}, and Ω_{back} are adjusted to ensure cruise equilibrium. The resulting power-consumption variation with respect to the baseline is shown in Figure 4.12.
Apart from some scattering due to numerical approximation, the curve trend is clear and consistent with the previous findings. A more nose-up configuration (decreased $|\alpha|$) causes an increase in both wing lift and drag, leading to lower total thrust demand and higher rotor pitch $|\gamma|$. Again, the power consumption reduction due to the decrement in thrust requirement prevails over the power increase due to rotor-performance degradation. The further $|\alpha|$ decreases, the greater the reduction in required power. This trend is bounded when the stall-margin constraint, enforced by *critical section theory*, is violated (red region in Figure 4.12, labeled as infeasible). The minimum power consumption for a feasible design corresponds to $\Delta P_{\%} = -2.8\%$.



(a) Power consumption percentage variation relative to the baseline.



(b) Wing and other vehicle components lift percentage variation relative to the baseline.

Figure 4.12: Cruise configuration comparison for the 10% wing design. Power and lift percentage variations, relative to the baseline, are analyzed over a global pitch angle α sweep. The remaining trim parameters γ, Ω_{front}, and Ω_{back} are adjusted to maintain cruise equilibrium at each α. All results correspond to trimmed configurations. The infeasible region, where the stall-margin constraint is violated, is highlighted in red.

This behavior may appear inconsistent with Section 4.5.1, where a further increase in wing lift from the +10% to the +15% configuration results in a power-consumption increment. However, Figure 4.12a shows that a further wing lift increase is followed by a net reduction in required power. The difference stems from the fact that the +15% configuration is studied for fixed $\alpha = -10.59°$, and the wing-lift increase induced by the twist

distribution negatively affects the performance of the remaining vehicle components. In the present sweep study, instead, α varies. Figure 4.12b shows that as $|\alpha|$ decreases, the lift contribution from the remaining vehicle components increases, resulting in a further reduction in thrust demand, with the minimum-power condition reached at $\alpha = -10.45°$ and $\gamma = -18.61°$.

In conclusion, the lift-based wing design strategy proves more effective than conventional lift-to-drag-ratio maximization (Section 4.4). When (α, Ω_{front}, Ω_{back}) are considered as trim parameters, both wing-lift increase and decrease can lead to a net reduction in power consumption, albeit modest. When (γ, Ω_{front}, Ω_{back}) are employed, the required power is reduced only if wing lift increases, such that the benefit due to reduced total thrust demand outweighs the penalty associated with degraded rotor performance. This second strategy leads to greater power savings.
Overall, when the two strategies are combined, with trim parameters (α, γ, Ω_{front}, Ω_{back}), a further reduction in power consumption can be achieved. The lower $|\alpha|$, the greater the benefit, owing to the combined lift contribution of the wing and the remaining vehicle components. However, a more nose-up configuration reduces the stall margin, such that the minimum-power configuration within the considered α range lies beyond the feasibility boundary enforced by *critical section theory*. Among all feasible configurations analyzed, the greatest power reduction is $\Delta P_{\%} = -2.8\%$, achieved by the $+10\%$ configuration at $\alpha = -10.45°$ and $\gamma = -18.61°$. The present analysis indicates that cruise efficiency is influenced more by the vertical-load sharing between the wing and the rotors and by the resulting trim condition than by local twist optimization alone. The main design variable is therefore not the detailed twist distribution itself, but the redistribution of lift under cruise equilibrium constraints. Further interpretation of the performance of this configuration is provided in Section 4.8.2, where the uncertainty associated with rotor modeling is assessed.

4.6. Wing Position Optimization

The primary phenomenon governing the flow field is rotor-wing interference, resulting in a local reduction in the effective angle of attack of the wing. Thus far, this has been assessed via twist-distribution modification, allowing an investigation of whether restoring the original wing lift distribution or redistributing it optimally could improve wing and vehicle performance. In this section, instead, the effect of wing longitudinal position is examined. The modification of this parameter is expected to affect wing-rotor interaction, leading to either performance enhancement or degradation. Moreover,

due to the related change in wing lift, wing positioning is coupled with wing incidence to widen the design space. The first step consists of assessing performance sensitivity to these variables without accounting for trim equilibrium. Should some configurations prove promising, considerations regarding the center-of-gravity shift, moment variation, and their effect on the trim configuration could be further investigated.

Figure 4.13 illustrates the displacement axis s and its range. In the baseline configuration, the wing is located in the most rearward position that allows a 5 cm clearance from the rear rotors. Consequently, in the present study, only forward positioning is investigated. The maximum displacement is $s = 0.28$ m, for which the wing is located 5 cm downstream of the front rotors. Table 4.9 reports the displacement and incidence ranges.
The design space is defined over two variables of limited ranges. Therefore, full-factorial sampling is employed to ensure adequate coverage of the parametric space without incurring prohibitive computational costs.

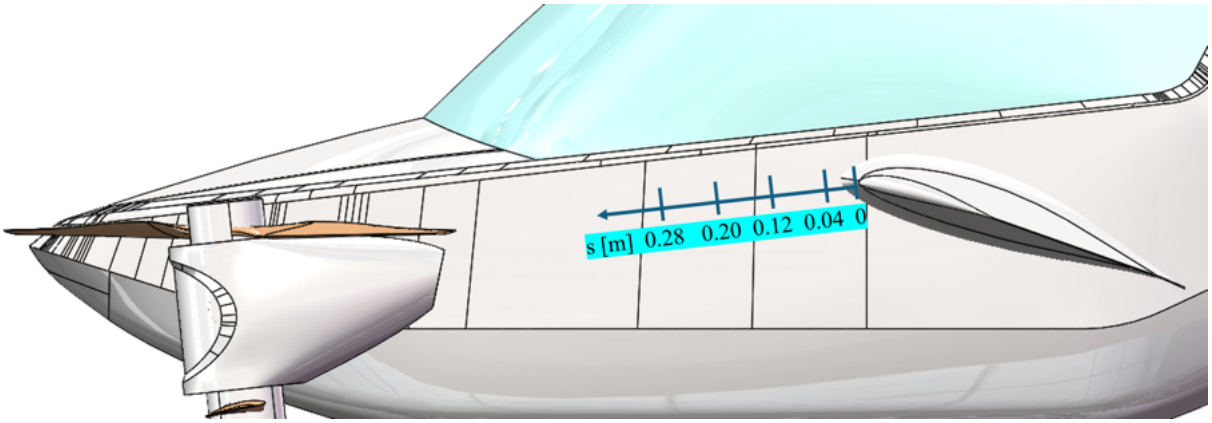


Figure 4.13: Definition of the displacement axis s and its feasible range within the vehicle geometry.

Variable	Range
Displacement s	$[0, 0.28]$ m
Incidence	$[16.5°,\ 18.5°]$

Table 4.9: Design variable ranges for the wing-position optimization study.

Prior to the design process, it is worthwhile to assess the effects of the front and rear rotors on the wing separately. Wing forward positioning increases the distance from the rear rotors and decreases it from the front rotors. Consequently, as the displacement s

increases, the wing is expected to experience more intense interference from the front rotors and less intense interference from the rear rotors, respectively.
Figure 4.14 compares the baseline load distributions obtained with the following rotor combinations: all rotors, front rotors only, rear rotors only, and no rotors. The local downwash effect is found to be induced solely by the front rotors. The rear rotors, instead, provide an upwash effect relative to the rotor-off configuration, thereby enhancing wing performance. While the front-rotor interference is spatially non-uniform, peaking near the mid-span, the rear-rotor interaction exhibits a more uniform spanwise distribution. The rear-rotor-induced upwash intensity is strongest inboard and gradually diminishes toward the tip. Overall, at inboard sections, the front-rotor-induced downwash is mitigated by the rear-rotor-induced upwash; toward the tip, both rotors induce upwash, which superimposes constructively in the all-rotors configuration.

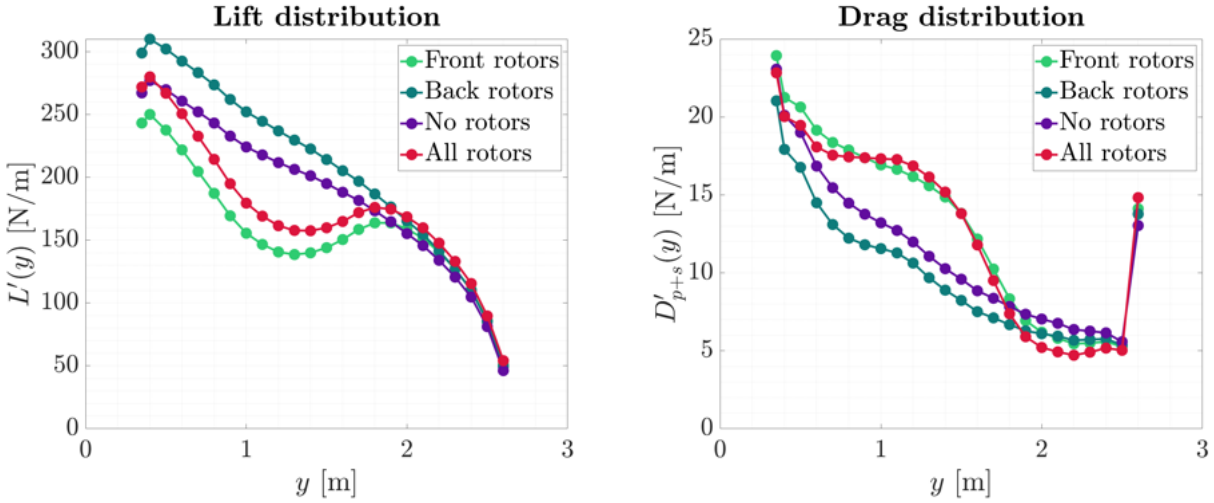


Figure 4.14: Spanwise lift and drag distributions for the baseline flight condition, obtained with four rotor combinations: all rotors, front rotors only, rear rotors only, and no rotors.

In summary, the front rotors mainly induce a local downwash effect, while the rear rotors induce a more uniform upwash, degrading and enhancing performance, respectively. At first glance, increasing the distance from the rear rotors, which benefit performance, while reducing it from the front rotors, which degrade it, would not be expected to yield any net gain. However, it cannot be assumed that the baseline wing position corresponds to the point of maximum adverse interaction with the front rotors nor to the point of maximum beneficial interaction with the rear rotors, within the displacement range of Table 4.9. This study is therefore justified to confirm or refute these preliminary expectations. A qualitatively similar investigation was conducted by Ai *et al.* [2], indicating that

performance improves as the wing–front-rotor distance increases and the wing–rear-rotor distance decreases. However, the operating conditions differ substantially from those of the present work, and the analysis was performed on a small-scale vehicle; accordingly, the results may differ. For this reason, it cannot be assumed a priori that the same behavior will be observed in the present case.

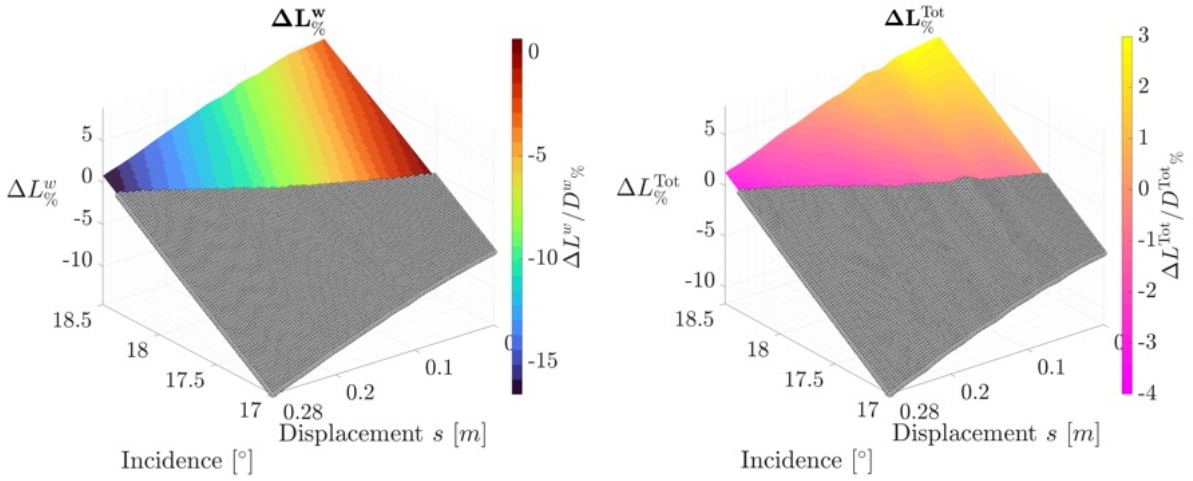


Figure 4.15: Wing and complete-vehicle lift percentage variations relative to the baseline, represented as surfaces over the incidence–displacement design space and colored by the respective lift-to-drag ratios. Results are reported in untrimmed conditions. The gray region identifies configurations with wing lift below the baseline value.

As anticipated, the limited size of the design space allows full-factorial sampling to be employed. The CFD simulations are employed to generate high-fidelity data over a grid in the parametric space. These data are interpolated via a Kriging surrogate model and presented in Figure 4.15. The surfaces represent wing and complete-vehicle lift percentage variation relative to the baseline, colored by the corresponding lift-to-drag ratios. The gray region delimits the configurations exhibiting a lower wing lift than the baseline.
Recalling that trim implications are set aside at this stage, wing lift monotonically decreases with increasing displacement s. Similarly, the wing lift-to-drag ratio peaks near the baseline configuration (null displacement, 17.5° incidence) and deteriorates with forward positioning. In particular, as s increases, the downwash region deepens, yielding a more pronounced lift decrease in the inboard sections, along with a drag increase, as shown in Figure 4.16. As discussed in previous sections, an increase in wing lift benefits the complete-vehicle lift-to-drag ratio, despite the associated drag increase. Consequently, from the $\Delta L^{\mathrm{Tot}}/D^{\mathrm{Tot}}{}_{\%}$ contours, the complete-vehicle lift-to-drag ratio monotonically increases with wing incidence. However, Sections 4.3 and 4.5 demonstrate that an indiscriminate increase in wing lift does not generally yield a substantial reduction in power

consumption. Furthermore, the $\Delta L^{\text{Tot}}_{\%}$ surface and the related contours show increasing degradation with wider displacement, with minor exceptions likely attributable to numerical approximation.

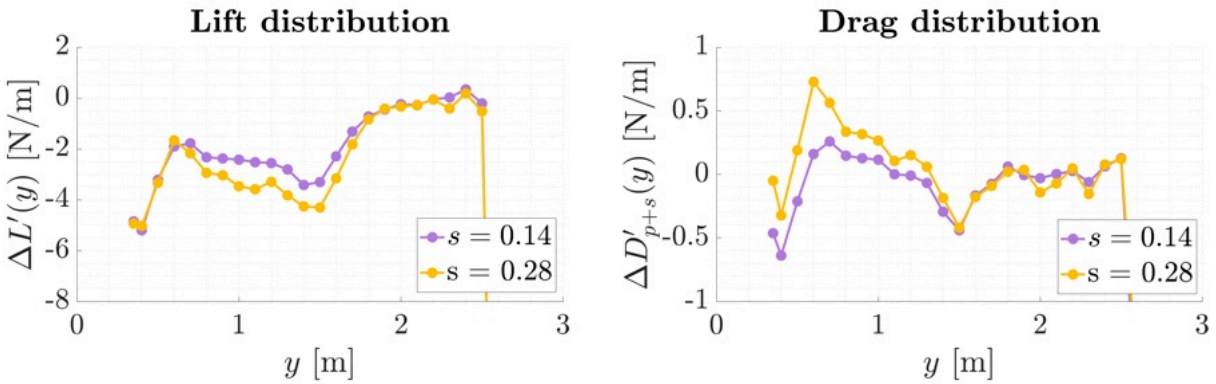


Figure 4.16: Spanwise lift and drag distribution variations relative to the baseline. While the wing incidence is fixed as the baseline value, the displacement s is modified.

In conclusion, and consistent with the work of Ai *et al.* [2], the study confirms the preliminary expectations. Forward positioning modifies the balance between front and rear rotor interference. As displacement s increases, the net effect is a stronger induced downwash, degrading both wing and complete-vehicle performance. Consequently, neglecting the center-of-gravity shift and moment variation, the wing position at $s = 0$ represents the most favorable configuration within the considered range, as it maximizes the distance from the front rotors while preserving the beneficial upwash generated by the rear rotors. Since all forward-position configurations already degrade the untrimmed aerodynamic performance, the present results do not justify further investigation accounting for trim considerations.

4.7. Alternative Design

The previous sections highlight both the potential and the limitations of the lift-increase strategy. While a higher lift force reduces the thrust magnitude required for trim, the associated drag penalty demands greater forward thrust. As a result, the trimming condition shifts toward higher rotor pitch angles $|\gamma|$, degrading rotor efficiency. The net reduction in power consumption is therefore modest or negligible.

A promising approach consists of addressing the increased drag through a modification of the vehicle design. Figure 4.17 presents two cargo UAV concepts based on a dual-system architecture, in which vertical and forward thrust are generated by dedicated,

separate rotor systems. The Cento design is similar to the present configuration, featuring eight rotors distributed over four axes, complemented by a rear pusher propeller for forward-thrust generation. The rationale underlying these configurations lies in rotor aerodynamics: in edgewise flow, a rotor operates more efficiently when its axis is perpendicular to the freestream. Specifically, reducing $|\gamma|$ increases thrust while leaving power consumption nearly unaffected [34]. By decoupling lift and propulsion, the lift rotors operate at low pitch angles in a favorable aerodynamic regime, while a dedicated pusher propeller, optimized for axial-flow conditions, handles forward thrust. This division of roles is expected to yield an overall reduction in power consumption compared with the baseline, where a single highly pitched rotor system must sustain lift and overcome drag.



(a) Cento [10, 12]. (b) Pipistrel [27].

Figure 4.17: Alternative design examples with forward pusher configurations.

In this section, the design concepts of the Cento and Pipistrel vehicles are applied as a low-fidelity investigation. For two configurations, the vehicle is trimmed in the vertical and pitching directions only. The forward equilibrium is satisfied by the pusher thrust T_{pusher}, which translates directly into a power requirement as

$$P_{\text{pusher}} = \frac{T_{\text{pusher}} V_\infty}{\eta_p}, \tag{4.5}$$

where V_∞ is the freestream velocity and η_p is the propulsive efficiency. Although $\eta_p = 0.8$ is a standard assumption in preliminary sizing [30], a lower value should be adopted to account for the aerodynamic interference introduced by the new component. Results are therefore presented for $\eta_p = 0.65$ and $\eta_p = 0.8$, to assess performance across a representative range of propulsive efficiency values.

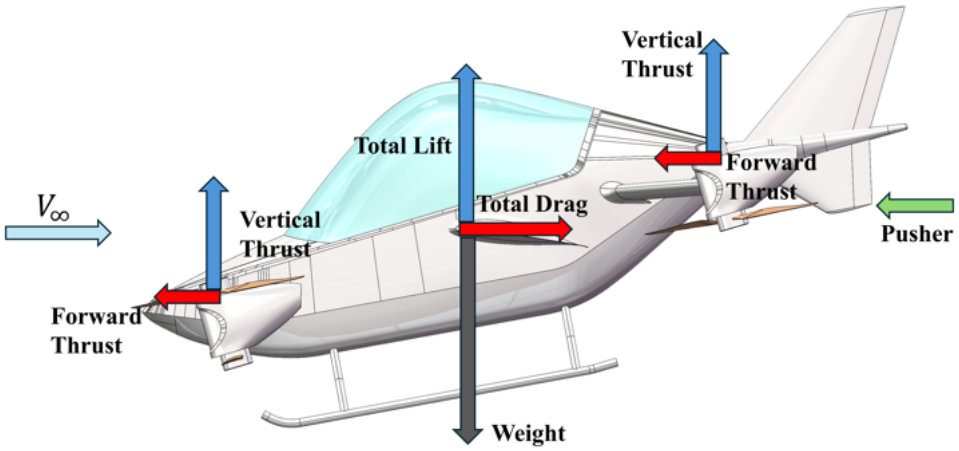


Figure 4.18: +15% configuration, alternative design.

4.7.1. +15% configuration

Figure 4.18 illustrates the +15% configuration introduced in Section 4.5, here combined with the alternative design concept featuring a pusher propeller.
The vehicle pitch angle and rotor shaft angle are kept fixed at their baseline values, namely $\alpha = -10.59°$ and $\gamma_m = -5.6°$. Vertical and pitching equilibrium are achieved by adjusting the rotor rotational speeds, Ω_{front} and Ω_{back}. By construction, the +15% configuration generates greater lift than the baseline. As a result, Ω_{front}, Ω_{back}, and consequently the rotor power consumption P_{rotors}, are reduced. Table 4.10 reports the rotor power required for vertical and pitching equilibrium, along with the thrust required from the pusher, T_{pusher}, which equals the residual forward force imbalance arising from the combined effect of increased drag and reduced rotor thrust. P_{pusher} is then estimated via Eq. (4.5) for $\eta_p = 0.65$ and $\eta_p = 0.8$. The total power consumption is $P_{\text{rotors+pusher}} = P_{\text{rotors}} + P_{\text{pusher}}$.
As expected, the alternative design yields lower power consumption for higher pusher efficiency. Comparing these results with Table 4.8 of Section 4.5, the +15% configuration yields −0.9% and −1.5% changes in required power with the global and rotor pitch trimming strategies, respectively. At first glance, the alternative design appears to yield a comparable reduction in power consumption. However, Section 4.8.2 reveals substantial uncertainty associated with the rotor modeling, particularly in terms of power consumption. The reader is referred to Section 4.8.2 for a thorough interpretation of the results of the alternative design strategy.

Configuration	T_{pusher} [N]	P_{rotors} [W]	P_{pusher} [W]		$P_{\text{rotors+pusher}}$ [W]	
			$\eta_p = 0.65$	$\eta_p = 0.8$	$\eta_p = 0.65$	$\eta_p = 0.8$
Baseline	/	5747	/	/	5747	5747
+15% altern.	20.55	4792	948	770	5740	5562
$\boldsymbol{\Delta}_{\%}$					−0.12%	−3.22%

Table 4.10: Rotor and pusher power consumption for the +15% alternative design configuration, compared with the baseline. Results are reported for two values of propulsive efficiency, $\eta_p = 0.65$ and $\eta_p = 0.8$.

4.7.2. Zero-pitch baseline

According to the VDM results, the forward pusher applied to the +15% configuration yields only marginal improvements in required power. The next step consists of applying the alternative design strategy to the baseline vehicle in its most straightforward form. Figure 4.19 shows the baseline configuration at global pitch angle $\alpha = 0°$ and rotor shaft angle $\gamma_m = -5.6°$. The rotor performance is expected to improve significantly, as the rotor pitch angle is substantially reduced in magnitude, $\gamma = \gamma_m = -5.6°$. The wing incidence is modified to achieve an adequate angle of attack, while its positioning, twist, and shape remain unchanged.

As in the previous section, vertical and pitch equilibrium are achieved by adjusting Ω_{front} and Ω_{back}. P_{pusher} is then estimated via Eq. (4.5) from the forward force imbalance T_{pusher}, for $\eta_p = 0.65$ and $\eta_p = 0.8$.

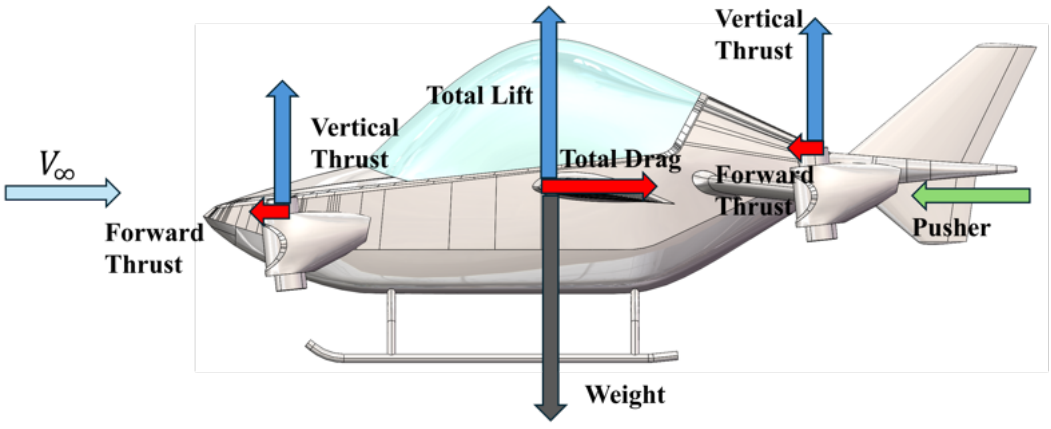


Figure 4.19: Zero-pitch baseline configuration, alternative design.

Table 4.11 reports the performance of the vehicle and the rotors. The zero-pitch configu-

ration yields better performance relative to the baseline. The wing generates more lift at a lower drag cost, while the remaining components of the vehicle exhibit an increase in both lift and drag. The net effect is a L^{Tot} and D^{Tot} increment, along with an improvement in the lift-to-drag ratio of $\sim 9\%$. Moreover, as expected, the rotor-system thrust-to-power ratio is increased due to the reduction in $|\gamma|$, consistent with the literature [34].
Despite a total drag increase, all considered performance metrics of the vehicle and the rotors show substantial improvements. Based on these observations, a reduction in total power consumption relative to the baseline would be expected. However, $P_{\text{rotors+pusher}}$ increases even when the most favorable value of $\eta_p = 0.8$ is considered. Specifically, the baseline would be outperformed only if $\eta_p \gtrsim 0.82$. This outcome is unlikely to be attributable to the D^{Tot} increment, whose penalty is outweighed by the lift gain, resulting in a higher lift-to-drag ratio. The most probable cause of the inconsistent result lies in the rotor modeling. In the present study, the VDM is validated in axial-flow conditions only and, as anticipated in Section 3.3.1, its reliability is to be questioned when a critical condition such as forward flight is simulated; its results should be interpreted with caution. Further discussion of the model uncertainty is provided in Section 4.8.2, where this configuration is analyzed with the SMM and compared with these results.

Configuration	L^w [N]	D^w_{p+s} [N]	L^w/D^w_{p+s} [-]	L^{Tot} [N]	D^{Tot}_{p+s} [N]	$L^{\text{Tot}}/D^{\text{Tot}}_{p+s}$ [-]	$(T/P)_{\text{rotors}}$ [N/W]
Baseline	408.7	29.7	13.74	524.9	95.5	5.50	0.089
Zero-pitch alt.	434.3	25.3	17.15	620.7	103.8	5.98	0.184
$\boldsymbol{\Delta}_{\%}$	+6.3%	−14.7%	+24.8%	+18.3%	+8.7%	+8.7%	+107%

Configuration	T_{pusher} [N]	P_{rotors} [W]	P_{pusher} [W]		$P_{\text{rotors+pusher}}$ [W]	
			$\eta_p = 0.65$	$\eta_p = 0.8$	$\eta_p = 0.65$	$\eta_p = 0.8$
Baseline	/	5747	/	/	5747	5747
Zero-pitch alt.	96.3	2220	4446	3613	6667	5833
$\boldsymbol{\Delta}_{\%}$					+16.0%	+1.5%

Table 4.11: Aerodynamic loads, rotor thrust-to-power ratio, and power budget for the zero-pitch alternative design configuration, compared with the baseline. Power results are reported for two values of propulsive efficiency, $\eta_p = 0.65$ and $\eta_p = 0.8$.

4.8. Model Uncertainty Assessment – VDM *vs.* SMM comparison

This section presents a comparison between the SMM and VDM results. Although the literature suggests that the SMM generally yields higher fidelity than the VDM, this cannot be assumed a priori in the present study due to the absence of validation against experimental data. Therefore, the following discussion should not be interpreted as a validation of the VDM against the SMM, but rather as a comparison between two different modeling techniques. Further conclusions about the fidelity of the two methods can be drawn only from a thorough and careful validation process.
First, the rotor physics is discussed, highlighting the unsteady nature of the flow. Then, the SMM is employed to simulate the three most relevant configurations of this study: the baseline configuration, the +10% configuration resulting from the four-parameter trim optimization (Section 4.5.2), and the zero-pitch baseline configuration from the alternative design strategy (Section 4.7.2).

4.8.1. Unsteady rotor flow characterization via SMM

The isolated rotors would exhibit a periodicity of 2/revolution, determined by the rotor blade count. However, the front and rear rotors must have different rotational speeds $\Omega_{\text{front}} \neq \Omega_{\text{back}}$ to ensure pitch equilibrium, breaking the periodicity of the rotor and vehicle loads. Consequently, the flow features a more complex periodicity that could only be observed over a sufficiently long time window, at a prohibitive computational cost. Hence, the results depend on the width of the averaging window.

$\Delta\tau_{\text{avg}}$	T_{z1} [N]	M_{z1} [Nm]	L^{Tot} [N]	L^{w} [N]
$\frac{1}{2}\tau_{\text{front}}$	138.0	12.80	491.7	410.1
τ_{front}	138.9	12.83	490.4	410.7
$2\tau_{\text{front}}$	138.5	12.82	492.1	411.6

Table 4.12: Effect of the averaging window width $\Delta\tau_{\text{avg}}$ on selected quantities. Results are averaged over different widths chosen as multiples of the front rotors' period τ_{front}.

Table 4.12 reports representative results averaged over different window widths $\Delta\tau_{\text{avg}}$ chosen as multiples of the front-rotor period τ_{front} (rotors 1 and 2, Figure 4.21), for the baseline configuration. Both rotor and vehicle loads show minimal sensitivity to the width of the averaging window. Therefore, although the periodicity of the complete vehicle can

be observed only over a sufficiently long time record, considering a window proportional to the rotor period still yields load estimates within acceptable uncertainty bounds.

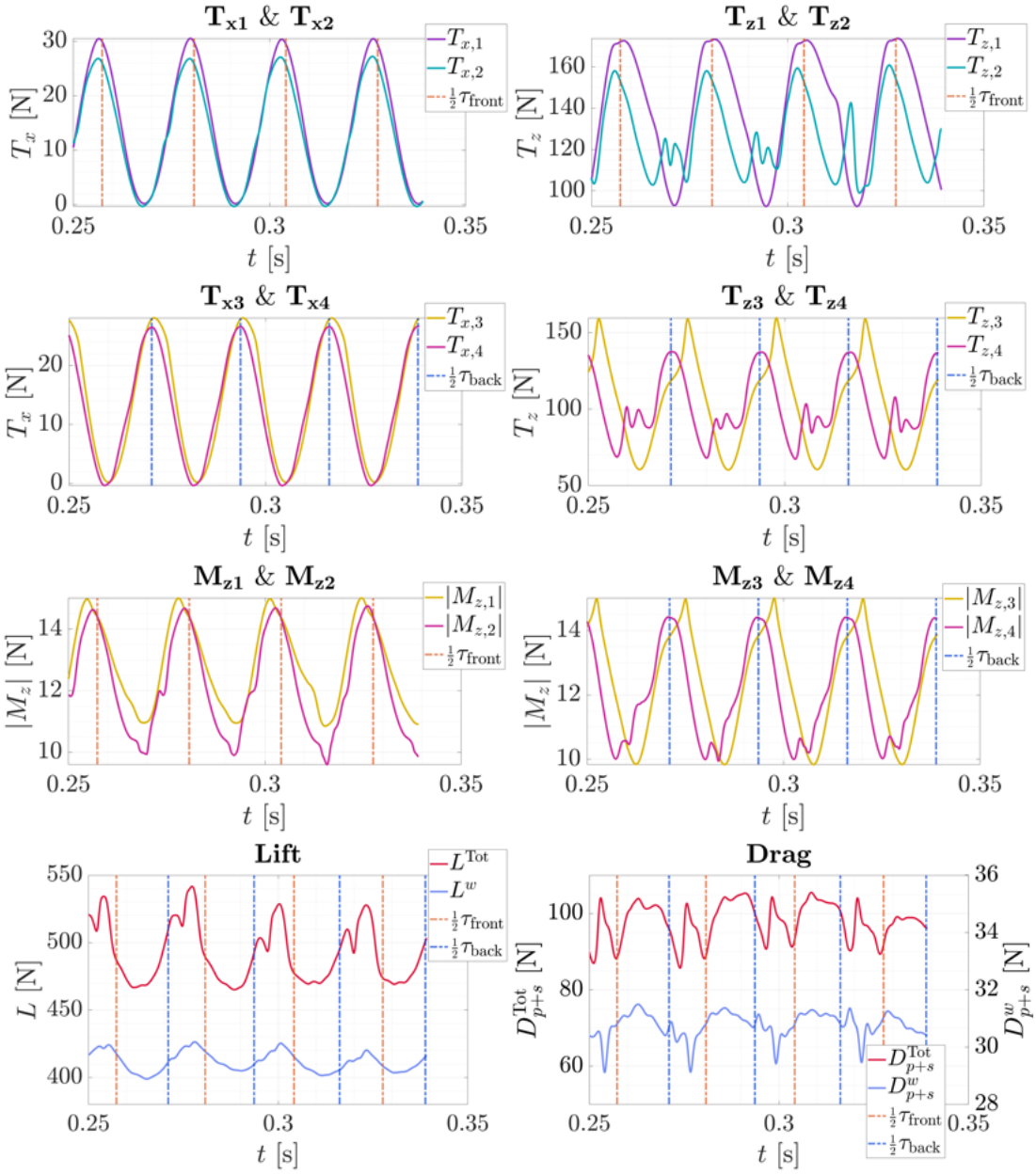


Figure 4.20: Time evolution of rotor thrust, torque, and vehicle aerodynamic loads for the baseline configuration (SMM).

Figure 4.20 shows the time evolution of the thrust components in the rotor-disk frame, the torque, and the lift and drag of the complete vehicle and the wing alone. All quantities exhibit large oscillations due to the inherent asymmetry of rotors in forward flight. In this condition, each blade experiences periodic variations in its loads, switching se-

quentially from advancing to retreating relative to the freestream. While this behavior is further assessed in the next paragraph, Figure 4.20 reveals the quasi-periodic nature of the oscillations, whose successive cycles are similar but not exactly superimposable. The rotor loads follow similar trends each half revolution, with approximately constant peak values but minor differences between successive cycles. The greatest impact is observed on the vehicle loads. The evident lack of periodicity stems from the rotor-vehicle interactions. At each revolution, the front and rear rotors occupy different relative azimuthal positions. Section 4.6 demonstrates that the front and rear rotors induce local downwash and upwash effects on the wing, respectively. Hence, the combination of the two wing-rotor interactions produces a varying net effect over each period, resulting in wider or narrower oscillations, both in duration and magnitude. However, Table 4.12 shows that the averaged loads have marginal sensitivity to this phenomenon, thus justifying the use of a limited averaging window $\Delta\tau_{\text{avg}}$ to characterize the mean loads.

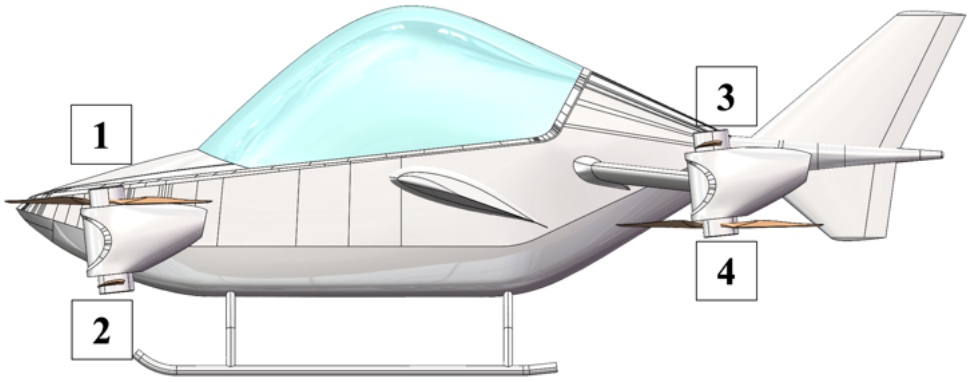


Figure 4.21: Rotor numbering convention used throughout the SMM analysis (rotors 1 to 4).

In the baseline configuration, the angle of attack of the vehicle is $\alpha = -10.59°$ and the shaft angle of the rotors is $\gamma_m = -5.6°$. Thus, the rotors operate at a pitch angle relative to the freestream of $\gamma = -16.19°$, in forward-flight conditions. Edgewise flow is a critical flight condition, in which, at each revolution, each blade experiences large load oscillations due to intrinsic asymmetry of the rotor disk. Periodically, each blade alternates between advancing and retreating relative to the freestream, resulting in a varying combination of freestream and rotation-induced velocities at each azimuthal position. This produces a perpetual variation in angle of attack and velocity magnitude during each revolution. The blade generates the most thrust when advancing, as the freestream and rotation-induced velocities add constructively, increasing the local dynamic pressure. Conversely, when retreating, the combination is destructive. The two contributions oppose each other,

reducing the local dynamic pressure to the point of generating reverse-flow regions near the root, where the rotational velocity is insufficient to overcome the freestream.

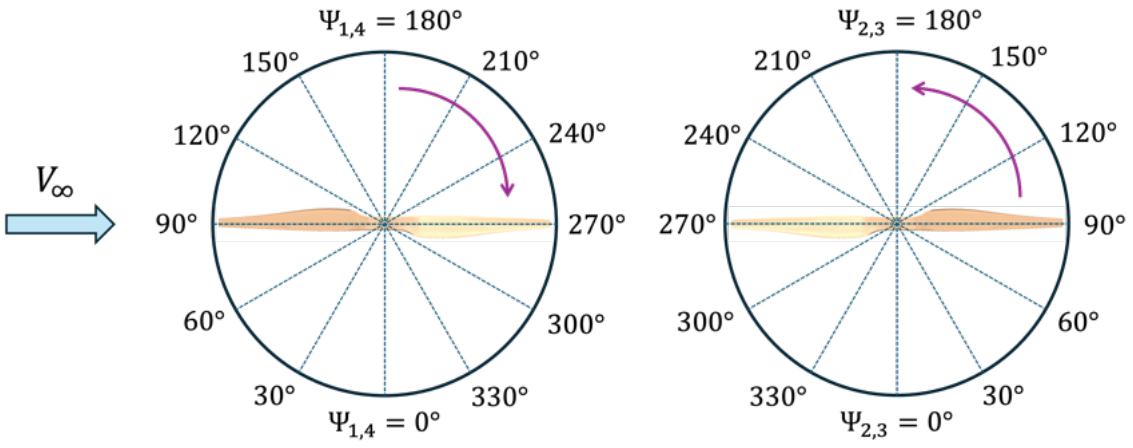


Figure 4.22: Definition of the rotor blade azimuthal position.

Typically, cyclic collective pitch is employed to periodically modify the blade pitch as a function of azimuthal position. This allows the blade sections to operate within an effective angle-of-attack range throughout most of the rotor azimuth, preserving aerodynamic effectiveness even during the retreating phase. This solution adds mechanical complexity and is not implemented in the vehicle design of the present study, rendering the blade largely ineffective over a substantial portion of each revolution. The twist distribution (Figure 3.3) is designed to maximize thrust during the advancing phase.

Figure 4.23 illustrates the contours of the sectional normal force coefficient multiplied by the local Mach number squared for the baseline configuration. The freestream flows from left to right, and rotors 1 and 4 rotate clockwise, while rotors 2 and 3 rotate counter-clockwise (see Figure 4.22). As a result, the blade is advancing over the lower half of the contours of rotors 1 and 4 and over the upper half of the remaining contours. The aforementioned phenomena are clearly visible in the plots. When the blade is advancing, the sectional normal force generated is at its maximum. Conversely, when the blade is retreating, the force is reduced, even reaching negative values over the inner sections near the root, where reverse flow occurs. Besides the clear asymmetry stemming from the advancing-retreating blade dichotomy, additional deviations from axial symmetry are observed and examined in the following paragraph.

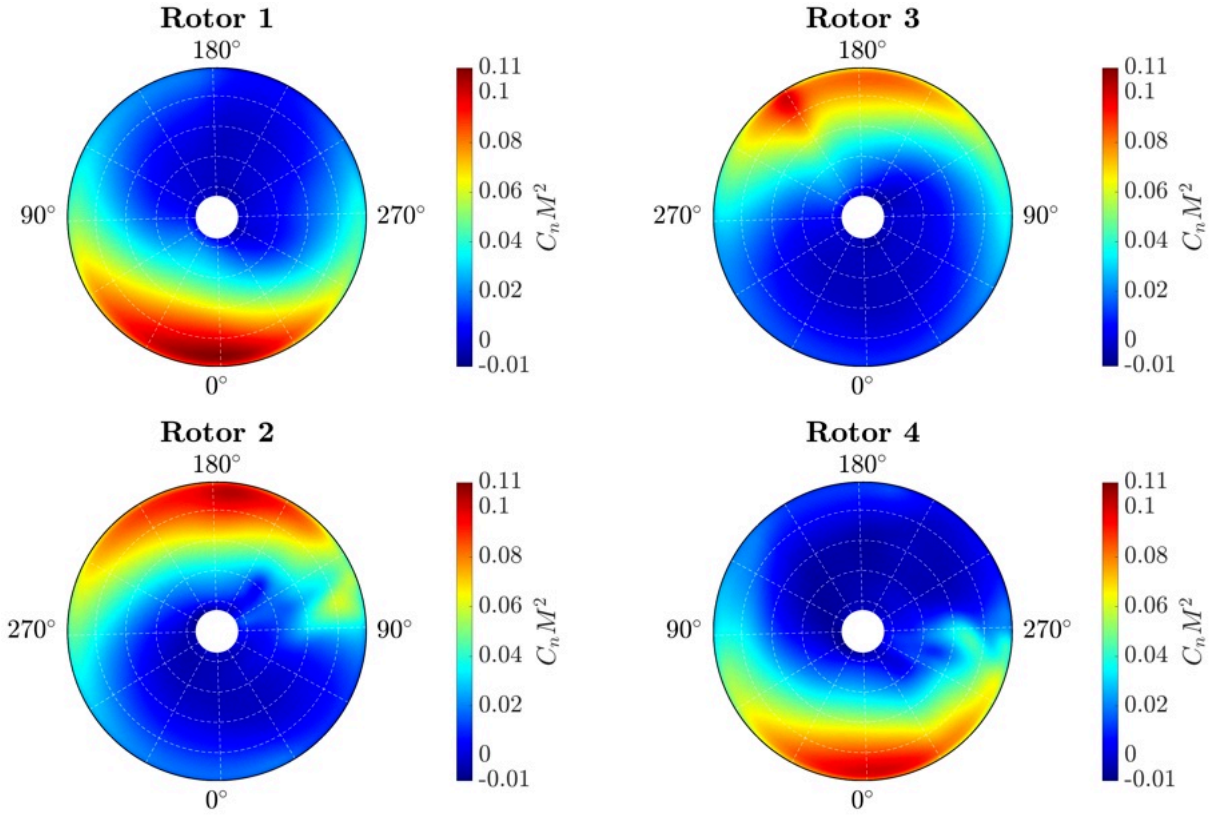


Figure 4.23: Contours of the sectional normal force coefficient C_nM^2 for the baseline configuration. The freestream flows from left to right.

Figure 4.24 shows the time derivative contours of the sectional normal-force coefficient multiplied by the local Mach number squared, for the baseline configuration. This visualization can be employed to highlight interaction events such as blade crossover and blade-vortex interaction [43]. The upper rotors predictably exhibit a more uniform distribution than the lower rotors, the latter being dominated by the vortex structures shed from the rotor hub mountings. Approximately, all rotors feature decreasing normal force over the left half-disk, where the local dynamic pressure decreases. The opposite holds over the right half-disk. Notably, the lower rotors experience larger positive gradients when the blade is aligned with the freestream direction than upper rotors. This behavior is likely attributable to the interaction of the blade with vortex structures shed from the rotor hub mountings. In the wake regions, the velocity magnitude is reduced due to the velocity defect and flow separation in the wake. This results in a less destructive combination with the retreating blade rotation-induced velocity and thus in a higher force derivative and force magnitude than in the upper rotors, as shown in Figures 4.23 and 4.24. Finally, because of the high complexity of the flow field, the vortex structures shed by the vehicle and the rotors combine, rendering the lower-rotor time-derivative contours

highly scattered and poorly informative. Although it is impossible to unambiguously attribute the local behavior of the lower blades to specific phenomena, rotors 3 and 4 both exhibit a region of sudden force reduction within the right half-disk, where the derivative is mostly positive. This is likely attributable to blade crossover, which is associated with a reduction in the normal sectional force [43].

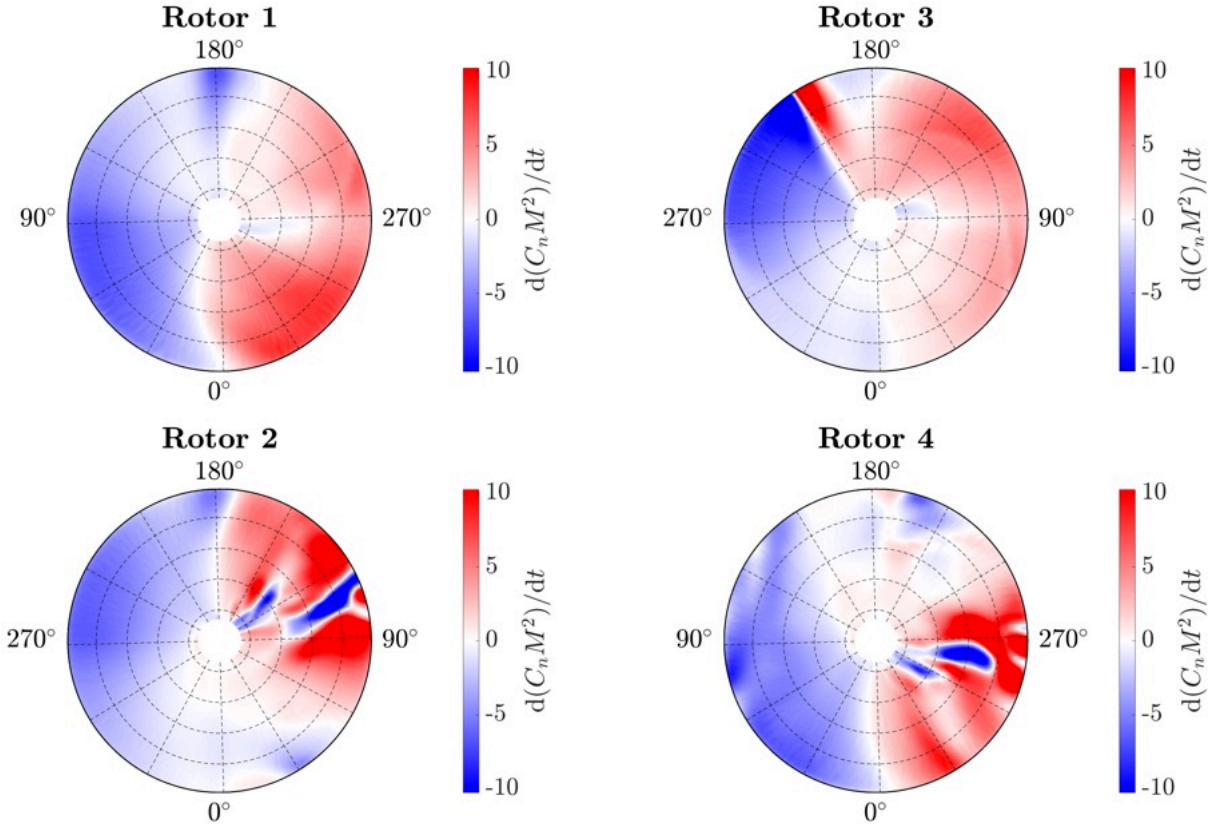


Figure 4.24: Time derivative contours of the sectional normal force coefficient C_nM^2 for the baseline configuration. The freestream flows from left to right.

This section is not meant to provide an in-depth characterization of the physics behind coaxial rotor interactions in forward flight, which is beyond the scope of the present study and impractical given the specificity of the present configuration. A thorough characterization of such phenomena is provided by Yuan *et al.* [43], to which the reader is referred for further detail.

4.8.2. VDM *vs.* SMM: load and power prediction uncertainty

Employing the SMM setup discussed in Sections 3.2.3 and 3.2.5, three characteristic configurations are simulated: the baseline configuration, the +10% configuration resulting

from the four-parameter trim optimization (Section 4.5.2), and the zero-pitch baseline configuration from the alternative design strategy (Section 4.7.2).

Baseline Starting from the baseline configuration, Table 4.13 reports the differences in the loads computed by VDM and SMM. The SMM results for the front rotors and the vehicle are averaged over one front-rotor period τ_{front}, while the performance of the rear rotors is averaged over τ_{back}. The integral aerodynamic loads of the vehicle show marginal sensitivity to the use of VDM or SMM. In particular, the wing lift is essentially unaffected, while the wing and total-vehicle drag values remain within a 5% variation. Notably, from Figure 4.25, the spanwise load distributions collapse neatly, suggesting that wing performance is equally described by the two methods in both integral and local terms. The major impact is on the lift generated by the vehicle components other than the wing, which shows a significant lift variation. This difference stems from the unsteady interaction between the blades and the vehicle, whose complexity apparently cannot be adequately captured by the steady-state VDM. Moreover, the VDM and SMM setups differ in several modeling aspects, such as the coupling of the pressure and velocity equations and the turbulence model. Given the substantial differences between the two methods, VDM and SMM results show excellent agreement in terms of vehicle aerodynamic loads.

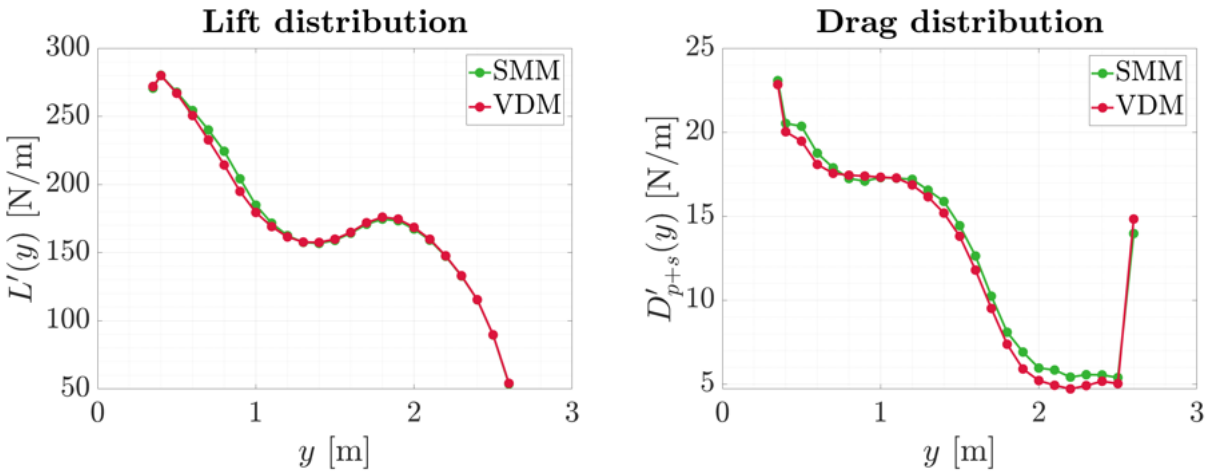


Figure 4.25: Baseline configuration spanwise lift and drag distributions comparison between VDM and SMM computations.

As expected, the most substantial deviations are within the rotor loads. The thrust magnitude shows a difference of up to 10%, with non-negligible uncertainty also in the thrust vector direction. δ_{Ti} is the tilt angle, i.e., the angle between the rotor axis and the thrust vector about the pitch axis. The high asymmetry observable in Figure 4.23 causes

the thrust vector to tilt upward, reducing the thrust component in the forward direction. Table 4.13 shows that the SMM predicts wider tilt angles, with a difference of up to 1.7° relative to the VDM. Finally, in percentage terms, the greatest variation lies in the torque computation, directly affecting power consumption, yielding differences between 15% and 25%.

Method	L^w [N]	D^w_{p+s} [N]	L^{Tot} [N]	D^{Tot}_{p+s} [N]	P [W]	δ_{T1} [°]	δ_{T2} [°]	δ_{T3} [°]	δ_{T4} [°]
VDM	408.7	29.7	524.9	95.5	5747	−4.4	−5.0	−5.9	−6.3
SMM	410.7	30.8	490.4	98.2	6755	−5.9	−5.8	−7.6	−7.2
$\boldsymbol{\Delta}_{\%}$	+0.5%	+3.5%	−6.6%	+2.8%	+17.5%				

Method	T_1 [N]	T_2 [N]	T_3 [N]	T_4 [N]	M_{z1} [N·m]	M_{z2} [N·m]	M_{z3} [N·m]	M_{z4} [N·m]
VDM	153.5	125.3	115.2	114.9	11.0	−9.7	−10.7	10.4
SMM	139.6	127.5	104.6	102.9	12.8	−12.2	−12.4	12.1
$\boldsymbol{\Delta}_{\%}$	−9.1%	+1.8%	−9.2%	−10.4%	+16.8%	+24.7%	+15.6%	+16.7%

Table 4.13: Baseline configuration: comparison of aerodynamic loads, rotor thrust, torque, and power consumption between VDM and SMM computations. Tilt angles δ_{Ti} denote the angle between the rotor axis and the thrust vector about the pitch axis.

From this brief analysis, it is inferred that a proper validation process is essential to reduce the uncertainty regarding the fidelity of the two methods when rotor performance is of concern. In particular, although for the baseline configuration the thrust magnitudes and their orientations lie within an acceptable uncertainty range, the power estimate is unreliable pending further experimental verification. However, both the VDM and SMM methods yield similar results in vehicle aerodynamic loads, especially over the wing, where the local effects also show marginal differences (Figure 4.25). Before drawing further conclusions, similar observations apply for the other two configurations tested with SMM.

10% configuration Table 4.14 reports the comparison between the VDM and SMM results for the +10% configuration retrieved from the four-parameter trim optimization (Section 4.5.2). The considerations made for the baseline configuration hold: the wing aerodynamic loads are only marginally affected by the choice of method, as confirmed by the load distributions (Figure 4.26), while the steady-state VDM solution is likely unable to capture the complex interaction between the blades and the remaining components of the vehicle. The differences in thrust computations are similar in magnitude to those

observed for the baseline configuration. However, the reduced thrust magnitude of the +10% configuration leads to a substantial increase in percentage deviation. Observations about the thrust tilt angle and torque are consistent with those in the previous paragraph.

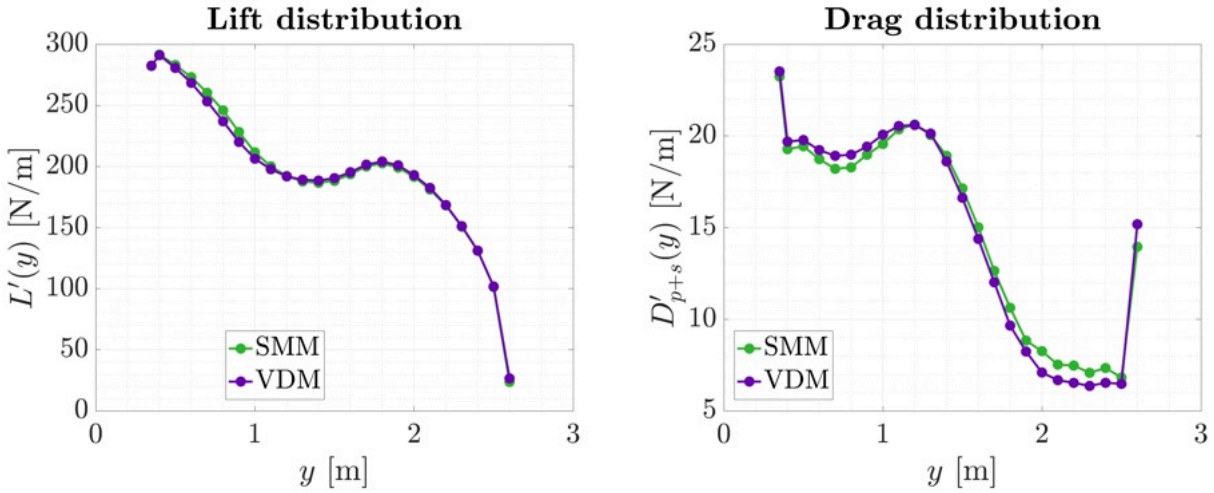


Figure 4.26: +10% configuration retrieved from the four-parameter trim optimization (Section 4.5.2). Spanwise lift and drag distributions comparison between VDM and SMM computations.

Method	L^w [N]	D^w_{p+s} [N]	L^{Tot} [N]	D^{Tot}_{p+s} [N]	P [W]	δ_{T1} [°]	δ_{T2} [°]	δ_{T3} [°]	δ_{T4} [°]
VDM	461.4	34.3	586.4	100.1	5585	−4.9	−5.5	−6.6	−6.9
SMM	463.6	34.9	548.6	104.2	6515	−6.7	−6.6	−9.1	−8.6
$\boldsymbol{\Delta}_{\%}$	+0.5%	+1.6%	−6.5%	+4.1%	+16.7%				

Method	T_1 [N]	T_2 [N]	T_3 [N]	T_4 [N]	M_{z1} [N·m]	M_{z2} [N·m]	M_{z3} [N·m]	M_{z4} [N·m]
VDM	139.3	113.8	98.7	102.6	10.9	−9.6	−10.1	10.1
SMM	121.6	108.8	84.9	85.3	12.6	−11.6	−11.7	11.7
$\boldsymbol{\Delta}_{\%}$	−12.7%	−4.4%	−14%	−16.8%	+16%	+20%	+15.2%	+15.4%

Table 4.14: +10% configuration retrieved from the four-parameter trim optimization (Section 4.5.2). Results comparison between VDM and SMM computations.

Owing to the large differences in the results of the two methods, none of the SMM configurations is correctly trimmed. The cruise-equilibrium search would require large computational resources, and would be minimally informative due to the lack of a validation

process. Therefore, within the SMM framework, it is not possible to assess whether the +10% configuration outperforms the baseline configuration in terms of power consumption. Without trimming, the SMM results yield a lower power consumption for the +10% configuration, with $\Delta P_{\%}^{\mathrm{SMM}} = -3.6\%$ relative to the baseline, compared with the VDM estimate of $\Delta P_{\%}^{\mathrm{VDM}} = -2.8\%$. Again, this result has no validity due to the absence of a trimming procedure and a validation process. Nevertheless, the qualitative agreement between the two estimates is noteworthy.

Zero-pitch baseline Table 4.15 reports the comparison between the VDM and SMM results for the zero-pitch baseline configuration from the alternative design strategy (Section 4.7.2). Compared with the other two configurations, the zero-pitch configuration exhibits larger deviations in wing performance, both in integral and local terms. In particular, Figure 4.27 reveals that the lift distributions computed by VDM and SMM collapse well, whereas the drag distributions exhibit non-negligible differences over the inboard sections. This is due to the more complex flow condition relative to the baseline: the angle of attack $\alpha = 0°$ causes the wakes shed from the front rotors and their hub to impinge on the wing, increasing the unsteady nature of the flow. It is reasonable that the distribution deviations occur in the inboard sections, where the rotor-induced effects peak. The steady-state simulation (VDM) smooths out unsteady effects, yielding a smoother drag distribution. It is worth noting that such a drag distribution does not depend on the averaging-window width $\Delta\tau$, as distributions computed over multiple values of $\Delta\tau$ exhibit consistent features.

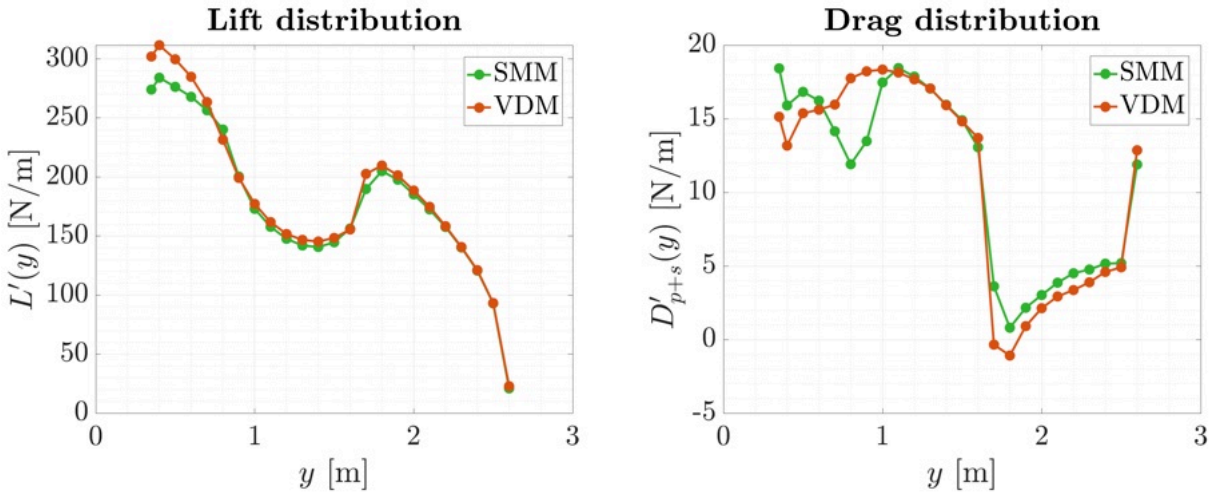


Figure 4.27: Zero-pitch baseline configuration from the alternative design strategy (Section 4.7.2). Spanwise lift and drag distributions comparison between VDM and SMM computations.

The uncertainty in the rotor loads is comparable to that of the other configurations tested. However, it offers a new interpretation of the results discussed in Section 4.7.2. The VDM computation leads to apparently inconsistent results, yielding increased power consumption despite enhanced vehicle and rotor performance relative to the baseline configuration. The SMM reveals a non-negligible uncertainty regarding thrust magnitude, its orientation, and power consumption. Assuming that the SMM better captures the latter features, the VDM yields a higher thrust magnitude, with a larger forward thrust component, at a lower power requirement. All of these elements reduce the rotor-related power consumption, thus limiting the advantage of installing a forward pusher. Were this assumption correct, the rotor power in trimmed conditions would be significantly increased such that, with the pusher power unchanged, its relative share would decrease, potentially reducing the overall power consumption relative to the baseline.
This reasoning is based on the apparent inconsistency of the results in Section 4.7.2. However, it has no validity without a thorough validation process. Moreover, with the shaft angle $\gamma_m = -5.6°$ and the rotor pitch $\gamma = \gamma_m + \alpha = -5.6°$, SMM results in Table 4.15 reveal a further limit to this alternative strategy. Three of the thrust tilt angles are larger than the rotor pitch $\delta_{Ti} > \gamma$, so that the thrust vectors are directed backward, with a component opposing the motion and contributing additional drag. Further studies must account for this phenomenon to select the optimal shaft angle, which balances the increased rotor performance associated with reduced rotor pitch with the possible drag increment due to the thrust tilt angle.

Method	L^w [N]	D^w_{p+s} [N]	L^{Tot} [N]	D^{Tot}_{p+s} [N]	P [W]	δ_{T1} [°]	δ_{T2} [°]	δ_{T3} [°]	δ_{T4} [°]
VDM	434.3	25.3	620.6	103.8	2220	−4.0	−4.0	−5.0	−5.4
SMM	422.6	25.8	593.7	103.6	2463	−5.3	−5.8	−7.1	−6.2
$\boldsymbol{\Delta}_{\%}$	−2.7%	+1.9%	−4.3%	−0.2%	+10.9%				

Method	T_1 [N]	T_2 [N]	T_3 [N]	T_4 [N]	M_{z1} [N·m]	M_{z2} [N·m]	M_{z3} [N·m]	M_{z4} [N·m]
VDM	117.0	102.6	100.5	88.0	5.8	−5.2	−6.8	6.0
SMM	103.7	92.4	80.7	81.1	6.2	−6.2	−7.0	7.0
$\boldsymbol{\Delta}_{\%}$	−11.4%	−9.9%	−19.7%	−7.8%	+6.7%	+19.5%	+3.1%	+16.9%

Table 4.15: Zero-pitch baseline configuration from the alternative design strategy (Section 4.7.2). Results comparison between VDM and SMM computations.

In conclusion, neither the VDM nor the SMM has been validated against experimental

data, yielding predictions of uncertain fidelity. The comparison between the two methods provides an estimate of the uncertainty associated with these approaches, revealing that the integrated vehicle aerodynamic loads are marginally affected by the choice of method. Local effects are consistently reproduced by both methods over the wing in configurations where the rotor wakes do not impinge directly on it. In the zero-pitch configuration, however, the drag distributions over the inboard sections differ between the two methods, suggesting that direct rotor-wake impingement on the wing introduces unsteady phenomena whose time-averaged effect cannot be adequately reproduced by the steady-state VDM. Predictably, greater discrepancies are observed in rotor loads, where the uncertainty in thrust-vector magnitude and orientation is non-negligible. The most pronounced deviation concerns torque estimation, resulting in power-consumption deviations of nearly 20% between the two methods.

In summary, based on these observations and given the prohibitive computational cost of the SMM, the VDM is preferable for predicting wing aerodynamic loads and local effects in the absence of direct rotor-wake interaction. Rotor load predictions, however, carry substantial uncertainty, and validation against experimental data is required to establish the relative fidelity of the two methods. For design-oriented studies involving rotor performance, the VDM should therefore be employed with awareness of its limitations, pending dedicated experimental validation.

5 | Conclusions and Future Developments

The present study investigated the aerodynamic performance of the W90 hybrid quad-rotor eVTOL in cruise conditions, aiming to reduce power consumption through modifications to the wing geometry and its position relative to the rotors. This work was motivated by the destructive interference between the front-rotor wakes and the wing, which locally causes a reduction in the effective angle of attack over most of the wingspan, degrading performance in terms of both lift and drag.

A CFD framework based on the RANS equations was set up using *STAR-CCM+* and employing the Virtual Disk Model (VDM) as the primary modeling tool, which offers an appealing compromise between computational cost and accuracy. This model was validated against available Sliding Mesh Method (SMM) results under axial-flow conditions, exhibiting good agreement near the operating rotational speed. Moreover, mesh and time independence studies were conducted to ensure solution reliability.

Consistent with the literature, the analysis of the baseline configuration confirmed that rotor-wing interference is the dominant phenomenon governing performance degradation. The front rotors induce a severe local downwash effect over most of the wingspan, peaking near mid-span, and a mild upwash effect outboard, resulting in a net reduction in lift and lift-to-drag ratio compared with the rotor-off configuration. In particular, the lift-to-drag ratio was reduced by 16.9% for the wing and by 10.6% for the complete vehicle.

Several design strategies were explored to mitigate this interference:

Downwash compensation strategy Increasing the twist in the rotor-wing interference region successfully restored the wing lift distribution toward the rotor-off condition. However, the associated drag penalty and the more nose-down trim condition degraded rotor efficiency, resulting in unchanged or increased power consumption. This strategy was therefore deemed ineffective.

Twist distribution optimization A surrogate-based optimization framework was constructed using Latin Hypercube Sampling (LHS), Kriging surrogate models, and a Genetic Algorithm within *Dakota*. The optimization objective was to maximize the lift-to-drag ratio under minimum-lift, geometric, stall-margin, and tip-loading constraints. The optimal twist distribution redistributed the loads toward the tip, reducing lift in the rotor-wing interference region and saturating the tip-loading constraint. The net improvement in the wing lift-to-drag ratio was only 2.4%, translating into negligible power savings.

Lift-based wing design strategy Six configurations with prescribed lift requirements ranging from -15% to $+15\%$ relative to the baseline were optimized. Due to the varied aerodynamic forces, a cruise-equilibrium search was required for each design and was performed using two different sets of trimming parameters. When trimmed via global pitch angle $(\alpha, \Omega_{\text{front}}, \Omega_{\text{back}})$, most configurations yielded modest power reductions. When trimmed via rotor pitch angle $(\gamma, \Omega_{\text{front}}, \Omega_{\text{back}})$, configurations with increased wing lift showed greater power savings. Since the $+10\%$ configuration exhibited a reduction with both trim strategies, it was natural to combine them through a four-parameter sweep $(\alpha, \gamma, \Omega_{\text{front}}, \Omega_{\text{back}})$. Overall, according to VDM predictions, a power reduction of $\Delta P_{\%} = -2.8\%$ for the $+10\%$ configuration at $\alpha = -10.45^\circ$ and $\gamma = -18.61^\circ$ was obtained. The key result is that cruise efficiency is governed more by vertical-load sharing between the wing and the rotors, together with the trim strategy, than by local twist optimization alone.

Wing position optimization Forward wing displacement increased front-rotor interference and reduced the beneficial effects of the rear rotors, degrading both wing and vehicle performance. The baseline position ($s = 0$) was confirmed as optimal, and no further investigation accounting for trim was deemed worthwhile.

Alternative design (forward pusher) A low-fidelity assessment of a dual-system configuration, with lift rotors operating at reduced pitch and a dedicated pusher propeller, was conducted. The $+15\%$ configuration, when modified with a forward pusher, exhibited a reduction in power consumption, confirming the potential of this strategy due to rotors operating at lower pitch angles and thus higher efficiency. To explore whether a more aggressive reduction in rotor pitch could yield even greater benefits, the zero-pitch baseline configuration was investigated. In this configuration, the rotors operate at $\gamma = \gamma_m = -5.6^\circ$, while the wing incidence is adjusted to maintain an adequate angle of attack. Despite improvements in both vehicle aerodynamic performance (lift-to-drag ratio increased by $\sim 9\%$) and rotor thrust-to-power ratio ($+107\%$ relative to the baseline), the

total power consumption increased compared with the baseline. This apparently inconsistent result was discussed within the assessment of the uncertainty associated with the rotor modeling.

Model Uncertainty assessment The comparison between VDM and SMM for three representative configurations reveals consistent predictions of wing aerodynamic loads and spanwise distributions, with deviations below 5% for the baseline and +10% configurations. Rotor thrust magnitudes, however, exhibit larger discrepancies of up to 10–15%, and thrust tilt angles differ by up to 1.7°. The largest uncertainty concerns torque estimation and, consequently, power consumption, where deviations range between 15% and 25%. Finally, in configurations where rotor wakes impinge directly on the wing – such as the zero-pitch case – drag distributions differ between the two methods, suggesting that the steady-state VDM may not fully capture the time-averaged effects of unsteady wake impingement.

Main conclusions In conclusion, rotor-wing interaction is the dominant cause of performance degradation in the W90 cruise configuration. The front rotors induce a local downwash that reduces wing lift and increases pressure drag, lowering the wing lift-to-drag ratio by 16.9% compared with the rotor-off condition. Twist distribution optimization alone is ineffective when wing lift is fixed, as the optimal twist redistributes loads toward the tip but the net gain in lift-to-drag ratio is marginal (+2.4%) and does not translate into a meaningful reduction in power consumption. The lift-based design strategy, combined with four-parameter trim optimization (α, γ, Ω_{front}, Ω_{back}), achieves a 2.8% reduction in power consumption, demonstrating that redistributing the vertical force burden toward the wing can yield efficiency improvements. Forward wing displacement amplifies front-rotor interference and reduces the beneficial upwash from the rear rotors, thus degrading performance; the baseline position (s = 0) is confirmed as optimal. The VDM is reliable for wing aerodynamic predictions in cruise, but rotor load and power estimates exhibit significant uncertainty. Torque predictions deviate by 15% to 25% compared with non-validated SMM results. Experimental validation is essential before drawing definitive conclusions on rotor performance. The alternative dual-system configuration shows potential, but the zero-pitch baseline case reveals an inconsistency: vehicle aerodynamic performance and rotor efficiency improve, yet total power increases. This apparent contradiction lies within the uncertainty inherent to the modeling approaches. The VDM was validated only in axial flow, and its reliability in edgewise flight is not guaranteed. The uncertainty assessment shows that VDM and SMM predictions differ substantially, but neither method has been validated against experimental data. Therefore, it cannot

be concluded whether the zero-pitch configuration truly increases power consumption or whether the VDM estimate is overly pessimistic.

Future developments Several directions for future work emerge from the present study. Given the uncertainty associated with rotor load and power estimates, experimental validation through wind-tunnel tests represents an essential next step. Such tests are required to validate both VDM and SMM predictions, particularly for rotor loads and power consumption in forward flight. Once the rotor modeling is validated, Bayesian optimization (e.g., Efficient Global Optimization) could refine the current surrogate-based results with fewer additional CFD evaluations. The present study kept the wing planform fixed; investigating the sweep angle or the taper ratio may yield further improvements. The present work introduced a forward pusher while retaining the original wing and rotor layout. More radical modifications, inspired by the Cento and Pipistrel concepts [10, 27], could be explored. The twist distribution of the D5620 rotor could be optimized specifically for the cruise condition to improve edgewise-flow efficiency. Finally, the present work focused exclusively on cruise; extending the analysis to hover and transition phases would provide a complete vehicle performance assessment.

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